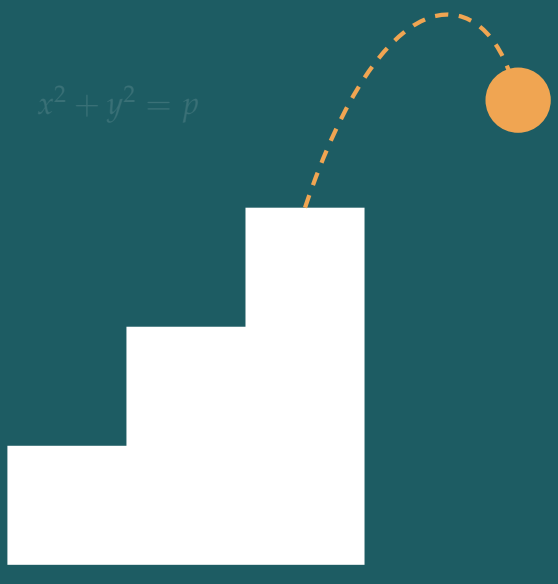
STEP PREPARATION · UNIVERSITY MATHEMATICS

Beyond STEP

STEP I, II AND III

2017 Worked Solutions



KEITH McNULTY

Beyond STEP

Past STEP questions as a doorway
to university mathematics

STEP I, II AND III

2017 Worked Solutions

Keith McNulty

Beyond STEP: Past STEP questions as a doorway to university mathematics

2017 Worked Solutions

Worked solutions to all 39 questions of the 2017 STEP papers (STEP I, II and III). Worked solutions to every STEP question since the specimen papers, the whole book *Beyond STEP* and every chapter on its own are free to read and download at `beyondstep.keithmcnulty.org`.

Have a proper go at a question before you read its solution. Every answer has been checked numerically, but with this many solutions some slips are inevitable: if something looks wrong, trust your own working and check it carefully.

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Corrections are welcome as issues at github.com/keithmcnulty/step_book_mcnulty.

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WORKED SOLUTION

STEP I 2017, Question 1

Pure • Integration by Substitution

STEP I 2017, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s1-q1

The key idea

Each of these integrands is secretly of the form $f'(x)/f(x)$ (or, in the last case, $f'(x)/f(x)^2$), once we clear the trigonometric fractions. The derivative of $x\sin x + \cos x$ is unusually tidy, because the two $\sin x$ terms cancel, and the other substitutions are built on the same cancellation. So the plan is: rewrite each integrand in terms of sin and cos (or sec and tan), spot the denominator whose derivative matches the numerator, and substitute.

Solution

Part (i)

With $u = x\sin x + \cos x$, the product rule gives

$$\frac{\mathrm{d}u}{\mathrm{d}x} = \sin x + x\cos x - \sin x = x\cos x.$$

Now multiply the top and bottom of the integrand by $\cos x$:

$$\frac{x}{x\tan x + 1} = \frac{x\cos x}{x\sin x + \cos x}.$$

The numerator is exactly $\mathrm{d}u/\mathrm{d}x$, so $x\cos x\,\mathrm{d}x = \mathrm{d}u$ and

$$\int \frac{x}{x\tan x + 1}\,\mathrm{d}x = \int \frac{1}{u}\,\mathrm{d}u = \ln|u| + C = \ln|x\sin x + \cos x| + C.$$

For the second integral we do the same thing with $\sin x$ in place of $\cos x$. Multiplying top and bottom by $\sin x$,

$$\frac{x}{x\cot x - 1} = \frac{x\sin x}{x\cos x - \sin x}.$$

Take $v = x\cos x - \sin x$. Then

$$\frac{\mathrm{d}v}{\mathrm{d}x} = \cos x - x\sin x - \cos x = -x\sin x,$$

so the numerator is $-\mathrm{d}v/\mathrm{d}x$ and

$$\int \frac{x}{x\cot x - 1}\,\mathrm{d}x = -\int \frac{1}{v}\,\mathrm{d}v = -\ln|x\cos x - \sin x| + C.$$

Don't lose the minus sign: it comes from differentiating $\cos x$.

Part (ii)

The first integrand already has a denominator worth trying. Let $w = x\sec^2 x - \tan x$. Using $\frac{\mathrm{d}}{\mathrm{d}x}\sec^2 x = 2\sec^2 x\tan x$,

$$\frac{\mathrm{d}w}{\mathrm{d}x} = \sec^2 x + 2x\sec^2 x\tan x - \sec^2 x = 2x\sec^2 x\tan x.$$

Once again the unwanted terms cancel, and the numerator is $\frac{1}{2}\,\mathrm{d}w/\mathrm{d}x$. Hence

$$\int \frac{x\sec^2 x\,\tan x}{x\sec^2 x - \tan x}\,\mathrm{d}x = \frac{1}{2}\int \frac{1}{w}\,\mathrm{d}w = \tfrac{1}{2}\ln|x\sec^2 x - \tan x| + C.$$

? **Ask yourself** The last integral has no sec or tan in it. How can it be related to $w = x\sec^2 x - \tan x$?

Factor $\cos^2 x$ out of the bracket in the denominator:

$$x - \sin x\cos x = \cos^2 x\left(x\sec^2 x - \frac{\sin x}{\cos x}\right) = \cos^2 x\,(x\sec^2 x - \tan x) = w\cos^2 x.$$

So $(x - \sin x\cos x)^2 = w^2\cos^4 x$, and

$$\frac{x\sin x\cos x}{(x-\sin x\cos x)^2} = \frac{x\sin x\cos x}{w^2\cos^4 x} = \frac{x\sec^2 x\tan x}{w^2},$$

since $\dfrac{\sin x\cos x}{\cos^4 x} = \dfrac{\sin x}{\cos x}\cdot\dfrac{1}{\cos^2 x}$. The numerator is again $\frac{1}{2}\,\mathrm{d}w/\mathrm{d}x$, so with the same substitution

$$\begin{aligned}\int \frac{x\sin x\cos x}{(x-\sin x\cos x)^2}\,\mathrm{d}x &= \frac{1}{2}\int \frac{1}{w^2}\,\mathrm{d}w = -\frac{1}{2w} + C\\ &= -\frac{1}{2(x\sec^2 x - \tan x)} + C = -\frac{\cos^2 x}{2(x-\sin x\cos x)} + C,\end{aligned}$$

where the last step multiplies top and bottom by $\cos^2 x$. If you prefer double angles, this is $-\dfrac{\cos^2 x}{2x - \sin 2x} + C$.

Checks

- Differentiate each answer and make sure you get the integrand back. For the last one, the quotient rule on $-\cos^2 x/\big(2(x-\sin x\cos x)\big)$ uses $\frac{\mathrm{d}}{\mathrm{d}x}(x-\sin x\cos x) = 1 - \cos 2x = 2\sin^2 x$, and the $\sin^2 x\cos^2 x$ terms combine neatly.

- Numerical evaluation: over $[0.3, 1.2]$ the first integral is $0.34953\ldots$, and $\ln|x\sin x + \cos x|$ changes by the same amount; the last integral over the same interval is $25.7364\ldots$, matching $-\cos^2 x/\big(2(x - \sin x\cos x)\big)$. All four answers were confirmed this way by computer.

Common slips

- Substituting without first clearing $\tan x$ or $\cot x$: the numerator only matches $\mathrm{d}u/\mathrm{d}x$ after multiplying through by $\cos x$ or $\sin x$.
- Dropping the minus sign in the second integral of part (i), or the factor $\frac{1}{2}$ in part (ii).
- Reaching for a logarithm in the last integral out of habit: $1/w^2$ integrates to $-1/w$.

Beyond STEP

The trick here is reverse-engineering: choose the substitution so that the derivative of the denominator, after cancellation, is exactly the numerator. Behind this lies the fact that $\int f'/f = \ln|f|$, the logarithmic derivative, which reappears in complex analysis (counting zeros of a function) and in differential equations as the integrating factor. Many STEP integrals are built by differentiating a carefully chosen function and hiding the answer; Siklos, *Advanced Problems in Mathematics*, has plenty more practice in spotting them.

WORKED SOLUTION

STEP I 2017, Question 2

Pure • Integrating Inequalities

STEP I 2017, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s1-q2

The key idea

Integration preserves inequalities: if $f(t) \leq g(t)$ on an interval $[a, b]$ with $a < b$, then $\int_a^b f \leq \int_a^b g$. The only care needed is with the direction of integration, because when $x < 1$ the interval $[1, x]$ runs backwards and the inequality flips. In the final part the same idea is used again: integrating a non-negative function from 1 to y gives something with the same sign as $y - 1$, and that sign is exactly what we need to divide through safely.

Solution

Part (i)

Case $x \geq 1$. For $1 \leq t \leq x$ we have $\frac{1}{t} \leq 1$, and the interval runs forwards, so

$$\int_1^x \frac{1}{t}\,\mathrm{d}t \leq \int_1^x 1\,\mathrm{d}t, \qquad \text{that is} \qquad \ln x \leq x - 1.$$

Case $0 < x \leq 1$. Now we integrate over $x \leq t \leq 1$, where $t \leq 1$ and so $\frac{1}{t} \geq 1$. Then

$$\int_x^1 \frac{1}{t}\,\mathrm{d}t \geq \int_x^1 1\,\mathrm{d}t, \qquad \text{that is} \qquad -\ln x \geq 1 - x.$$

Multiplying by -1 reverses the inequality and gives $\ln x \leq x - 1$ again. So $(*)$ holds for all $x > 0$.

Part (ii)

Case $x \geq 1$. For $t \geq 1$ we have $\frac{1}{t^2} \leq \frac{1}{t}$, so

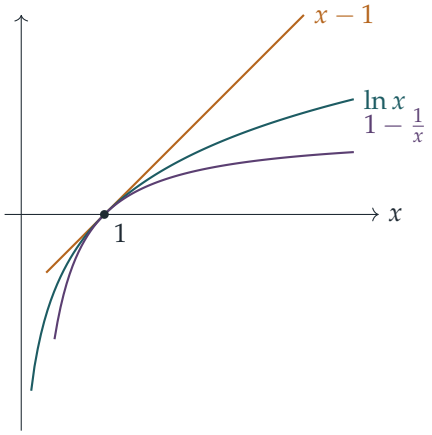
$$\int_1^x \frac{1}{t^2}\,\mathrm{d}t \leq \int_1^x \frac{1}{t}\,\mathrm{d}t, \qquad \text{that is} \qquad 1 - \frac{1}{x} \leq \ln x.$$

Case $0 < x \leq 1$. For $x \leq t \leq 1$ the inequality reverses, $\frac{1}{t^2} \geq \frac{1}{t}$ (multiply $t \leq 1$ by the positive number $\frac{1}{t^2}$). Integrating over $[x, 1]$,

$$\frac{1}{x} - 1 \geq -\ln x,$$

and multiplying by -1 gives $\ln x \geq 1 - \frac{1}{x}$. So $(**)$ holds for all $x > 0$.

(Alternatively, $(**)$ follows from $(*)$ at once: put $1/x$ in place of x in $(*)$ to get $-\ln x \leq \frac{1}{x} - 1$.)



All three curves touch at $x = 1$, where each has gradient 1; $\ln x$ is squeezed between the other two.

Part (iii)

Both $(*)$ and $(**)$ say that a certain function is non-negative on $x > 0$:

$$f(x) = x - 1 - \ln x \geq 0, \qquad g(x) = \ln x - 1 + \frac{1}{x} \geq 0.$$

We need a simple fact about integrating such a function.

? **Ask yourself** If $h(x) \geq 0$ for all $x > 0$, what is the sign of $\int_1^y h(x)\,\mathrm{d}x$ when $y < 1$?

If $y > 1$ the integral is over a forward interval, so it is ≥ 0. If $0 < y < 1$, then $\int_1^y h = -\int_y^1 h \leq 0$. In either case

$$(y - 1)\int_1^y h(x)\,\mathrm{d}x \;\geq\; 0.$$

This lets us handle both $y > 1$ and $0 < y < 1$ together.

We need $\int \ln x \, \mathrm{d}x = x \ln x - x$ (integrating by parts with $u = \ln x$), so

$$\int_1^y \ln x \, \mathrm{d}x = y \ln y - y + 1.$$

The right-hand inequality. Integrating f,

$$\begin{aligned}\int_1^y f(x) \, \mathrm{d}x &= \frac{(y-1)^2}{2} - \big(y \ln y - y + 1\big) = \frac{(y-1)^2}{2} + (y-1) - y \ln y \\ &= \frac{(y-1)(y+1)}{2} - y \ln y.\end{aligned}$$

By the fact above,

$$(y-1)\left[\frac{(y-1)(y+1)}{2} - y \ln y\right] \geq 0 \quad \Longrightarrow \quad \frac{(y-1)^2(y+1)}{2} \geq y(y-1) \ln y.$$

Since $y > 0$ and $y \neq 1$, the number $y(y-1)^2$ is positive, and dividing by it keeps the direction:

$$\frac{y+1}{2y} \geq \frac{\ln y}{y-1}.$$

The left-hand inequality. Integrating g,

$$\int_1^y g(x) \, \mathrm{d}x = \big(y \ln y - y + 1\big) - (y-1) + \ln y = (y+1) \ln y - 2(y-1).$$

By the fact above,

$$(y-1)\big[(y+1) \ln y - 2(y-1)\big] \geq 0 \quad \Longrightarrow \quad (y+1)(y-1) \ln y \geq 2(y-1)^2.$$

Now divide by $(y+1)(y-1)^2 > 0$:

$$\frac{\ln y}{y-1} \geq \frac{2}{y+1}.$$

Putting the two together,

$$\frac{2}{y+1} \leq \frac{\ln y}{y-1} \leq \frac{y+1}{2y} \qquad \text{for } y > 0,\ y \neq 1,$$

as required.

Ask yourself Why multiply by $y-1$ before dividing, rather than dividing straight away?

Dividing an inequality by $y-1$ would reverse it when $y < 1$, forcing two separate cases. Multiplying by $y-1$ first produces an expression containing $(y-1)^2$, which is positive whichever side of 1 we are on, so one argument covers both cases.

Checks

- At $y = 2$: $\frac{2}{3} \approx 0.667 \leq \ln 2 \approx 0.693 \leq \frac{3}{4} = 0.75$.
- At $y = \frac{1}{2}$: $\frac{4}{3} \approx 1.333 \leq 2\ln 2 \approx 1.386 \leq \frac{3}{2}$.
- As $y \to 1$ all three expressions tend to 1 (the middle one is the gradient of ln at 1), consistent with the curves touching in the diagram.
- Replacing y by $1/y$ and then dividing through by y gives back exactly the same pair of inequalities, so the result for $0 < y < 1$ is equivalent to the result for $y > 1$.

Common slips

- Integrating over $[1, x]$ when $x < 1$ without noticing that the inequality reverses.
- Dividing by $y - 1$ without considering its sign, and so proving part (iii) only for $y > 1$.

Beyond STEP

The middle quantity $\dfrac{\ln y}{y-1}$ is the reciprocal of the *logarithmic mean* of 1 and y, so the left-hand inequality says that the logarithmic mean is at most the arithmetic mean $\frac{1+y}{2}$. In fact the logarithmic mean is also at least the geometric mean $\sqrt{y}$, which gives $\frac{\ln y}{y-1} \leq \frac{1}{\sqrt{y}}$, a sharper upper bound than $\frac{y+1}{2y}$. The technique of integrating an inequality, and the monotonicity of the integral that justifies it, are basic tools in a first analysis course. Abbott's *Understanding Analysis* treats the integral carefully.

WORKED SOLUTION

STEP I 2017, Question 3

Pure • Tangents to a Parabola

STEP I 2017, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s1-q3

The key idea

Everything here can be written in terms of the parameters p and q, so the plan is to find the coordinates of R, S and T and then compute both areas directly. Triangle RST is especially friendly because its side ST lies along the y-axis: its height is simply the distance of R from that axis. For triangle OPQ, with a vertex at the origin, the formula $\frac{1}{2}|x_1y_2 - x_2y_1|$ does the work in one line.

Solution

The tangent at P

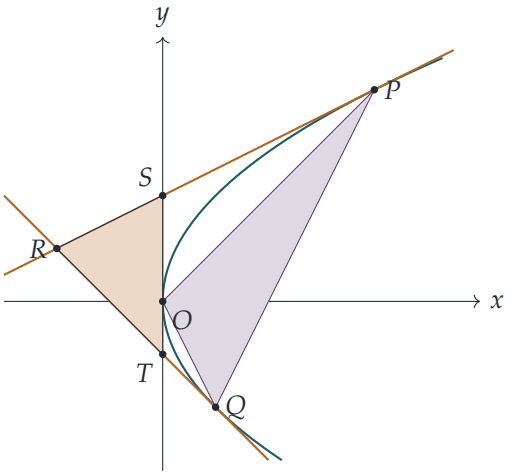
Differentiating $y^2 = 4ax$ implicitly gives $2y\dfrac{\mathrm{d}y}{\mathrm{d}x} = 4a$, so $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{2a}{y}$. At P, where $y = 2ap \neq 0$ (since $a, p > 0$), the gradient is

$$\frac{2a}{2ap} = \frac{1}{p}.$$

The tangent through $P(ap^2, 2ap)$ is therefore

$$y - 2ap = \frac{1}{p}\left(x - ap^2\right) \quad \Longrightarrow \quad y = \frac{1}{p}x - ap + 2ap = \frac{1}{p}x + ap,$$

as required. In the same way, since Q has the same form with q in place of p (and $q \neq 0$), the tangent at Q is $y = \dfrac{1}{q}x + aq$.



The sketch shows the case $a = 1$, $p = 2$, $q = -1$: R lies to the left of the y-axis, between S and T in height.

The points R, S and T

Putting $x = 0$ in each tangent gives

$$S = (0,\, ap), \qquad T = (0,\, aq).$$

At R the two tangents meet, so

$$\frac{x}{p} + ap = \frac{x}{q} + aq \quad \Longrightarrow \quad x\left(\frac{1}{p} - \frac{1}{q}\right) = a(q-p) \quad \Longrightarrow \quad x \cdot \frac{q-p}{pq} = a(q-p).$$

Since $p > 0 > q$ we have $q - p \neq 0$, so we may divide by it (this is also why the tangents are not parallel and R exists). Hence $x = apq$, and then $y = \dfrac{apq}{p} + ap = a(p+q)$:

$$R = \big(apq,\ a(p+q)\big).$$

Ask yourself Which side of the y-axis is R on, and why does it matter for the area?

Because $p > 0$ and $q < 0$, the product pq is negative, so the x-coordinate apq of R is negative: R is to the left of the axis, at distance $a|pq| = -apq$ from it. Areas must be positive, so we keep track of signs rather than writing apq as a "height".

Area of triangle RST

Take ST as the base. It lies along the y-axis with length $ap - aq = a(p-q)$, which is positive since $p > q$. The perpendicular height from R to the y-axis is $-apq$. So

$$[RST] = \tfrac{1}{2} \cdot a(p-q) \cdot (-apq) = -\tfrac{1}{2}a^2pq(p-q).$$

Area of triangle OPQ

For a triangle with vertices O, (x_1, y_1) and (x_2, y_2), the area is $\frac{1}{2}|x_1y_2 - x_2y_1|$. With $P(ap^2, 2ap)$ and $Q(aq^2, 2aq)$,

$$x_1y_2 - x_2y_1 = ap^2 \cdot 2aq - aq^2 \cdot 2ap = 2a^2pq(p-q).$$

Here $a^2 > 0$, $pq < 0$ and $p - q > 0$, so this is negative, and

$$[OPQ] = \tfrac{1}{2} \cdot \left(-2a^2pq(p-q)\right) = -a^2pq(p-q).$$

Conclusion

Comparing the two results,

$$[OPQ] = -a^2pq(p-q) = 2\left(-\tfrac{1}{2}a^2pq(p-q)\right) = 2\,[RST],$$

so the area of triangle OPQ is twice the area of triangle RST, as required.

Checks

In the sketched case $a = 1$, $p = 2$, $q = -1$ we get $P(4,4)$, $Q(1,-2)$, $R(-2,1)$, $S(0,2)$ and $T(0,-1)$. Then $[RST] = \frac{1}{2} \cdot 3 \cdot 2 = 3$ and $[OPQ] = \frac{1}{2}|4 \cdot (-2) - 1 \cdot 4| = 6$, twice as big. You can also check that R satisfies both tangent equations: $1 = \frac{-2}{2} + 2$ and $1 = \frac{-2}{-1} - 1$. The identity was also confirmed by computer, in exact arithmetic, for 2000 random choices of a, p and q.

Common slips

- Writing the height of triangle RST as apq: this is negative, because R is on the far side of the y-axis from the curve.
- Dropping the modulus in $\frac{1}{2}|x_1y_2 - x_2y_1|$ and ending up with a negative area for OPQ.
- Dividing by $q - p$ without saying why it is non-zero.

Beyond STEP

The parametrisation $(at^2, 2at)$ makes every point of the parabola correspond to one number t, and the tangents at $t = p$ and $t = q$ always meet at $(apq, a(p+q))$: the "product" and "sum" of the parameters. Results like this, together with the $2:1$ area ratio here, are part of the classical projective geometry of conics, where tangents and chords are studied using parameters and determinants. The determinant formula for the area of a triangle is the starting point for seeing area as a signed, linear quantity, which is developed in Strang, *Introduction to Linear Algebra*; Siklos, *Advanced Problems in Mathematics*, has more STEP-style practice with parametric curves.

WORKED SOLUTION

STEP I 2017, Question 4

Pure • Quadratics and Geometric Series

STEP I 2017, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s1-q4

The key idea

Everything depends on how many values of r in the open interval $(-1,1)$ give a particular value of p (or q). That is a question about a parabola restricted to $-1 < r < 1$: a horizontal line meets it once, twice or not at all, and the sketch tells us which. Once we know r, the sum is fixed, and because the sum is a one-to-one function of r, two values of r always give two different sums. To find the equation satisfied by the sums, we write r in terms of the sum and substitute.

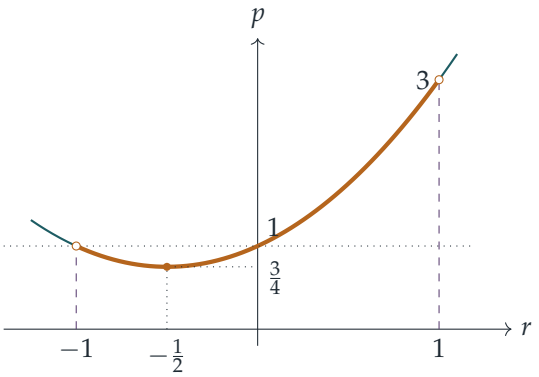
Solution

Part (i)

Completing the square,

$$1 + r + r^2 = \left(r + \tfrac{1}{2}\right)^2 + \tfrac{3}{4},$$

so the graph is a parabola with its minimum point at $\left(-\tfrac{1}{2}, \tfrac{3}{4}\right)$. It passes through $(-1,1)$, $(0,1)$ and $(1,3)$.



On $-1 < r < 1$ (the thick arc) the function decreases from 1 (not attained, since $r = -1$ is excluded) to its minimum $\frac{3}{4}$ at $r = -\frac{1}{2}$, then increases towards 3 (again not attained, since $r = 1$ is excluded). Hence

$$\tfrac{3}{4} \leq p < 3.$$

When $1 < p < 3$. The values of r giving this p are the roots of

$$r^2 + r + (1 - p) = 0.$$

The product of the roots is $1 - p < 0$, so there is one positive root and one negative root. Their sum is -1, so the negative root equals -1 minus the positive root, which is less than -1 and therefore not allowed. The positive root is

$$r = \frac{-1 + \sqrt{4p - 3}}{2},$$

and since $1 < 4p - 3 < 9$ we have $0 < r < 1$, so it is allowed. (On the graph: the line at height $p \in (1, 3)$ meets the thick arc only on the rising part $0 < r < 1$.) So p determines r uniquely, and hence $S = 1/(1 - r)$ uniquely.

When $\frac{3}{4} < p < 1$. Now the discriminant $1 - 4(1 - p) = 4p - 3$ is positive, so the quadratic has two distinct real roots. Their product $1 - p$ is positive and their sum is -1, so both are negative. Each root equals -1 minus the other, and the other is negative, so each root is greater than -1. Both roots therefore lie in $(-1, 0)$ and both are allowed.

> **Ask yourself** Two values of r, fine; but why must they give two *different* values of S?

Because $S = \dfrac{1}{1 - r}$ can be inverted: $r = 1 - \dfrac{1}{S}$. Different values of r therefore give different values of S, so there are exactly two possible values of S.

To find the equation they satisfy, substitute $r = 1 - u$ with $u = 1/S$ into $r^2 + r + 1 - p = 0$:

$$\begin{aligned}(1 - u)^2 + (1 - u) + 1 - p &= 0 \\ u^2 - 3u + (3 - p) &= 0.\end{aligned}$$

Putting $u = 1/S$ and multiplying by S^2 (which is non-zero) gives

$$(3 - p)S^2 - 3S + 1 = 0,$$

as required. Both values of S satisfy it, since both come from allowed roots r.

Part (ii)

Completing the square again,

$$q = 1 + 2r + 3r^2 = 3\left(r + \tfrac{1}{3}\right)^2 + \tfrac{2}{3}.$$

The minimum is $\frac{2}{3}$ at $r = -\frac{1}{3}$; at $r = -1$ the value is 2, and at $r = 1$ it is 6. So on $-1 < r < 1$:

- on the falling part, $-1 < r \leq -\frac{1}{3}$, q takes each value in $\left[\frac{2}{3}, 2\right)$ exactly once;
- on the rising part, $-\frac{1}{3} \leq r < 1$, q takes each value in $\left[\frac{2}{3}, 6\right)$ exactly once.

The two parts share only the point $r = -\frac{1}{3}$. Hence a value of q comes from

- exactly one r if $q = \frac{2}{3}$ (namely $r = -\frac{1}{3}$) or $2 \leq q < 6$ (rising part only);
- exactly two values of r if $\frac{2}{3} < q < 2$ (one on each part);
- no r at all otherwise.

For instance, at $q = 2$ the equation $3r^2 + 2r - 1 = 0$ factorises as $(3r - 1)(r + 1) = 0$, and the root $r = -1$ is excluded, leaving only $r = \frac{1}{3}$.

As before, T determines r: since $1 - r > 0$, we have $1 - r = 1/\sqrt{T}$. So different r give different T. Therefore

$$T \text{ is determined uniquely when } q = \tfrac{2}{3} \text{ or } 2 \leq q < 6,$$

and

$$T \text{ has two possible values when } \tfrac{2}{3} < q < 2.$$

The quadratic. Write $r = 1 - u$, where $u = 1 - r = 1/\sqrt{T} > 0$. Then

$$q = 1 + 2(1 - u) + 3(1 - u)^2 = 3u^2 - 8u + 6.$$

Since $u^2 = 1/T$, this reads $\dfrac{3}{T} + 6 - q = \dfrac{8}{\sqrt{T}}$. Multiplying by T,

$$(6 - q)T + 3 = 8\sqrt{T}.$$

This is already a quadratic in $\sqrt{T}$. To get one in T itself, square both sides:

$$(6 - q)^2 T^2 + 6(6 - q)T + 9 = 64T,$$

that is,

$$(6 - q)^2 T^2 - 2(3q + 14) T + 9 = 0.$$

Both possible values of T satisfy this. Squaring can't have brought in a stray root here. With $q < 6$ and $T > 0$, the left-hand side $(6 - q)T + 3$ is positive, so the alternative $(6 - q)T + 3 = -8\sqrt{T}$ has no solutions with $T > 0$. Also, the quadratic in T has positive product $9/(6 - q)^2$ and positive sum, so its roots are both positive. So its two roots are exactly the two values of T.

Checks

- Take $p = \frac{7}{8}$. Then $r^2 + r + \frac{1}{8} = 0$ gives $r = \frac{-2 \pm \sqrt{2}}{4}$, both in $(-1, 0)$. The corresponding sums $S \approx 0.873$ and $S \approx 0.540$ satisfy $\frac{17}{8}S^2 - 3S + 1 = 0$, whose roots have sum $\frac{24}{17} \approx 1.412$ and product $\frac{8}{17}$.

- Take $q = 1$. Then $3r^2 + 2r = 0$ gives $r = 0$ or $r = -\frac{2}{3}$, so $T = 1$ or $T = \frac{9}{25}$. The quadratic becomes $25T^2 - 34T + 9 = (T-1)(25T-9)$, as it should.
- At the boundary $q = \frac{2}{3}$ the quadratic is $\frac{256}{9}T^2 - 32T + 9 = 0$, with discriminant $1024 - 1024 = 0$. It has a double root $T = \frac{9}{16} = 1/(1+\frac{1}{3})^2$, matching the single value $r = -\frac{1}{3}$.
- Thousands of random values of p and q in each range were checked by computer: the root counts and both quadratics agree.

Common slips

- Forgetting that $r = \pm 1$ are excluded, which wrongly puts $p = 1$ in the "two values" case and $q = 2$ out of the "unique" set.
- Missing the isolated value $q = \frac{2}{3}$ (the vertex), where T is also unique.
- Squaring without checking that no extra roots have been introduced.

Beyond STEP

The question is really about when a function is invertible: the map $r \mapsto p$ is one-to-one on part of its domain and two-to-one on the rest, and the answer depends on the domain, not just on the formula. The series $\sum r^n$ and $\sum nr^{n-1}$ are a power series and its term-by-term derivative. Why differentiating term by term is allowed inside the radius of convergence is a standard first-year analysis result. Abbott's *Understanding Analysis* treats power series and uniform convergence carefully and readably.

WORKED SOLUTION

STEP I 2017, Question 5

Pure • Optimisation by Differentiation

STEP I 2017, Question 5

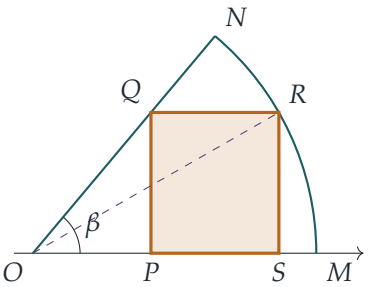
Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s1-q5

The key idea

Once you place the vertices, everything is fixed by the single variable x: Q sits directly above P on the line $y = x\tan\beta$, and R is level with Q on the circle. Pythagoras gives s in terms of x, and the area becomes an ordinary one-variable function to differentiate. The final angle condition then drops out of the half-angle formula for tan.

Solution



The area

Since $PQRS$ is a rectangle with P and S on the x-axis, the sides PQ and SR are vertical. So Q has x-coordinate x. It lies on ON, the line $y = x\tan\beta$, so $Q = (x,\, x\tan\beta)$, and $R = (s,\, x\tan\beta)$. The rectangle has width $PS = s - x$ (positive, as S is further from O than P) and height $PQ = x\tan\beta$, so

$$A = x(s - x)\tan\beta,$$

as required.

Finding s and $\frac{\mathrm{d}A}{\mathrm{d}x}$

Since R lies on the circle, $OR = a$, so $s^2 + x^2\tan^2\beta = a^2$. As $s > 0$,

$$s = \sqrt{a^2 - x^2\tan^2\beta}.$$

Differentiating $s^2 = a^2 - x^2\tan^2\beta$ implicitly gives $2s\dfrac{\mathrm{d}s}{\mathrm{d}x} = -2x\tan^2\beta$, so $\dfrac{\mathrm{d}s}{\mathrm{d}x} = -\dfrac{x\tan^2\beta}{s}$. Now write $A = (xs - x^2)\tan\beta$ and use the product rule on xs:

$$\begin{aligned}\frac{\mathrm{d}A}{\mathrm{d}x} &= \left(s + x\frac{\mathrm{d}s}{\mathrm{d}x} - 2x\right)\tan\beta \\ &= (s - 2x)\tan\beta - \frac{x^2}{s}\tan^3\beta,\end{aligned}$$

which is the required result.

The greatest area

First the range of x. As x increases from 0, Q moves up ON and R moves round the arc towards N; the rectangle shrinks to a vertical segment when R reaches N, that is when $x = a\cos\beta$ (then $s = \sqrt{a^2 - a^2\sin^2\beta} = a\cos\beta = x$). So $0 < x < a\cos\beta$, and A is a continuous function on $[0, a\cos\beta]$ with $A = 0$ at both ends and $A > 0$ in between. Its greatest value is therefore at a stationary point inside the interval.

Setting $\frac{\mathrm{d}A}{\mathrm{d}x} = 0$, dividing by $\tan\beta > 0$ and multiplying by $s > 0$:

$$\begin{aligned}s^2 - 2xs &= x^2\tan^2\beta \\ s^2 - 2xs + x^2 &= x^2(1 + \tan^2\beta) = x^2\sec^2\beta \\ (s - x)^2 &= x^2\sec^2\beta.\end{aligned}$$

Ask yourself Which square root should we take, and how do we know this is a maximum?

Both $s - x$ and $x\sec\beta$ are positive (since $0 < \beta < \frac{\pi}{2}$), so $s - x = x\sec\beta$; the other root would make $s - x$ negative. So there is exactly one stationary point in $0 < x < a\cos\beta$, given by

$$s = x(1 + \sec\beta),$$

and by the previous paragraph it must give the greatest area.

To find that area, substitute into $s^2 + x^2\tan^2\beta = a^2$. Using $\tan^2\beta = \sec^2\beta - 1$,

$$(1 + \sec\beta)^2 + \tan^2\beta = 2\sec\beta + 2\sec^2\beta = 2\sec\beta\,(1 + \sec\beta),$$

so

$$x^2 = \frac{a^2}{2\sec\beta\,(1 + \sec\beta)} = \frac{a^2\cos^2\beta}{2(1 + \cos\beta)}.$$

At this point $s - x = x\sec\beta$, so

$$A = x \cdot x\sec\beta \cdot \tan\beta = x^2\,\frac{\sin\beta}{\cos^2\beta} = \frac{a^2\sin\beta}{2(1+\cos\beta)}$$
$$= \frac{a^2 \cdot 2\sin\tfrac{1}{2}\beta\cos\tfrac{1}{2}\beta}{2\cdot 2\cos^2\tfrac{1}{2}\beta} = \tfrac{1}{2}a^2\tan\tfrac{1}{2}\beta,$$

using $\sin\beta = 2\sin\frac{1}{2}\beta\cos\frac{1}{2}\beta$ and $1+\cos\beta = 2\cos^2\frac{1}{2}\beta$.

The angle ROS

In the right-angled triangle ORS, $\tan\angle ROS = \dfrac{SR}{OS} = \dfrac{x\tan\beta}{s}$. At the maximum,

$$\tan\angle ROS = \frac{\tan\beta}{1+\sec\beta} = \frac{\sin\beta}{\cos\beta+1} = \tan\tfrac{1}{2}\beta,$$

by the same two half-angle identities. Both $\angle ROS$ and $\frac{1}{2}\beta$ lie in $(0, \frac{\pi}{2})$, where tan is one-to-one, so $\angle ROS = \frac{1}{2}\beta$: at the best rectangle, OR bisects the sector.

Checks

- Take $a = 1$, $\beta = \frac{\pi}{3}$. Then $x^2 = \frac{1/4}{3} = \frac{1}{12}$, $s = 3x$, and $A = x\cdot 2x\cdot\sqrt{3} = \frac{\sqrt{3}}{6} \approx 0.2887$, which equals $\frac{1}{2}\tan\frac{\pi}{6}$. A computer search over a fine grid of x, for many random a and β, found the same maximum and the same optimal x.
- As $\beta \to 0$ the sector collapses and $\frac{1}{2}a^2\tan\frac{1}{2}\beta \to 0$, as it should.
- Units: A is a^2 times a pure number, so it is an area.

Common slips

- Forgetting that s depends on x, and differentiating $x(s-x)\tan\beta$ as if s were constant.
- Taking $s - x = -x\sec\beta$ as well, or not justifying that the stationary point is a maximum.

Beyond STEP

The tidy answer, with OR along the bisector of the sector, has a neat explanation: if $R = (a\cos\theta, a\sin\theta)$, a short calculation gives $A = \frac{1}{2}a^2\big(\cos(2\theta-\beta) - \cos\beta\big)/\sin\beta$, which is plainly greatest when $\theta = \frac{1}{2}\beta$. Choosing the right variable can make an optimisation almost trivial. At university you meet constrained optimisation in several variables, where instead of eliminating s you could maximise $x(s-x)$ subject to $s^2 + x^2\tan^2\beta = a^2$ using Lagrange multipliers. Siklos, *Advanced Problems in Mathematics*, has many more questions in this spirit.

WORKED SOLUTION

STEP I 2017, Question 6

Pure • Integrals and Sign Changes

STEP I 2017, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s1-q6

The key idea

A continuous function that never changes sign, and isn't identically zero, has a non-zero integral: it stays away from zero on some small interval, and that interval alone contributes something. Part (i) makes this precise, and parts (ii) and (iii) apply it to cleverly chosen functions. In (ii) the integrand $(x-\alpha)^2 g(x)$ has the same sign as g; in (iii) integration by parts turns the data about h into the conditions of (ii) for h'.

Solution

Part (i)

Suppose, for contradiction, that f does *not* take both positive and negative values on $[0,1]$. Then either $f(x) \geq 0$ for all x in $[0,1]$, or $f(x) \leq 0$ for all such x. Replacing f by $-f$ if necessary (this changes neither the hypothesis $\int_0^1 f(x)\,\mathrm{d}x = 0$ nor the conclusion), we may assume $f(x) \geq 0$ throughout.

We are told $f(c) \neq 0$ for some $c \in [0,1]$, so $f(c) > 0$. Since f is continuous at c, there is a $\delta > 0$ such that $f(x) > \frac{1}{2} f(c)$ for all x in $[0,1]$ with $|x-c| < \delta$. This set contains an interval $[p,q] \subseteq [0,1]$ of positive length $q-p$ (take $[c,\, c+\eta]$ if $c \leq \frac{1}{2}$ and $[c-\eta,\, c]$ if $c > \frac{1}{2}$, where $0 < \eta < \delta$ and $\eta \leq \frac{1}{2}$). Since $f \geq 0$ everywhere,

$$\int_0^1 f(x)\,\mathrm{d}x \;\geq\; \int_p^q f(x)\,\mathrm{d}x \;\geq\; \tfrac{1}{2} f(c)\,(q-p) \;>\; 0,$$

which contradicts $\int_0^1 f(x)\,\mathrm{d}x = 0$. Hence f takes both positive and negative values on $[0,1]$.

Ask yourself Why isn't "$f \geq 0$ and $f(c) > 0$, so the integral is positive" enough on its own?

A function can be positive at a single point without its integral noticing; a single point

has no width. It's continuity that spreads the positivity at c over an interval of positive length, and that is what forces the integral to be positive.

Part (ii)

Expanding the square and using the three conditions $(*)$,

$$\begin{aligned}\int_0^1 (x-\alpha)^2 g(x)\,\mathrm{d}x &= \int_0^1 x^2 g(x)\,\mathrm{d}x - 2\alpha\int_0^1 x g(x)\,\mathrm{d}x + \alpha^2\int_0^1 g(x)\,\mathrm{d}x \\ &= \alpha^2 - 2\alpha\cdot\alpha + \alpha^2\cdot 1 = 0.\end{aligned}$$

Now suppose, for contradiction, that $g(x)\neq 0$ for every $x\in[0,1]$. Then g cannot take both signs on $[0,1]$ (otherwise, by the assumption at the start of the question, it would vanish somewhere), so g is strictly positive throughout or strictly negative throughout.

Let $f(x)=(x-\alpha)^2 g(x)$, which is continuous. Since $(x-\alpha)^2\geq 0$, f has the same sign as g wherever it is non-zero, so f does not take both signs. But f is non-zero somewhere in $[0,1]$: $(x-\alpha)^2$ vanishes only at $x=\alpha$, so $f(x)\neq 0$ for every $x\in[0,1]$ other than (possibly) α. And $\int_0^1 f(x)\,\mathrm{d}x = 0$. By part (i), f must take both signs, a contradiction. Hence $g(x)=0$ for some x in $[0,1]$.

A linear example. With $g(x)=a+bx$ the conditions $(*)$ read

$$a+\tfrac{1}{2}b=1,\qquad \tfrac{1}{2}a+\tfrac{1}{3}b=\alpha,\qquad \tfrac{1}{3}a+\tfrac{1}{4}b=\alpha^2.$$

From the first, $a=1-\frac{1}{2}b$; substituting into the second gives $\frac{1}{2}+\frac{1}{12}b=\alpha$, so

$$b=12\alpha-6,\qquad a=4-6\alpha.$$

The third condition then becomes

$$\tfrac{1}{3}(4-6\alpha)+\tfrac{1}{4}(12\alpha-6)=\alpha-\tfrac{1}{6}=\alpha^2,$$

that is, $6\alpha^2-6\alpha+1=0$, so

$$\alpha=\frac{3\pm\sqrt{3}}{6}.$$

Notice that α isn't free: a linear g only works for these two values. Taking $\alpha=\frac{3+\sqrt{3}}{6}$ gives $b=2\sqrt{3}$ and $a=1-\sqrt{3}$, so

$$g(x)=1-\sqrt{3}+2\sqrt{3}\,x,$$

which is zero at

$$x=\frac{\sqrt{3}-1}{2\sqrt{3}}=\frac{3-\sqrt{3}}{6}\approx 0.211,$$

and this lies in $[0,1]$, as part (ii) says it must. (The other choice, $\alpha=\frac{3-\sqrt{3}}{6}$, gives $g(x)=1+\sqrt{3}-2\sqrt{3}\,x$, which is zero at $x=\frac{3+\sqrt{3}}{6}\approx 0.789$, also in $[0,1]$.)

Part (iii)

We show that $g = h'$ satisfies $(*)$ with $\alpha = 1 - \beta$. Here h' is continuous, as part (ii) requires. First,

$$\int_0^1 h'(x)\,\mathrm{d}x = h(1) - h(0) = 1.$$

Integrating by parts with $u = x$, $\mathrm{d}v = h'(x)\,\mathrm{d}x$,

$$\int_0^1 xh'(x)\,\mathrm{d}x = \Big[xh(x)\Big]_0^1 - \int_0^1 h(x)\,\mathrm{d}x = 1 - \beta.$$

Integrating by parts again, now with $u = x^2$,

$$\begin{aligned}\int_0^1 x^2h'(x)\,\mathrm{d}x &= \Big[x^2h(x)\Big]_0^1 - 2\int_0^1 xh(x)\,\mathrm{d}x \\ &= 1 - \beta(2-\beta) = 1 - 2\beta + \beta^2 = (1-\beta)^2.\end{aligned}$$

? **Ask yourself** Where does the strange value $\frac{1}{2}\beta(2-\beta)$ come from?

It's chosen precisely so that the third integral comes out as the square of the second. After the two integrations by parts, the second integral is $1 - \beta$, so we need the third to be $(1-\beta)^2$, and that is exactly what $\frac{1}{2}\beta(2-\beta)$ delivers.

So h' is continuous and satisfies $(*)$ with $\alpha = 1 - \beta$. By part (ii), $h'(x) = 0$ for some x in $[0, 1]$.

Checks

- For $g(x) = 1 - \sqrt{3} + 2\sqrt{3}\,x$, direct integration gives $\int_0^1 g = 1$, $\int_0^1 xg = \frac{3+\sqrt{3}}{6} \approx 0.7887$ and $\int_0^1 x^2 g \approx 0.6220 = (0.7887)^2$.
- Part (iii) can be tested on $h(x) = (1-\sqrt{3})x + \sqrt{3}\,x^2$, whose derivative is the g above. Then $h(0) = 0$, $h(1) = 1$, $\beta = \frac{3-\sqrt{3}}{6} \approx 0.2113$, and $\int_0^1 xh = \frac{1}{2}\beta(2-\beta) \approx 0.1890$. Its derivative vanishes at $x \approx 0.211$, so h has a turning point inside $[0, 1]$, as predicted.

Common slips

- In (i), arguing "$f \geq 0$, so $\int f > 0$" without using continuity. A non-negative function can have zero integral without being zero everywhere (if it is non-zero at only one point), so continuity is essential.
- In (ii), applying (i) to g itself. We know nothing about $\int g$ except that it is 1; part (i) has to be applied to $(x-\alpha)^2 g$, and we must check that this function is non-zero somewhere.
- In (iii), forgetting that h' must be continuous for (ii) to apply. That's why the question says h has a *continuous* derivative.

Beyond STEP

The conditions $(*)$ say that if g were a probability density, its mean would be α and its variance $\int (x-\alpha)^2 g = 0$. A genuine density with zero variance is impossible, and that's why g must dip below zero somewhere. Part (i) is a standard lemma of real analysis: a continuous, non-negative function with zero integral vanishes identically. It is used constantly, for example in the calculus of variations. Abbott's *Understanding Analysis* develops continuity and the Riemann integral carefully enough to prove it, and to prove the intermediate value theorem that the question lets you assume.

WORKED SOLUTION

STEP I 2017, Question 7

Pure • Napoleon's Theorem

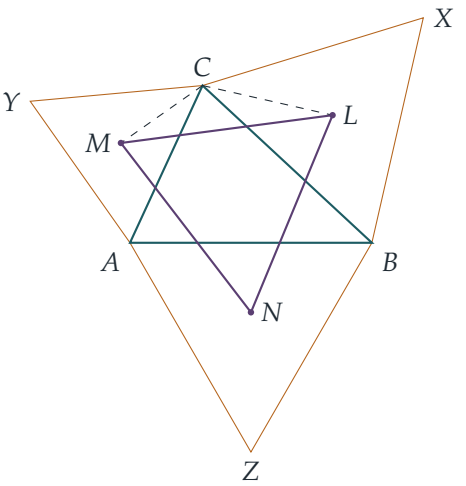
STEP I 2017, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s1-q7

The key idea

Everything happens at the vertices of ABC. The centre of an equilateral triangle sits on the bisector of each of its angles, so the lines from C to L and to M each make $30°$ with a side of ABC. That gives the angle LCM in terms of C. The cosine rule in triangle LCM then gives $|LM|$. The answer turns out to be symmetric in a, b, c, and that symmetry is what makes LMN equilateral.



Solution

Part (i)

In the equilateral triangle CYA of side b, the centre M lies on all three medians, which are also the altitudes and angle bisectors. The median from C has length $\frac{\sqrt{3}}{2}b$, and the

centre is two-thirds of the way along it from the vertex. So

$$|CM| = \tfrac{2}{3} \cdot \tfrac{\sqrt{3}}{2}\, b = \frac{b}{\sqrt{3}}.$$

In the same way, triangle BXC has side a, so

$$|CL| = \frac{a}{\sqrt{3}}.$$

Part (ii)

Since CM bisects the 60° angle YCA, the line CM makes 30° with CA, on the side away from B. Likewise CL makes 30° with CB, on the side away from A. Going round C from L to M we pass through 30°, then the angle C of the triangle, then another 30°. So

$$\angle LCM = C + 60^\circ.$$

? **Ask yourself** What if $C > 120^\circ$, so that $C + 60^\circ$ is more than 180°?

Then the angle of triangle LCM at C is the other way round, $360^\circ - (C + 60^\circ)$. But $\cos(360^\circ - \theta) = \cos\theta$, so the cosine rule below gives the same result in every case.

Applying the cosine rule in triangle LCM, with the lengths from part (i):

$$\begin{aligned} |LM|^2 &= \frac{a^2}{3} + \frac{b^2}{3} - \frac{2ab}{3}\cos(C + 60^\circ) \\ &= \frac{a^2 + b^2}{3} - \frac{2ab}{3}\left(\tfrac{1}{2}\cos C - \tfrac{\sqrt{3}}{2}\sin C\right) \\ &= \frac{a^2 + b^2}{3} - \frac{ab\cos C}{3} + \frac{\sqrt{3}\,ab\sin C}{3}. \end{aligned}$$

Now we use two facts about triangle ABC. The cosine rule gives $2ab\cos C = a^2 + b^2 - c^2$, and the area is $\Delta = \frac{1}{2}ab\sin C$, so $ab\sin C = 2\Delta$. Multiplying by 6,

$$\begin{aligned} 6|LM|^2 &= 2(a^2 + b^2) - 2ab\cos C + 2\sqrt{3}\,ab\sin C \\ &= 2a^2 + 2b^2 - (a^2 + b^2 - c^2) + 4\sqrt{3}\,\Delta \\ &= a^2 + b^2 + c^2 + 4\sqrt{3}\,\Delta, \end{aligned}$$

as required.

The right-hand side is unchanged if we permute a, b, c, and Δ does not depend on how we label the triangle. Repeating the argument at vertex A (for $|MN|$) and at vertex B (for $|NL|$) gives the same expression. So

$$|LM| = |MN| = |NL|,$$

and LMN is equilateral.

Equal areas. An equilateral triangle of side s has area $\frac{\sqrt{3}}{4}s^2$. Write $S = a^2 + b^2 + c^2$. Then

$$[LMN] = \frac{\sqrt{3}}{4}|LM|^2 = \frac{\sqrt{3}}{24}\left(S + 4\sqrt{3}\,\Delta\right) = \frac{\sqrt{3}}{24}S + \frac{\Delta}{2}.$$

Hence

$$[LMN] = \Delta \iff \frac{\sqrt{3}}{24}S = \frac{\Delta}{2} \iff S = \frac{12\Delta}{\sqrt{3}} = 4\sqrt{3}\,\Delta.$$

Every step is reversible, so this is the required "if and only if".

Part (iii)

Write everything in terms of a, b and C, using $c^2 = a^2 + b^2 - 2ab\cos C$ and $\Delta = \frac{1}{2}ab\sin C$. Each line below is equivalent to the one before:

$$\begin{aligned} & a^2 + b^2 + c^2 = 4\sqrt{3}\,\Delta \\ \iff \quad & 2a^2 + 2b^2 - 2ab\cos C = 2\sqrt{3}\,ab\sin C \\ \iff \quad & a^2 + b^2 = ab\left(\cos C + \sqrt{3}\sin C\right) \\ \iff \quad & a^2 + b^2 = 2ab\cos(C - 60^\circ), \end{aligned}$$

because $\cos C + \sqrt{3}\sin C = 2\left(\cos C\cos 60^\circ + \sin C\sin 60^\circ\right) = 2\cos(C - 60^\circ)$. Subtracting $2ab$ from both sides,

$$a^2 - 2ab + b^2 = 2ab\cos(C - 60^\circ) - 2ab \iff (a-b)^2 = -2ab\left(1 - \cos(C - 60^\circ)\right).$$

So the two conditions are equivalent.

> **?** **Ask yourself** How can a square equal something with a minus sign in front?

Only if both sides are zero. The left-hand side $(a-b)^2$ is at least 0. On the right, $ab > 0$ and $1 - \cos(C - 60^\circ) \geq 0$, so the right-hand side is at most 0. Equality therefore holds exactly when

$$a = b \quad \text{and} \quad \cos(C - 60^\circ) = 1.$$

Since $0^\circ < C < 180^\circ$, we have $-60^\circ < C - 60^\circ < 120^\circ$, and the only angle in this range with cosine 1 is 0°. So $C = 60^\circ$. A triangle with $a = b$ and an apex angle of 60° has its other two angles equal, summing to 120°, so all three angles are 60° and the triangle is equilateral. Conversely, an equilateral triangle has $a = b$ and $C = 60^\circ$, so it satisfies the condition.

Combining this with part (ii): $[LMN] = [ABC] \iff a^2 + b^2 + c^2 = 4\sqrt{3}\,\Delta \iff ABC$ is equilateral.

Checks

- Equilateral ABC with side 1: then $\Delta = \frac{\sqrt{3}}{4}$ and $6|LM|^2 = 3 + 3 = 6$, so $|LM| = 1$, and the areas are equal, as part (iii) predicts.

- The 3–4–5 right-angled triangle: $S = 50$ and $\Delta = 6$, so $|LM|^2 = (50 + 24\sqrt{3})/6 \approx 15.26$. Then $[LMN] \approx 6.61$, which is bigger than $[ABC] = 6$. A computer check with coordinates for the points L, M, N gave the same values, and confirmed the formula for $6|LM|^2$ on thousands of random triangles, including obtuse ones.
- Part (iii) also shows $a^2 + b^2 + c^2 \geq 4\sqrt{3}\,\Delta$ for every triangle, so $[LMN] \geq [ABC]$ always.

Common slips

- Taking $|CM|$ to be the whole median $\frac{\sqrt{3}}{2}b$, or one-third of it, instead of two-thirds.
- Using $\angle LCM = C + 30^\circ$ or C, forgetting the 30° on each side.
- In part (iii), stopping at "$a = b$". You also need $C = 60^\circ$, and the range of C to rule out other angles with cosine 1.

Beyond STEP

This is Napoleon's theorem: the centres of equilateral triangles erected outwards on the sides of any triangle form an equilateral triangle. The inequality $a^2 + b^2 + c^2 \geq 4\sqrt{3}\,\Delta$ found in part (iii) is Weitzenböck's inequality. Complex numbers give a very slick proof of Napoleon's theorem: rotating by 60° is multiplication by $e^{i\pi/3}$. Needham's *Visual Complex Analysis* develops this way of doing geometry beautifully.

WORKED SOLUTION

STEP I 2017, Question 8

Pure • Recurrence Relations and Induction

STEP I 2017, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s1-q8

The key idea

The expression $a_n^2 + 2a_nb_n - b_n^2$ is unchanged by one step of the recurrence, so it stays at its starting value for ever. Dividing it by b_n^2 turns it into an exact equation for the ratio c_n, with a small error term $1/b_n^2$. Completing the square in that equation does almost all the work: it gives the limit, the side of $\sqrt{2} - 1$ the ratio lies on, and the rational bounds for $\sqrt{2}$.

Solution

The induction

For $n = 1$ we have $a_1^2 + 2a_1b_1 - b_1^2 = 1 + 4 - 4 = 1$, so $(*)$ holds.

Suppose $(*)$ holds for some $n \geq 1$. Writing $a = a_n$, $b = b_n$,

$$\begin{aligned} a_{n+1}^2 &= a^2 + 4ab + 4b^2, \\ 2a_{n+1}b_{n+1} &= 2(a + 2b)(2a + 5b) = 4a^2 + 18ab + 20b^2, \\ b_{n+1}^2 &= 4a^2 + 20ab + 25b^2. \end{aligned}$$

Adding the first two and subtracting the third,

$$a_{n+1}^2 + 2a_{n+1}b_{n+1} - b_{n+1}^2 = a^2 + 2ab - b^2 = 1.$$

So $(*)$ holds for $n + 1$, and by induction it holds for all $n \geq 1$.

Part (i)

First, every a_n and b_n is positive: this is true for $n = 1$, and the recurrence builds a_{n+1} and b_{n+1} from a_n and b_n with positive coefficients. Hence

$$b_{n+1} = 2a_n + 5b_n > 5b_n,$$

since $a_n > 0$. Starting from $b_1 = 2 = 2 \times 5^0$ and multiplying by at least 5 at each step gives $b_n \geq 2 \times 5^{n-1}$ for all $n \geq 1$ (a one-line induction).

Now divide $(*)$ by b_n^2, which is not zero:

$$c_n^2 + 2c_n - 1 = \frac{1}{b_n^2}, \qquad \text{that is} \qquad (c_n+1)^2 = 2 + \frac{1}{b_n^2}. \tag{†}$$

Ask yourself Which square root do we take?

Since $c_n > 0$, the number $c_n + 1$ is positive, so we take the positive root:

$$c_n + 1 = \sqrt{2 + \frac{1}{b_n^2}}.$$

As $n \to \infty$, $0 < 1/b_n^2 \leq \frac{1}{4}\,25^{-(n-1)} \to 0$, so $c_n + 1 \to \sqrt{2}$ (the square root function is continuous). Therefore $c_n \to \sqrt{2} - 1$, as required.

Part (ii)

From (†), $(c_n+1)^2 = 2 + 1/b_n^2 > 2$. Since $c_n + 1 > 0$ and $\sqrt{2} > 0$, this gives $c_n + 1 > \sqrt{2}$, that is, $c_n > \sqrt{2} - 1$.

For the other bound, divide 2 by both sides of $c_n + 1 > \sqrt{2}$; both sides are positive, so the inequality reverses:

$$\frac{2}{c_n+1} < \frac{2}{\sqrt{2}} = \sqrt{2}.$$

Together,

$$\frac{2}{c_n+1} < \sqrt{2} < c_n + 1.$$

Ask yourself Which value of n produces $99/70$?

We need $c_n + 1 = (a_n + b_n)/b_n$ to be $99/70$, so we look for $b_n = 70$. Running the recurrence:

$$(a_1, b_1) = (1, 2), \quad (a_2, b_2) = (5, 12), \quad (a_3, b_3) = (29, 70).$$

So $c_3 = \frac{29}{70}$ and $c_3 + 1 = \frac{99}{70}$, while

$$\frac{2}{c_3+1} = \frac{140}{99}.$$

Putting $n = 3$ into the double inequality gives

$$\frac{140}{99} < \sqrt{2} < \frac{99}{70}.$$

Checks

- $(a_3, b_3) = (29, 70)$ satisfies $(*)$: $841 + 4060 - 4900 = 1$.

- $\frac{140}{99} \approx 1.414141$ and $\frac{99}{70} \approx 1.414286$, with $\sqrt{2} \approx 1.414214$ between them. The interval has width $\frac{1}{6930}$.
- The next term, $(a_4, b_4) = (169, 408)$, gives $c_4 \approx 0.4142157$, already within 2.2×10^{-6} of $\sqrt{2} - 1$, which matches the error falling roughly like $1/b_n^2$.
- Squaring: $140^2 = 19600 < 2 \times 99^2 = 19602$ and $99^2 = 9801 > 2 \times 70^2 = 9800$.

Common slips

- Expanding $(*)$ for $n+1$ and not collecting terms carefully: the ab and b^2 coefficients cancel only just ($4 + 18 - 20 = 2$ and $4 + 20 - 25 = -1$).
- Claiming $b_{n+1} \geq 5b_n$ without noting that $a_n > 0$.
- Forgetting to justify the positive square root, or reversing an inequality on taking reciprocals without saying both sides are positive.

Beyond STEP

The recurrence is multiplication by the matrix $\begin{pmatrix} 1 & 2 \\ 2 & 5 \end{pmatrix}$, which has determinant 1 and preserves the quadratic form $x^2 + 2xy - y^2$; finding invariant forms like this is the heart of the theory of Pell's equation. The numbers $c_n + 1$, namely $\frac{3}{2}, \frac{17}{12}, \frac{99}{70}, \ldots$, are every other continued-fraction convergent of $\sqrt{2}$; convergents are the best rational approximations for the size of their denominators. Davenport's *The Higher Arithmetic* covers continued fractions and Pell's equation beautifully.

WORKED SOLUTION

STEP I 2017, Question 9

Mechanics • Projectile Motion

STEP I 2017, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

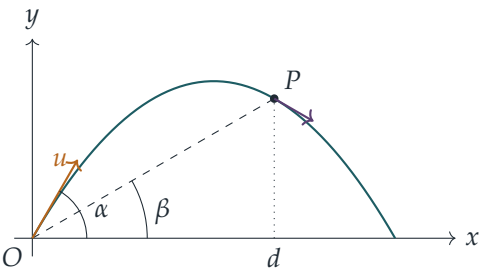
step.maths.org/questions/17-s1-q9

The key idea

Write down the trajectory equation and make it pass through P. That gives u^2 as a function of the projection angle alone, and the cleanest variable to work with is $T = \tan\alpha$, because $\sec^2\alpha = 1 + T^2$ turns everything into a rational function of T. Minimising that function is ordinary calculus, and the stationary-point equation turns out to be a double-angle formula in disguise.

Solution

Model the particle as moving freely under gravity alone (no air resistance), with O as origin, x horizontal towards P and y vertically up. Write $T = \tan\alpha$ and $k = \tan\beta > 0$.



Part (i)

With horizontal velocity $u\cos\alpha$, the particle reaches horizontal distance x at time $t = x/(u\cos\alpha)$, and its height then is

$$y = x\tan\alpha - \frac{gx^2}{2u^2\cos^2\alpha} = xT - \frac{gx^2(1+T^2)}{2u^2}.$$

Passing through $P = (d, dk)$ means

$$dk = dT - \frac{gd^2(1+T^2)}{2u^2} \quad \Longrightarrow \quad u^2 = \frac{gd}{2} \cdot \frac{1+T^2}{T-k}.$$

The right-hand side must be positive, so only angles with $T > k$, that is $\alpha > \beta$, are possible. That makes sense: we must aim above P. So we must minimise

$$f(T) = \frac{1+T^2}{T-k} \qquad (T > k).$$

By the quotient rule,

$$f'(T) = \frac{2T(T-k) - (1+T^2)}{(T-k)^2} = \frac{T^2 - 2kT - 1}{(T-k)^2}.$$

The numerator is zero at $T = k \pm \sqrt{k^2+1}$. Only the $+$ root exceeds k. Since $f(T) \to \infty$ both as $T \to k^+$ and as $T \to \infty$, and there is only one stationary point in between, that point gives the minimum. So the optimal α satisfies

$$T^2 - 2kT - 1 = 0. \qquad (*)$$

Ask yourself How do we simplify $f(T)$ at the minimum without substituting the surd?

Use $(*)$ to rewrite things. It gives $k = \dfrac{T^2-1}{2T}$, so

$$T - k = \frac{2T^2 - T^2 + 1}{2T} = \frac{1+T^2}{2T}, \qquad \text{hence} \qquad f(T) = \frac{1+T^2}{(1+T^2)/(2T)} = 2T.$$

Therefore

$$u^2 = \frac{gd}{2} \cdot 2T = gd\tan\alpha,$$

as required.

For the angle, $(*)$ in the form $k = \dfrac{T^2-1}{2T}$ reads

$$\tan\beta = -\frac{1-\tan^2\alpha}{2\tan\alpha} = -\cot 2\alpha = \tan(2\alpha - 90^\circ),$$

using $\tan 2\alpha = \dfrac{2\tan\alpha}{1-\tan^2\alpha}$ and $-\cot\theta = \tan(\theta - 90^\circ)$. (If $\alpha = 45^\circ$ then $(*)$ gives $k = 0$, which is excluded because $\beta > 0$, so $\cot 2\alpha$ is defined.) Since $0^\circ < \alpha < 90^\circ$, the angle $2\alpha - 90^\circ$ lies in $(-90^\circ, 90^\circ)$, as does β, and tan is one-to-one on that interval. So $\beta = 2\alpha - 90^\circ$, that is,

$$2\alpha = \beta + 90^\circ.$$

In words, the best direction of projection bisects the angle between the line OP and the upward vertical.

Part (ii)

At P the horizontal velocity is still $u\cos\alpha$. The time taken is $d/(u\cos\alpha)$, so the vertical velocity is

$$v_y = u\sin\alpha - \frac{gd}{u\cos\alpha}.$$

If the particle is moving at angle θ above the horizontal at P, then

$$\tan\theta = \frac{v_y}{u\cos\alpha} = \tan\alpha - \frac{gd}{u^2\cos^2\alpha} = T - \frac{gd(1+T^2)}{gdT} = T - \frac{1+T^2}{T} = -\frac{1}{T} = -\cot\alpha,$$

using $u^2 = gdT$ from part (i). So $\tan\theta = \tan(\alpha - 90^\circ)$, and since θ lies between -90° and 90°, $\theta = \alpha - 90^\circ$.

The particle is moving *downwards* as it passes through P, at an angle of $90^\circ - \alpha$ below the horizontal. Its direction at P is perpendicular to its direction of projection.

Checks

- Take $\beta = 30^\circ$. Then $\alpha = 60^\circ$ and $u^2 = \sqrt{3}\,gd$. Directly, $f(T)$ at $T = \sqrt{3}$ is $4/(\sqrt{3} - \frac{1}{\sqrt{3}}) = 2\sqrt{3}$, which matches $2T$. The final direction is 30° below the horizontal, as in the diagram.
- Letting $\beta \to 0$ gives $\alpha \to 45^\circ$ and $u^2 \to gd$. This is the familiar fact that the greatest range on level ground, u^2/g, comes from projecting at 45°. The particle then lands at 45° below the horizontal, again perpendicular to its launch direction.
- A numerical search over α for 200 random choices of d and β found the minimising angle at $\frac{1}{2}(\beta + 90^\circ)$ every time, with minimum value $gd\tan\alpha$.

Common slips

- Dropping the condition $T > k$ and keeping the root $k - \sqrt{k^2+1}$. That root is negative, which would mean projecting downwards.
- Concluding $\beta = 2\alpha - 90^\circ$ from equal tangents without saying why both angles lie in an interval where tan is one-to-one.
- Giving the angle in (ii) as $90^\circ - \alpha$ without saying that the particle is descending.

Beyond STEP

For a fixed speed u, the points you can reach lie inside a parabola, the *envelope* or "parabola of safety" of the family of trajectories. The minimum-speed trajectory to P is the one for which P lies on that envelope, and the perpendicularity in (ii) is one of that curve's geometric properties. Envelopes of families of curves, and optimisation with constraints, are developed in first-year courses. Kleppner and Kolenkow's *An*

Introduction to Mechanics is a good place to see projectile problems handled with this kind of care.

WORKED SOLUTION

STEP I 2017, Question 10

Mechanics • Collisions in a Chain

STEP I 2017, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s1-q10

The key idea

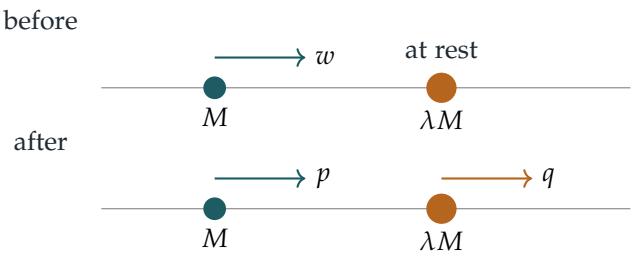
Every collision in the chain looks the same: a particle of mass M moving at speed w hits a particle of mass λM at rest. So we solve that one collision once, in terms of the ratio λ, and the whole chain follows by repeating it, giving geometric sequences. After that, each part is a matter of comparing the velocities the particles are left with, to see whether anyone catches up with the particle in front.

Solution

Model all the particles as points moving along the x-axis, with velocities measured in the positive x-direction.

One typical collision

A particle A of mass M moving at speed w hits a particle B of mass λM at rest. Let their velocities afterwards be p and q.



Conservation of momentum and Newton's experimental law give

$$Mw = Mp + \lambda Mq, \qquad q - p = ew.$$

Substituting $p = q - ew$ into the first equation, $w = q - ew + \lambda q$, so

$$q = \frac{1+e}{1+\lambda}\,w, \qquad p = q - ew = \frac{1-\lambda e}{1+\lambda}\,w.$$

Write $r = \dfrac{1+e}{1+\lambda}$ and $s = \dfrac{1-\lambda e}{1+\lambda}$. Note that

$$q - p = ew > 0,$$

so straight after any such collision the two particles separate.

Part (i)

The first collision is P (mass m, speed u) hitting P_1 (mass λm). Taking $w = u$ and $M = m$,

$$u_1 = \frac{1+e}{1+\lambda}\,u,$$

as required, and P is left with velocity su.

P_1 then moves off and hits P_2, of mass $\lambda^2 m = \lambda \cdot (\lambda m)$. The mass ratio is again λ, so this is the same collision with $w = u_1$. In general P_n (mass $\lambda^n m$) hits P_{n+1} (mass $\lambda^{n+1} m$) at speed u_n, so $u_{n+1} = ru_n$ and P_n is left with velocity su_n. Hence

$$u_n = \left(\frac{1+e}{1+\lambda}\right)^n u.$$

Here the first collision of P_n is being hit by P_{n-1}, and its second is hitting P_{n+1}. (Straight after its first collision P_n is moving away from P_{n-1}, so the next thing that can happen to it is reaching P_{n+1}, provided nothing behind has caught it up first. This is the chain of events the question intends, and in parts (ii)–(iv) we check it really happens.) So

$$v_n = su_n = \frac{(1-\lambda e)(1+e)^n}{(1+\lambda)^{n+1}}\,u,$$

as a velocity in the positive x-direction. As a speed it is the modulus of this, which matters only if $\lambda e > 1$, when P_n bounces back.

Part (ii)

Let $e > \lambda$. Then $\lambda < e < 1$, so $\lambda e < 1$ and $s > 0$; also $r > 1$.

The collisions take place in order: P with P_1, then P_1 with P_2, then P_2 with P_3, and so on. After P_n has struck P_{n+1}, the velocities from the back of the line are

$$su,\ su_1,\ su_2,\ \ldots,\ su_n,\ u_{n+1},\ 0,\ 0,\ \ldots$$

for $P, P_1, \ldots, P_n, P_{n+1}, P_{n+2}, \ldots$. Since $u_k = r^k u$ with $r > 1$ and $s > 0$,

$$0 < su < su_1 < su_2 < \cdots < su_n,$$

and $u_{n+1} > su_n$ because $u_{n+1} - su_n = (r-s)u_n = eu_n > 0$. So among $P, P_1, \ldots, P_{n+1}$, each particle is moving strictly faster than the one behind it, and no two of them can

ever meet again. The only collision still to come is P_{n+1} reaching the stationary P_{n+2}, and the same argument then applies with $n+1$ in place of n.

So each P_n ($n \geq 1$) is hit once by P_{n-1}, then hits P_{n+1} once, and is never involved in another collision: exactly two collisions each. (This also confirms the order of events assumed in (i).)

Part (iii)

If $e = \lambda$ then $r = 1$ and $s = \dfrac{1-e^2}{1+e} = 1-e$. So

$$u_n = u \quad \text{and} \quad v_n = (1-e)u \text{ for every } n,$$

and P is also left with velocity $(1-e)u$. The motion is a pulse travelling along the line: the particle at the front always moves at speed u and, on hitting the next particle, hands on speed u and drops to $(1-e)u$. Everything behind moves together at $(1-e)u$, so no further collisions happen among them, and the particle in front, at speed u, pulls away from them. Each P_n is involved in exactly two collisions.

After N collisions, $P, P_1, \ldots, P_{N-1}$ move at $(1-e)u$ and P_N (mass $\lambda^N m = e^N m$) at u. The kinetic energy is

$$\begin{aligned}\tfrac{1}{2}m(1-e)^2u^2\left(1+e+\cdots+e^{N-1}\right) + \tfrac{1}{2}e^N mu^2 &= \tfrac{1}{2}mu^2\left[(1-e)^2\,\frac{1-e^N}{1-e} + e^N\right] \\ &= \tfrac{1}{2}mu^2\left[(1-e)(1-e^N) + e^N\right] \\ &= \tfrac{1}{2}mu^2\left[1-e+e^{N+1}\right].\end{aligned}$$

Dividing by the initial energy $\frac{1}{2}mu^2$, the fraction lost after N collisions is

$$1 - \left(1-e+e^{N+1}\right) = e\left(1-e^N\right) \to e \quad \text{as } N \to \infty,$$

since $0 < e < 1$. This is the required result.

Part (iv)

If $\lambda e = 1$ then $s = 0$ and

$$r = \frac{1+e}{1+1/e} = \frac{e(1+e)}{e+1} = e.$$

So every collision brings the moving particle dead to rest: P stops, then P_1 moves off at eu, hits P_2 and stops, and so on, with $u_n = e^n u$ and $v_n = 0$. Each P_n has exactly two collisions, and the particles end up at rest, each touching the next, while a single particle carries the motion forward ever more slowly. As P_n takes time $1/(e^n u)$ to reach P_{n+1}, the gaps between collisions grow without bound.

Ask yourself Momentum is conserved, so how can the energy go?

After N collisions only P_N moves, with mass $\lambda^N m = e^{-N}m$ and speed $e^N u$. Its momentum is $e^{-N}m \cdot e^N u = mu$, as it must be, but its kinetic energy is

$$\tfrac{1}{2}\,e^{-N}m\,(e^N u)^2 = e^N \cdot \tfrac{1}{2}mu^2.$$

The momentum is carried by ever heavier, ever slower particles, and kinetic energy $p^2/(2M)$ for fixed momentum p tends to zero as the mass M grows. The fraction remaining, e^N, tends to 0, so eventually *all* of the initial kinetic energy is lost.

Checks

- With $e = 1$ and $\lambda = 1$ (equal masses, perfectly elastic), $u_1 = u$ and P stops: the familiar Newton's cradle exchange.
- In (iii), a single collision ($N = 1$) loses the fraction $e(1 - e)$. Directly: P ends at $(1 - e)u$ and P_1, of mass em, at u, so the energy left is $(1 - e)^2 + e = 1 - e + e^2$ of the original, which agrees.
- A step-by-step computer simulation of the collisions, with exact fractions, confirmed the formulae for u_n and v_n, the two-collisions-each pattern for $e > \lambda$, the fraction lost $e(1 - e^N)$ (for $e = \frac{1}{2}$, 0.49999998 after 25 collisions) and the remaining fraction e^N when $\lambda e = 1$.

Common slips

- Writing Newton's law as $q - p = -ew$ or the wrong way round: the speed of separation is e times the speed of approach, and here both are positive.
- Forgetting that the mass ratio is λ at *every* collision, which is what makes the answers geometric.
- Assuming the order of events without checking it. When $e < \lambda$ and $\lambda e < 1$, a particle left behind can catch up with the one in front before that one reaches the next particle, and the simple pattern breaks down.

Beyond STEP

Part (iv) shows momentum and energy behaving very differently: momentum is conserved exactly, yet the energy drains away completely because it is carried by ever heavier particles. Chains of colliding particles are a simple model of how a pulse travels through a medium of varying density, and they lead on to wave propagation in discrete lattices. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats collisions and the centre-of-mass frame with this kind of care.

WORKED SOLUTION

STEP I 2017, Question 11

Mechanics • Rigid Body Statics with Friction

STEP I 2017, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s1-q11

The key idea

Everything happens in one vertical cross-section of the box, which is a square tilted through α. The natural axes are therefore along the bottom of the box and up its side, not horizontal and vertical: then the two normal reactions are along the axes, and only the weight needs splitting. Two resolutions give one relation between the normal reactions, the moment equation about the midpoint (which kills the weight) gives a second, and comparing them with the compound-angle formula for tan produces the answer.

Solution

The model and the forces

Treat the rod as a uniform rigid rod, and work in the vertical plane containing it. In that plane the bottom of the box lies along the line of greatest slope, at angle α to the horizontal, and the side through O is perpendicular to it. Take axes at O:

- x along the bottom of the box, pointing up the slope (towards A);
- y along the side of the box, perpendicular to the plane (towards B).

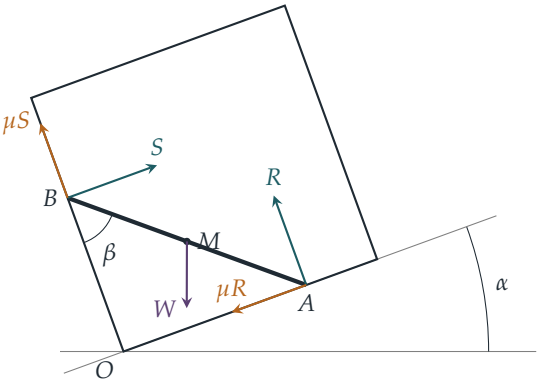
The side is perpendicular to the bottom, so triangle OAB has a right angle at O, and the angle at B is β. Hence

$$OA = 2L\sin\beta, \qquad OB = 2L\cos\beta.$$

Write $\mu = \tan\gamma$. The forces on the rod are:

- at A: normal reaction R in the $+y$ direction, and friction μR towards O (the $-x$ direction), since A is about to slip away from O and friction is limiting;
- at B: normal reaction S in the $+x$ direction, and friction μS away from O (the $+y$ direction), since B is about to slip towards O;

- the weight W at the midpoint M, vertically down. The y-axis makes angle α with the upward vertical, so the weight has components $-W\sin\alpha$ along x and $-W\cos\alpha$ along y.



Resolving

Along the bottom of the box (x) and up the side (y):

$$S - \mu R - W\sin\alpha = 0, \tag{1}$$
$$R + \mu S - W\cos\alpha = 0. \tag{2}$$

Since $R + \mu S = W\cos\alpha \neq 0$, dividing (1) by (2) gives

$$\frac{S - \mu R}{R + \mu S} = \tan\alpha. \tag{3}$$

Moments about the midpoint

The weight acts at M, so it has no moment about M. Relative to M, the point A is at $(L\sin\beta,\ -L\cos\beta)$ and B is at $(-L\sin\beta,\ L\cos\beta)$. Taking anticlockwise moments in the diagram (the moment of a force (F_x, F_y) acting at relative position (p, q) is $pF_y - qF_x$):

$$\text{at } A: \quad L\sin\beta\cdot R - (-L\cos\beta)(-\mu R) = LR(\sin\beta - \mu\cos\beta),$$
$$\text{at } B: \quad -L\sin\beta\cdot\mu S - L\cos\beta\cdot S = -LS(\cos\beta + \mu\sin\beta).$$

These must add to zero, so

$$R(\sin\beta - \mu\cos\beta) = S(\cos\beta + \mu\sin\beta).$$

Now put $\mu = \tan\gamma$ and multiply through by $\cos\gamma$ (which is positive):

$$R(\sin\beta\cos\gamma - \cos\beta\sin\gamma) = S(\cos\beta\cos\gamma + \sin\beta\sin\gamma),$$
$$\text{that is,} \quad R\sin(\beta - \gamma) = S\cos(\beta - \gamma).$$

? **Ask yourself** Can we divide by R and by $\cos(\beta-\gamma)$?

Both normal reactions are pushes, so $R>0$ and $S>0$. Then $\sin(\beta-\gamma)$ and $\cos(\beta-\gamma)$ cannot both be zero, and since $S\cos(\beta-\gamma)=R\sin(\beta-\gamma)$ with $R,S>0$ they have the same sign, so neither is zero and $\tan(\beta-\gamma)>0$. As β and γ both lie in $(0,\frac{1}{2}\pi)$, $\beta-\gamma$ lies in $(-\frac{1}{2}\pi,\frac{1}{2}\pi)$, and so in fact

$$\frac{S}{R}=\tan(\beta-\gamma),\qquad 0<\beta-\gamma<\tfrac{1}{2}\pi. \tag{4}$$

Putting them together

Divide the top and bottom of the left-hand side of (3) by R, and write $k=S/R=\tan(\beta-\gamma)$ and $\mu=\tan\gamma$:

$$\tan\alpha=\frac{k-\mu}{1+k\mu}=\frac{\tan(\beta-\gamma)-\tan\gamma}{1+\tan(\beta-\gamma)\tan\gamma}=\tan\big((\beta-\gamma)-\gamma\big)=\tan(\beta-2\gamma),$$

using the compound-angle formula (the denominator $1+k\mu$ is positive, so nothing is undefined). By (4) and $0<\gamma<\frac{1}{2}\pi$,

$$\beta-2\gamma=(\beta-\gamma)-\gamma\in\left(-\tfrac{1}{2}\pi,\ \tfrac{1}{2}\pi\right),$$

and $\alpha\in(0,\frac{1}{2}\pi)$ too. The tangent function is one-to-one on $(-\frac{1}{2}\pi,\frac{1}{2}\pi)$, so $\tan\alpha=\tan(\beta-2\gamma)$ forces

$$\beta=\alpha+2\gamma,$$

as required. This is consistent with the condition $\alpha<\beta$ stated in the question; it also tells us that this configuration is only possible when $\alpha+2\gamma<\frac{1}{2}\pi$.

Checks

- *Smooth box:* with $\gamma=0$ the result says $\beta=\alpha$. Check this directly: with no friction the two normal reactions meet at P, the fourth corner of the rectangle with diagonal AB and sides along the box. Three forces in equilibrium must be concurrent, so the weight's line through M must pass through P, i.e. MP is vertical. The diagonal MP is parallel to OB turned through β away from O, while OB itself leans at α to the vertical towards O's side; so MP is at $\beta-\alpha$ to the vertical, and is vertical exactly when $\beta=\alpha$.
- *A numerical case:* take $\alpha=20^\circ$, $\gamma=15^\circ$, so $\beta=50^\circ$. Then (1) and (2) give $R\approx 0.791W$ and $S\approx 0.554W$ (both positive), $S/R\approx 0.700\approx\tan 35^\circ$, and the total moment about M comes out as zero.

Common slips

- Resolving horizontally and vertically: it works, but every one of the four contact forces then needs splitting, and the algebra becomes much heavier.

- Getting a friction direction wrong. Friction opposes the impending motion: towards O at A, away from O (up the side) at B.
- Concluding from $\tan\alpha = \tan(\beta - 2\gamma)$ without saying why the two angles lie in an interval where tan is one-to-one.

Beyond STEP

Writing the friction as μR with $\mu = \tan\gamma$ is the idea of the *angle of friction*: in limiting equilibrium the total contact force lies on the edge of a "friction cone" of half-angle γ about the normal. Problems about rigid bodies resting against rough surfaces become geometry problems about these cones and about three forces meeting at a point, and the same picture underlies wedges, ladders and the stability of stacked objects. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops rigid-body statics and torque carefully from first principles.

WORKED SOLUTION

STEP I 2017, Question 12

Probability and Statistics • Expectation and Conditioning

STEP I 2017, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s1-q12

The key idea

The organiser always collects the same takings, Nc; the only uncertainty is whether the prize has to be paid out at all. So the expected profit is the takings minus J times the probability that *somebody* picked the winning number, and everything comes down to the complementary event that *nobody* did. That probability is a power $(1-p)^N$, which is exactly what the approximation $(*)$ is built to handle.

Solution

Part (i)

Let W be the winning number and let E be the event that no participant picks W. Whatever W turns out to be, each participant independently avoids it with probability $1-\frac{1}{N}$, so

$$\mathrm{P}(E) = \left(1-\frac{1}{N}\right)^N.$$

The organiser receives Nc in every case. If E happens, nobody wins and the profit is Nc; otherwise the prize is shared among the winners, but the total paid out is J however many winners there are, so the profit is $Nc-J$. Hence

$$\text{expected profit} = Nc - J\,\mathrm{P}(E^c) = Nc - J\left[1-\left(1-\frac{1}{N}\right)^N\right].$$

Using $(*)$ with $a=1$, $(1-\frac{1}{N})^N \approx \mathrm{e}^{-1}$, so the expected profit is approximately $Nc - J(1-\mathrm{e}^{-1})$. If $J=2Nc$ this becomes

$$Nc - 2Nc\left(1-\mathrm{e}^{-1}\right) = Nc\left(\frac{2}{\mathrm{e}}-1\right).$$

Since $\mathrm{e} > 2$, we have $2/\mathrm{e} < 1$, so this is negative: the organiser expects to make a loss.

Part (ii)

There are γN popular numbers and $(1-\gamma)N$ unpopular ones. Each participant picks exactly one number, so the probabilities over all N numbers sum to 1:

$$\gamma N \cdot \frac{a}{N} + (1-\gamma)N \cdot \frac{b}{N} = 1 \quad \Longrightarrow \quad \gamma a + (1-\gamma)b = 1.$$

The organiser still chooses W uniformly, so W is popular with probability γ and unpopular with probability $1-\gamma$.

> **Ask yourself** The participants are no longer choosing uniformly. Does that change how we find $\mathrm{P}(E)$?

Only through the probability of avoiding W, which now depends on which kind of number W is. So we condition on that. If W is popular, each participant independently avoids it with probability $1-\frac{a}{N}$; if unpopular, with probability $1-\frac{b}{N}$. By the law of total probability and $(*)$,

$$\mathrm{P}(E) = \gamma\left(1-\frac{a}{N}\right)^N + (1-\gamma)\left(1-\frac{b}{N}\right)^N \approx \gamma\,\mathrm{e}^{-a} + (1-\gamma)\,\mathrm{e}^{-b}.$$

As in part (i), the expected profit is $Nc - J\big(1-\mathrm{P}(E)\big)$, so

$$\text{expected profit} \approx \gamma J\,\mathrm{e}^{-a} + (1-\gamma)J\,\mathrm{e}^{-b} + (Nc - J),$$

which is of the required form with

$$A = \gamma J, \qquad B = (1-\gamma)J, \qquad C = Nc - J.$$

The case $\gamma = \frac{1}{8}$, $a = 9b$. The relation gives $\frac{1}{8}\cdot 9b + \frac{7}{8}b = 1$, that is $2b = 1$, so

$$b = \tfrac{1}{2}, \qquad a = \tfrac{9}{2}.$$

With $J = 2Nc$ we have $A = \frac{1}{4}Nc$, $B = \frac{7}{4}Nc$ and $C = -Nc$, so the expected profit is approximately

$$Nc\left(\tfrac{1}{4}\mathrm{e}^{-9/2} + \tfrac{7}{4}\mathrm{e}^{-1/2} - 1\right).$$

The first term in the bracket is positive, so it is enough to show that $\frac{7}{4}\mathrm{e}^{-1/2} > 1$, that is $\sqrt{\mathrm{e}} < \frac{7}{4}$. Since $\mathrm{e} < 3$,

$$\sqrt{\mathrm{e}} < \sqrt{3} < \tfrac{7}{4},$$

where the last step holds because both sides are positive and $3 < \frac{49}{16}$ (that is, $48 < 49$). Hence the bracket is positive, and the organiser expects to make a profit. The prize is the same as in part (i), but because the winning number is often an unpopular one, it goes unclaimed more often.

Checks

- With $J = 2Nc$, part (i) gives an expected profit of about $(2/\mathrm{e} - 1)Nc \approx -0.264\,Nc$, while the final part gives $\left(\frac{1}{4}\mathrm{e}^{-4.5} + \frac{7}{4}\mathrm{e}^{-0.5} - 1\right)Nc \approx 0.064\,Nc$. A small profit, as claimed.
- For $N = 800$ the exact powers $(1 - a/N)^N$ give $0.0640\,Nc$, so the approximation $(*)$ is very good here.
- If $\gamma = 1$ (every number is "popular"), the relation forces $a = 1$ and the part (ii) formula reduces to part (i)'s $J\mathrm{e}^{-1} + Nc - J$.
- An exact enumeration by computer for $N = 4$ (all 4^4 ways the participants can choose) agrees with both unapproximated formulas.

Common slips

- Worrying about how the prize is shared. The organiser pays out J in total if there is at least one winner, whatever the number of winners.
- Assuming the winning number is popular with probability depending on a or b. The organiser picks uniformly, so it is popular with probability γ.
- Writing the relation as $a + b = 1$ or $\gamma a + b = 1$, forgetting to count how many numbers of each kind there are.

Beyond STEP

The number of participants who pick the winning number is binomial with parameters N and p, and $(*)$ is the first step in seeing that for large N and Np fixed it is approximately Poisson: $(1 - p)^N \to \mathrm{e}^{-Np}$ is just the Poisson probability of zero. Real lotteries with roll-overs are analysed in exactly this way, and players who choose unpopular numbers really do have a higher expected return. Grimmett and Welsh, *Probability: An Introduction*, develops the Poisson limit and conditioning arguments like the one used here.

WORKED SOLUTION

STEP I 2017, Question 13

Probability and Statistics • Recurrence Relations for Probabilities

STEP I 2017, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s1-q13

The key idea

Look at each slice and ask how it is used: it can be toast, the first slice of a sandwich, or the second slice of a sandwich. Exactly one of these happens, and the slice before it decides which kinds are possible. If slice $r-1$ finishes a snack, slice r starts a new one; if it doesn't, slice r must be the second half of a sandwich. That gives a first-order recurrence, which we solve by the usual trick of subtracting off the fixed point.

Solution

Setting up

Number the slices $1, 2, \ldots, n$ in the order they are used. For each slice r exactly one of the following happens:

- T_r: slice r is used for toast;
- A_r: slice r is the first of the two slices in a sandwich;
- S_r: slice r is the second of the two slices in a sandwich.

So $t_r = \mathrm{P}(T_r)$, $s_r = \mathrm{P}(S_r)$ and $\mathrm{P}(A_r) = 1 - s_r - t_r$. Say that slice r *ends a snack* if T_r or S_r happens; this has probability $s_r + t_r$.

The first slice starts the first snack, so it cannot be the second slice of a sandwich:

$$s_1 = 0.$$

(Also $t_1 = p$ if $n \geq 2$, and $t_1 = 1$ if $n = 1$.)

The two equations

The equation for t_r. Slice r starts a new snack exactly when slice $r-1$ ends a snack, which has probability $s_{r-1} + t_{r-1}$. Given that slice r starts a snack, slices $r, r+1, \ldots, n$ are left, and when $r \leq n-1$ that is at least two slices, so I choose toast with probability

p, whatever happened earlier. If slice $r-1$ does not end a snack, slice r completes a sandwich and is not toast. Hence

$$t_r = (s_{r-1} + t_{r-1})\, p \qquad (2 \le r \le n-1).$$

This fails for $r = n$ because a snack starting at the last slice is toast for certain.

The equation for s_r. Slice r is the second slice of a sandwich exactly when slice $r-1$ is the first slice of one, that is, when slice $r-1$ does *not* end a snack. So

$$s_r = 1 - (s_{r-1} + t_{r-1}) \qquad (2 \le r \le n).$$

No condition on the number of slices left is needed here, so this holds right up to $r = n$.

The recurrence for s_r

The second equation, applied with r replaced by $r+1$, gives $s_r + t_r = 1 - s_{r+1}$, so the first equation can be written as

$$t_r = p\,(1 - s_r) \qquad (2 \le r \le n-1).$$

This is also true for $r = 1$ when $n \ge 2$, as $t_1 = p$ and $s_1 = 0$. Substituting $t_{r-1} = p(1 - s_{r-1})$ into the equation for s_r, valid when $1 \le r-1 \le n-1$:

$$\begin{aligned} s_r &= 1 - s_{r-1} - p(1 - s_{r-1}) \\ &= (1-p)(1 - s_{r-1}) \\ &= q\,(1 - s_{r-1}), \end{aligned}$$

for $2 \le r \le n$, and in particular for $2 \le r \le n-1$, as required.

Solving the recurrence

> **Ask yourself** How do we solve $s_r = q - q s_{r-1}$? What constant sequence satisfies it?

A constant solution $s_r = c$ needs $c = q - qc$, so $c = \dfrac{q}{1+q}$. Subtracting $c = q - qc$ from the recurrence,

$$s_r - \frac{q}{1+q} = -q\left(s_{r-1} - \frac{q}{1+q}\right),$$

so $s_r - \frac{q}{1+q}$ is multiplied by $-q$ at each step. Starting from $s_1 - \frac{q}{1+q} = -\frac{q}{1+q}$,

$$s_r - \frac{q}{1+q} = (-q)^{r-1}\left(-\frac{q}{1+q}\right) = \frac{(-q)^r}{1+q},$$

which gives

$$s_r = \frac{q + (-q)^r}{1+q} \qquad (1 \le r \le n-1).$$

(Since the recurrence holds up to $r = n$, the same formula actually holds for $r = n$ too.)

For t_r, we use $t_r = p(1 - s_r)$ and

$$1 - s_r = \frac{1 + q - q - (-q)^r}{1 + q} = \frac{1 - (-q)^r}{1 + q},$$

so

$$t_r = \frac{p\big(1 - (-q)^r\big)}{1 + q} \qquad (1 \le r \le n - 1).$$

The last slice

As noted above, the equation for s_r and the recurrence $s_r = q(1 - s_{r-1})$ hold for $r = n$ as well, so the formula continues:

$$s_n = \frac{q + (-q)^n}{1 + q}.$$

The last slice cannot be the first slice of a sandwich, as there is nothing after it. So it is either toast or the second slice of a sandwich, and $t_n = 1 - s_n$:

$$t_n = \frac{1 - (-q)^n}{1 + q}.$$

Compared with t_r for $r < n$, the factor p has gone: once a snack starts on the last slice, toast is certain. (These also hold when $n = 1$: they give $s_1 = 0$ and $t_1 = 1$.)

Checks

- Take $n = 3$ and $p = q = \frac{1}{2}$. The possible orders are toast–toast–toast (probability $\frac{1}{4}$), toast then a sandwich (probability $\frac{1}{4}$), and a sandwich then toast (probability $\frac{1}{2}$). So $s_3 = \frac{1}{4}$ and $t_3 = \frac{3}{4}$. The formulae give $s_3 = \dfrac{\frac{1}{2} - \frac{1}{8}}{\frac{3}{2}} = \frac{1}{4}$ and $t_3 = \dfrac{1 + \frac{1}{8}}{\frac{3}{2}} = \frac{3}{4}$.
- With $q = 0$ (always toast) we get $s_r = 0$ and $t_r = 1$ for every r, as we should.
- As r grows, $s_r \to \frac{q}{1+q}$. This makes sense: in the long run a fraction q of snacks are sandwiches, each with one "second slice", and a snack uses $1 + q$ slices on average, so a proportion $\frac{q}{1+q}$ of slices are second slices.
- We confirmed all the formulae by computer, using exact enumeration of every snack sequence for $n \le 12$ and several values of p.

Common slips

- Using $t_n = p(s_{n-1} + t_{n-1})$: when a snack starts on the last slice it is toast with probability 1, not p.

- Thinking the events "slice $r-1$ is toast" and "slice r is toast" are independent. They are not: what matters is whether slice r starts a snack, and that depends on the history.
- Losing the sign when solving the recurrence: the ratio is $-q$, so the answers alternate about $\frac{q}{1+q}$.

Beyond STEP

Whether slice r starts a new snack forms a simple *renewal process*: the snack lengths are independent, each 1 or 2, and the limit $\frac{q}{1+q}$ is an instance of the renewal theorem (the long-run rate of renewals is one over the mean gap). Equivalently, the type of each slice is a Markov chain, and $(-q)^r$ shows how fast it forgets its starting state. Grimmett and Welsh, *Probability: An Introduction*, covers both Markov chains and recurrences of this kind.

WORKED SOLUTION

STEP II 2017, Question 1

Pure • Reduction Formulae

STEP II 2017, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s2-q1

The key idea

The arctan x is the awkward part of the integrand, but its derivative is a rational function, so integration by parts with $u = \arctan x$ swaps it for $\frac{1}{1+x^2}$. Then the trick in (ii) is to pair two integrals so that the numerators combine to $x^{n+1}(1+x^2)$ and the troublesome denominator cancels. That gives a reduction formula stepping n by 2; using it twice steps n by 4, which is exactly what the induction in (iii) needs.

Solution

Part (i)

Integrate by parts with $u = \arctan x$ and $\mathrm{d}v = x^n\,\mathrm{d}x$, so $v = \frac{x^{n+1}}{n+1}$ (fine, since $n \geq 0$):

$$\begin{aligned} I_n &= \left[\frac{x^{n+1}}{n+1}\arctan x\right]_0^1 - \int_0^1 \frac{x^{n+1}}{n+1}\cdot\frac{1}{1+x^2}\,\mathrm{d}x \\ &= \frac{1}{n+1}\cdot\frac{\pi}{4} - \frac{1}{n+1}\int_0^1 \frac{x^{n+1}}{1+x^2}\,\mathrm{d}x, \end{aligned}$$

using $\arctan 1 = \frac{\pi}{4}$ and the fact that the boundary term vanishes at $x = 0$. Multiplying by $n+1$ gives the required result:

$$(n+1)I_n = \frac{\pi}{4} - \int_0^1 \frac{x^{n+1}}{1+x^2}\,\mathrm{d}x.$$

For $n = 0$ the remaining integral is $\int_0^1 \frac{x}{1+x^2}\,\mathrm{d}x = \left[\frac{1}{2}\ln(1+x^2)\right]_0^1 = \frac{1}{2}\ln 2$, so

$$I_0 = \frac{\pi}{4} - \frac{1}{2}\ln 2.$$

Part (ii)

Apply (i) with $n+2$ in place of n and add it to (i) as it stands:

$$\begin{aligned}(n+3)I_{n+2}+(n+1)I_n &= \frac{\pi}{2}-\int_0^1 \frac{x^{n+3}+x^{n+1}}{1+x^2}\,\mathrm{d}x\\ &= \frac{\pi}{2}-\int_0^1 x^{n+1}\,\mathrm{d}x\\ &= \frac{\pi}{2}-\frac{1}{n+2},\end{aligned}$$

because $x^{n+3}+x^{n+1}=x^{n+1}(1+x^2)$.

Ask yourself The formula links I_4 to I_2, and I_2 to I_0. Which values of n do we need?

We need $n=0$ and $n=2$:

$$\begin{aligned}n=0:&\quad 3I_2+I_0=\frac{\pi}{2}-\frac{1}{2},\\ n=2:&\quad 5I_4+3I_2=\frac{\pi}{2}-\frac{1}{4}.\end{aligned}$$

Subtracting the first from the second eliminates I_2:

$$5I_4=I_0+\frac{1}{4}=\frac{\pi}{4}+\frac{1}{4}-\frac{1}{2}\ln 2,\qquad\text{so}\qquad I_4=\frac{\pi+1-2\ln 2}{20}.$$

Part (iii)

Write $S_n=\sum_{r=1}^{2n}\frac{(-1)^r}{r}$ for short.

Finding A. For $n=1$, $S_1=-1+\frac{1}{2}=-\frac{1}{2}$, so the claim reads $5I_4=A+\frac{1}{4}$. From (ii), $5I_4=\frac{\pi}{4}+\frac{1}{4}-\frac{1}{2}\ln 2$, so we must take

$$A=\frac{\pi}{4}-\frac{1}{2}\ln 2\ (=I_0),$$

and with this value the claim is true for $n=1$.

Inductive step. Suppose $(4k+1)I_{4k}=A-\frac{1}{2}S_k$ for some $k\geq 1$. We want to reach I_{4k+4}, which is two steps of the formula from (ii). With $n=4k+2$ and $n=4k$ it gives

$$\begin{aligned}(4k+5)I_{4k+4}+(4k+3)I_{4k+2}&=\frac{\pi}{2}-\frac{1}{4k+4},\\ (4k+3)I_{4k+2}+(4k+1)I_{4k}&=\frac{\pi}{2}-\frac{1}{4k+2}.\end{aligned}$$

Ask yourself How do we get rid of the unwanted I_{4k+2}?

It appears with the same coefficient in both equations, so subtracting the second from the first removes it:

$$(4k+5)I_{4k+4} = (4k+1)I_{4k} + \frac{1}{4k+2} - \frac{1}{4k+4}.$$

On the other side, S_{k+1} has two more terms than S_k, namely $r = 2k+1$ (odd, so negative) and $r = 2k+2$ (even, so positive):

$$-\frac{1}{2}S_{k+1} = -\frac{1}{2}S_k - \frac{1}{2}\Big(-\frac{1}{2k+1} + \frac{1}{2k+2}\Big) = -\frac{1}{2}S_k + \frac{1}{4k+2} - \frac{1}{4k+4}.$$

Using the inductive hypothesis,

$$(4k+5)I_{4k+4} = A - \frac{1}{2}S_k + \frac{1}{4k+2} - \frac{1}{4k+4} = A - \frac{1}{2}S_{k+1},$$

which is the claim for $n = k+1$. Since it holds for $n = 1$, by induction it holds for all $n \geq 1$, with $A = \frac{\pi}{4} - \frac{1}{2}\ln 2$.

Checks

- Numerically, $I_0 \approx 0.438825$ and $I_4 \approx 0.137765$; Simpson's rule on the original integrals agrees to ten decimal places.
- A equals I_0, so the formula in (iii) is also true for $n = 0$ (the sum is then empty), a good sign.
- As $n \to \infty$, the integral in (i) tends to 0 (the integrand is at most x^{n+1}), so $(4n+1)I_{4n} \to \frac{\pi}{4}$. Meanwhile $S_n \to -\ln 2$, so the right-hand side tends to $\frac{\pi}{4} - \frac{1}{2}\ln 2 + \frac{1}{2}\ln 2 = \frac{\pi}{4}$. The two sides agree in the limit.

Common slips

- Writing $v = \frac{x^{n+1}}{n+1}$ but forgetting to divide the second integral by $n+1$ too.
- Getting the signs of the two new terms of the alternating sum wrong: $r = 2k+1$ is odd, so $(-1)^r = -1$.
- Proving the inductive step without first fixing A from the base case.

Beyond STEP

The final check uses the alternating harmonic series $1 - \frac{1}{2} + \frac{1}{3} - \cdots = \ln 2$, and this question in effect proves it: the identity in (iii) plus the fact that $(4n+1)I_{4n} \to \frac{\pi}{4}$ forces the partial sums to converge to $\ln 2$. Working with integrals like $\int_0^1 \frac{x^{n+1}}{1+x^2}\,\mathrm{d}x$ that shrink to zero is a standard way to control the error in a series, and you will meet it in a first analysis course. Abbott's *Understanding Analysis* is a friendly place to see how such limits are made rigorous.

WORKED SOLUTION

STEP II 2017, Question 2

Pure • Periodic Sequences

STEP II 2017, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s2-q2

The key idea

Compose the map with itself and simplify: two steps of the recurrence give another map of exactly the same "linear over linear" shape, so four steps are just two steps applied twice. Each time we ask whether $x_{n+k} = x_n$, clearing the denominator produces a quadratic in x_n, and that quadratic factorises. One of its factors is always $x^2 + (b - a)x + 1$, whose roots are precisely the *fixed points* of the recurrence, the values that give a constant sequence. Spotting that is what makes both parts work.

Solution

Write $f(x) = \dfrac{ax - 1}{x + b}$, so that $x_{n+1} = f(x_n)$. Then

$$x_{n+2} = \frac{a\,\dfrac{ax_n - 1}{x_n + b} - 1}{\dfrac{ax_n - 1}{x_n + b} + b}.$$

Multiplying top and bottom by $x_n + b$ (non-zero, by assumption) gives

$$x_{n+2} = \frac{a(ax_n - 1) - (x_n + b)}{(ax_n - 1) + b(x_n + b)} = \boxed{\frac{(a^2 - 1)x_n - (a + b)}{(a + b)x_n + b^2 - 1}}$$

It is worth recording one more fact now. A value x is a fixed point ($f(x) = x$) exactly when $ax - 1 = x^2 + bx$, that is, when

$$q(x) \equiv x^2 + (b - a)x + 1 = 0. \qquad (*)$$

If x_0 is a root of q then $x_1 = x_0$, and so on: the sequence is constant, with period 1.

Part (i)

Suppose the sequence has period 2. Then in particular $x_2 = x_0$. Putting $x = x_0$ in the formula for x_{n+2} and clearing the denominator,

$$\begin{aligned} 0 &= (a^2-1)x - (a+b) - x\big[(a+b)x + b^2 - 1\big] \\ &= -(a+b)x^2 + (a^2-b^2)x - (a+b) \\ &= -(a+b)\big[x^2 + (b-a)x + 1\big], \end{aligned}$$

using $a^2 - b^2 = (a+b)(a-b)$. So $(a+b)\,q(x_0) = 0$.

> **Ask yourself** Why can't we finish by saying "$q(x_0) = 0$ is only one value of x_0"?

Because x_0 is a single number, and nothing stops it being a root of q. What rules it out is the meaning of $(*)$: if $q(x_0) = 0$ then x_0 is a fixed point, so $x_1 = x_0$ and the sequence has $x_{n+1} = x_n$ for all n. That is period 1, which the definition of period 2 forbids (take $m = 1$). Hence $q(x_0) \neq 0$, and therefore

$$a + b = 0$$

is necessary for period 2.

(Conversely, if $b = -a$ the two-step formula collapses to $x_{n+2} = \dfrac{(a^2-1)x_n}{a^2-1} = x_n$. The case $a^2 = 1$ cannot occur here, since then $x_1 = 1$ when $a = 1$, $b = -1$, or $x_1 = -1$ when $a = -1$, $b = 1$; either way $x_1 + b = 0$, which is excluded.)

Part (ii)

Write $A = a^2 - 1$, $B = b^2 - 1$ and $c = a + b$, so that $x_{n+2} = g(x_n)$ with $g(x) = \dfrac{Ax - c}{cx + B}$. Applying g twice, and again multiplying top and bottom by $cx + B$,

$$x_{n+4} = \frac{A(Ax_n - c) - c(cx_n + B)}{c(Ax_n - c) + B(cx_n + B)} = \frac{(A^2 - c^2)x_n - c(A+B)}{c(A+B)x_n + B^2 - c^2}.$$

Now $x_{n+4} = x_n$ holds exactly when

$$\begin{aligned} 0 &= (A^2-c^2)x_n - c(A+B) - x_n\big[c(A+B)x_n + B^2 - c^2\big] \\ &= -c(A+B)x_n^2 + (A^2 - B^2)x_n - c(A+B) \\ &= -(A+B)\big[c\,x_n^2 + (B-A)x_n + c\big]. \end{aligned}$$

Since $B - A = b^2 - a^2 = (b-a)(a+b) = (b-a)c$, the bracket is $c\,q(x_n)$, so

$$x_{n+4} = x_n \iff (a^2 + b^2 - 2)\,(a+b)\,q(x_n) = 0.$$

Necessity. Suppose the period is 4. The sequence is not constant, so x_0 is not a fixed point and $q(x_0) \neq 0$. If $a + b = 0$, part (i)'s converse gives $x_2 = x_0$, and indeed $x_{n+2} = x_n$ for

all n, contradicting period 4. So $a+b \neq 0$, and the factorisation with $n=0$ forces

$$a^2+b^2=2.$$

Sufficiency. Suppose $a^2+b^2=2$. Then $A+B=0$, so $x_{n+4}=x_n$ for every n. We must check that no smaller m works.

- $a+b \neq 0$: otherwise $a^2=b^2=1$ and, as in part (i), $x_1+b=0$, which is excluded.
- There are no real fixed points: the discriminant of q is $(b-a)^2-4=(a^2+b^2)-2ab-4=-2(1+ab)$, and $|ab| \leq \frac{1}{2}(a^2+b^2)=1$, with $ab=-1$ only when $(a,b)=\pm(1,-1)$, which we have just excluded. So $ab>-1$, the discriminant is negative, and $q(x_n) \neq 0$ for every real x_n. Hence $m=1$ fails.
- $m=2$: from part (i), $x_2=x_0$ would need $(a+b)q(x_0)=0$, which is false. So $m=2$ fails.
- $m=3$: if $x_{n+3}=x_n$ for all n, then $x_{n+1}=x_{(n+1)+3}=x_{n+4}=x_n$ for all n, giving period 1, which we've just ruled out.

Therefore, for real a, b and x_0 (with the denominators non-zero, as the question allows), the sequence has period 4 if and only if

$$\boxed{a^2+b^2=2.}$$

The starting value x_0 plays no part in the condition.

Checks

Take $a=\sqrt{2}$, $b=0$ and $x_0=1$. The sequence runs

$$1, \quad \sqrt{2}-1, \quad -1, \quad \sqrt{2}+1, \quad 1, \ \ldots$$

which has period exactly 4. For example, $f(-1)=(-\sqrt{2}-1)/(-1)=\sqrt{2}+1$. A computer check confirmed the two-step and four-step formulas exactly for thousands of random rational a, b, x_0. It also confirmed period 4 for about two thousand random points on the circle $a^2+b^2=2$, and found no return after four steps for random (a,b) off that circle.

Common slips

- Concluding from $(a+b)q(x_0)=0$ that $a+b=0$ without explaining why $q(x_0) \neq 0$. The fixed-point interpretation is the reason.
- Giving $a^2+b^2=2$ as the answer for period 4 without checking that the period isn't really 1 or 2.
- Losing the factor $a+b$ in the bracket of part (ii), so that the period-2 case $a+b=0$ wrongly appears to give period 4.

Beyond STEP

A map $x \mapsto \dfrac{\alpha x + \beta}{\gamma x + \delta}$ is a *Möbius transformation*. Composing two such maps corresponds to multiplying their 2×2 matrices, here $\begin{pmatrix} a & -1 \\ 1 & b \end{pmatrix}$. The map has period k exactly when the kth power of the matrix is a multiple of the identity, which is governed by the ratio $(\text{trace})^2/\det = (a+b)^2/(ab+1)$. Period 4 corresponds to the value 2, a "rotation by $\pi/4$" in disguise. Needham's *Visual Complex Analysis* has a beautiful chapter on Möbius transformations, and Strang's *Introduction to Linear Algebra* covers the eigenvalue ideas behind it.

WORKED SOLUTION

STEP II 2017, Question 3

Pure • Curve Sketching

STEP II 2017, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s2-q3

The key idea

The equation $\sin y = k$ never has just one solution: if y_0 is one, so is $\pi - y_0$, and so is anything differing from these by 2π. So every sketch here is a handful of copies of one basic curve, placed by the symmetries of sine. Part (i) shows the pattern with straight lines; in (ii) the calculus pins down the shape of the basic piece $y = \arcsin(\frac{1}{2}\sin x)$, and the symmetries do the rest. Part (iii) is (ii) again after writing $\cos y$ as a sine.

Solution

Part (i)

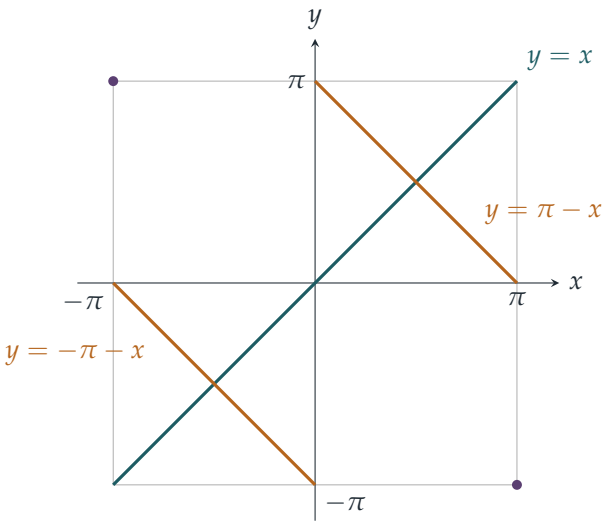
We have $\sin y = \sin x$ exactly when

$$y = x + 2k\pi \qquad \text{or} \qquad y = \pi - x + 2k\pi \qquad (k \in \mathbb{Z}).$$

Now we see which of these lines meet the square $-\pi \leq x, y \leq \pi$.

- $y = x$ runs along the whole diagonal, from $(-\pi, -\pi)$ to (π, π).
- $y = x \pm 2\pi$ need $y - x = \pm 2\pi$. In the square, $|y - x| \leq 2\pi$ with equality only at the corners, so these give just the two isolated points $(-\pi, \pi)$ and $(\pi, -\pi)$.
- $x + y = \pi$ is the segment from $(0, \pi)$ to $(\pi, 0)$, and $x + y = -\pi$ (the case $k = -1$) is the segment from $(-\pi, 0)$ to $(0, -\pi)$.
- $x + y = 3\pi$ or -3π is impossible, since $|x + y| \leq 2\pi$.

So the set is the diagonal $y = x$, the two segments of $x + y = \pm\pi$, and the two corner points $(-\pi, \pi)$ and $(\pi, -\pi)$.



The isolated corner points are easy to miss. They are genuine solutions, since $\sin(\pm\pi) = 0$.

Part (ii)

Differentiating $\sin y = \frac{1}{2}\sin x$ implicitly gives $y'\cos y = \frac{1}{2}\cos x$. For $0 \le y \le \frac{1}{2}\pi$ we have $\cos y \ge 0$, so

$$\cos y = \sqrt{1-\sin^2 y} = \sqrt{1-\tfrac{1}{4}\sin^2 x} = \tfrac{1}{2}\sqrt{4-\sin^2 x},$$

which is never zero. Hence

$$y' = \frac{\cos x}{\sqrt{4-\sin^2 x}}.$$

Write $s = \sin x$. By the quotient rule, using $\dfrac{\mathrm{d}}{\mathrm{d}x}\sqrt{4-s^2} = \dfrac{-s\cos x}{\sqrt{4-s^2}}$,

$$y'' = \frac{-\sin x\sqrt{4-s^2} - \cos x\cdot\dfrac{-s\cos x}{\sqrt{4-s^2}}}{4-s^2} = \frac{-s(4-s^2)+s\cos^2 x}{(4-s^2)^{3/2}}$$
$$= \frac{s\left(-4+s^2+1-s^2\right)}{(4-s^2)^{3/2}} = -\frac{3\sin x}{(4-\sin^2 x)^{3/2}},$$

using $\cos^2 x = 1 - s^2$. This is the required result.

The basic piece. In this quarter-square the curve is $y = f(x)$, where $f(x) = \arcsin(\frac{1}{2}\sin x)$. It starts at $(0,0)$ and ends at $(\frac{1}{2}\pi, \frac{1}{6}\pi)$, because $\arcsin\frac{1}{2} = \frac{1}{6}\pi$, so it never gets near the top edge $y = \frac{1}{2}\pi$. Its gradient is $\frac{1}{2}$ at the origin and falls to 0 at $x = \frac{1}{2}\pi$. Also $y'' < 0$ for $0 < x \le \frac{1}{2}\pi$ and $y'' = 0$ at $x = 0$, so the curve is concave and increasing, with a point of inflection at the origin.

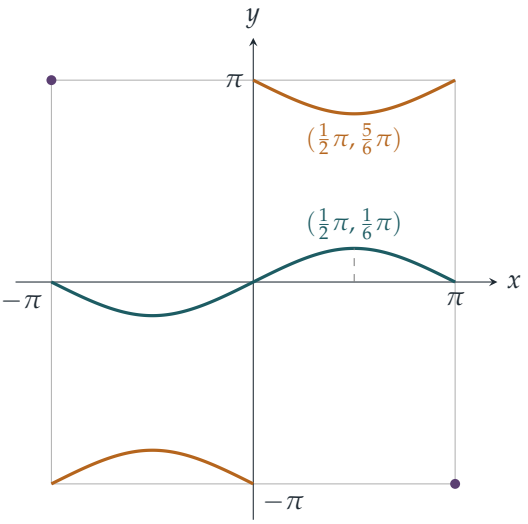
Ask yourself How do we get the whole square from this little arc without doing any more calculus?

We use symmetry. Since $\sin(\pi - x) = \sin x$ and f is odd, the graph of $y = f(x)$ on $[-\pi, \pi]$ is symmetric about $x = \frac{1}{2}\pi$ and has rotational symmetry about the origin. It is a gentle wave through $(-\pi, 0)$, $(0, 0)$ and $(\pi, 0)$, with maximum $(\frac{1}{2}\pi, \frac{1}{6}\pi)$ and minimum $(-\frac{1}{2}\pi, -\frac{1}{6}\pi)$. Its gradient is $\frac{1}{2}$ at the origin and $-\frac{1}{2}$ at $x = \pm\pi$.

As in (i), the full solution set is given by $y = f(x) + 2k\pi$ or $y = \pi - f(x) + 2k\pi$. Within the square this leaves:

- the wave $y = f(x)$ for all $-\pi \leq x \leq \pi$;
- $y = \pi - f(x)$, which lies in the square only where $f(x) \geq 0$, that is for $0 \leq x \leq \pi$. This is an arch hanging down from $(0, \pi)$ to (π, π), lowest at $(\frac{1}{2}\pi, \frac{5}{6}\pi)$;
- $y = -\pi - f(x)$, which lies in the square only where $f(x) \leq 0$, that is for $-\pi \leq x \leq 0$. This is an arch rising from $(-\pi, -\pi)$ to $(0, -\pi)$, highest at $(-\frac{1}{2}\pi, -\frac{5}{6}\pi)$;
- the isolated corner points $(-\pi, \pi)$ and $(\pi, -\pi)$, from $y = \pi - f(x)$ at $x = -\pi$ and $y = -\pi - f(x)$ at $x = \pi$.

The other values of k give $|y| > \pi$. The arches meet the lines $y = \pm\pi$ at gradient $\pm\frac{1}{2}$, and are the wave rotated through a half-turn and moved up or down. Compare (i): the diagonal has flattened into the wave and the two segments have bent into arches.



Part (iii)

Ask yourself How can we turn $\cos y$ into something we've already sketched?

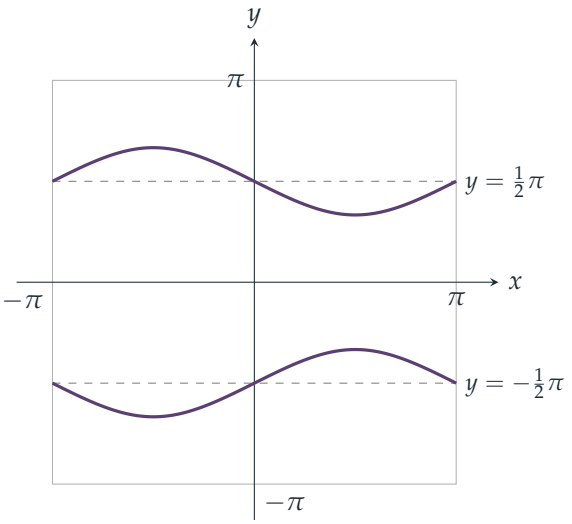
Write $\cos y = \sin(\frac{1}{2}\pi - y)$. Then $\cos y = \frac{1}{2}\sin x$ says that $u = \frac{1}{2}\pi - y$ satisfies $\sin u = \frac{1}{2}\sin x$, so $u = f(x) + 2k\pi$ or $u = \pi - f(x) + 2k\pi$. That is,

$$y = \tfrac{1}{2}\pi - f(x) - 2k\pi \qquad \text{or} \qquad y = f(x) - \tfrac{1}{2}\pi - 2k\pi.$$

Since $|f(x)| \le \frac{1}{6}\pi$, only $k = 0$ gives values in $[-\pi, \pi]$, and then it does so for every x. So the set consists of exactly two complete waves:

- $y = f(x) - \frac{1}{2}\pi$: the wave from (ii) moved down by $\frac{1}{2}\pi$, oscillating between $-\frac{2}{3}\pi$ and $-\frac{1}{3}\pi$;
- $y = \frac{1}{2}\pi - f(x)$: its mirror image in the x-axis (as it must be, since $\cos y$ is even in y). It has minimum $(\frac{1}{2}\pi, \frac{1}{3}\pi)$ and maximum $(-\frac{1}{2}\pi, \frac{2}{3}\pi)$.

There are no isolated points this time.



Checks

- Sample points: $(\frac{1}{2}\pi, \frac{1}{6}\pi)$ gives $\sin\frac{1}{6}\pi = \frac{1}{2} = \frac{1}{2}\sin\frac{1}{2}\pi$, and $(\frac{1}{2}\pi, \frac{5}{6}\pi)$ works too. In (iii), $(\frac{1}{2}\pi, \frac{1}{3}\pi)$ gives $\cos\frac{1}{3}\pi = \frac{1}{2}$.
- At $x = 0$ the formula gives $y' = \frac{1}{2}$, matching the small-angle approximation $y \approx \frac{1}{2}x$. At $x = \frac{1}{2}\pi$ it gives $y' = 0$, as it must at the maximum.
- A computer check confirmed the formulas for y' and y'' by finite differences at random points of $(0, \frac{1}{2}\pi)$. It also found every solution $y \in [-\pi, \pi]$ for many random x, and they matched the branches above: two in (ii) for $x \neq 0, \pm\pi$, and always two in (iii).

Common slips

- Missing the corner points $(-\pi, \pi)$ and $(\pi, -\pi)$ in (i) and (ii).
- Drawing the arches in (ii) over the whole of $[-\pi, \pi]$: $y = \pi - f(x)$ leaves the square as soon as $f(x) < 0$.
- Taking $\cos y = -\frac{1}{2}\sqrt{4 - \sin^2 x}$: the sign is fixed by $0 \leq y \leq \frac{1}{2}\pi$.

Beyond STEP

A set like $\{(x, y) : F(x, y) = 0\}$ is a *level set*. The implicit function theorem says it is locally a smooth graph wherever $\partial F/\partial y \neq 0$. Here that fails exactly where $\cos y = 0$, and since that never happens when $|\sin y| \leq \frac{1}{2}$, the curves in (ii) are smooth graphs over x. In (i), the crossing points are where both partial derivatives of $\sin y - \sin x$ vanish, and that is where the level set fails to be a curve. Bartle and Sherbert, *Introduction to Real Analysis*, builds the single-variable groundwork; Abbott, *Understanding Analysis*, is a lively companion.

WORKED SOLUTION

STEP II 2017, Question 4

Pure • Cauchy-Schwarz Integral Inequality

STEP II 2017, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s2-q4

The key idea

The Schwarz inequality bounds the integral of a product, $\int fg$, above by the integrals of the squares, $\int f^2$ and $\int g^2$. So the whole skill is choosing f and g so that every integral is one we can compute, except possibly the single one we want to bound. If the integral we want is $\int fg$, we get an *upper* bound. If it is $\int f^2$ or $\int g^2$, dividing through gives a *lower* bound.

Solution

Throughout, write $(*)$ as $\left(\int fg\right)^2 \le \int f^2 \cdot \int g^2$.

Part (i)

The target has e^t in it, so we take $f(x) = 1$, $g(x) = \mathrm{e}^x$, $a = 0$ and $b = t$. Then

$$\int_0^t \mathrm{e}^x \,\mathrm{d}x = \mathrm{e}^t - 1, \qquad \int_0^t 1 \,\mathrm{d}x = t, \qquad \int_0^t \mathrm{e}^{2x} \,\mathrm{d}x = \tfrac{1}{2}\left(\mathrm{e}^{2t} - 1\right),$$

so $(*)$ says

$$(\mathrm{e}^t - 1)^2 \le \frac{t}{2}\left(\mathrm{e}^{2t} - 1\right) = \frac{t}{2}(\mathrm{e}^t - 1)(\mathrm{e}^t + 1).$$

Since $t > 0$ we have $\mathrm{e}^t - 1 > 0$, and also $\mathrm{e}^t + 1 > 0$. Dividing by the positive quantity $(\mathrm{e}^t - 1)(\mathrm{e}^t + 1)$ keeps the direction of the inequality:

$$\frac{\mathrm{e}^t - 1}{\mathrm{e}^t + 1} \le \frac{t}{2},$$

as required.

Part (ii)

We want $\int_0^1 \mathrm{e}^{-\frac{1}{2}x^2}\,\mathrm{d}x$ to appear as $\int g^2$, so we take $g(x) = \mathrm{e}^{-\frac{1}{4}x^2}$ with $f(x) = x$ on $[0,1]$. The product $x\,\mathrm{e}^{-\frac{1}{4}x^2}$ is exactly the kind of thing we can integrate (the power of x outside is the derivative of the exponent, up to a constant):

$$\int_0^1 x\,\mathrm{e}^{-\frac{1}{4}x^2}\,\mathrm{d}x = \left[-2\,\mathrm{e}^{-\frac{1}{4}x^2}\right]_0^1 = 2\left(1-\mathrm{e}^{-\frac{1}{4}}\right), \qquad \int_0^1 x^2\,\mathrm{d}x = \frac{1}{3}.$$

So $(*)$ gives

$$4\left(1-\mathrm{e}^{-\frac{1}{4}}\right)^2 \le \frac{1}{3}\int_0^1 \mathrm{e}^{-\frac{1}{2}x^2}\,\mathrm{d}x,$$

and multiplying by 3,

$$\int_0^1 \mathrm{e}^{-\frac{1}{2}x^2}\,\mathrm{d}x \ge 12\left(1-\mathrm{e}^{-\frac{1}{4}}\right)^2.$$

Part (iii)

Write $J = \int_0^{\frac{1}{2}\pi} \sqrt{\sin x}\,\mathrm{d}x$. All integrals here are over $[0, \frac{1}{2}\pi]$, where $\sin x \ge 0$ and $\cos x \ge 0$, so all the roots are real.

Upper bound. Here J should be $\int fg$. Take $f(x) = 1$ and $g(x) = \sqrt{\sin x}$, so that $g^2 = \sin x$ is easy to integrate:

$$J^2 \le \int_0^{\frac{1}{2}\pi} 1\,\mathrm{d}x \cdot \int_0^{\frac{1}{2}\pi} \sin x\,\mathrm{d}x = \frac{\pi}{2}\cdot 1.$$

Since $J > 0$, taking positive square roots gives $J \le \sqrt{\pi/2}$.

Lower bound. Now J must be one of the squared integrals, so we take $f(x) = (\sin x)^{1/4}$, giving $f^2 = \sqrt{\sin x}$.

? **Ask yourself** Which g makes both $\int fg$ and $\int g^2$ computable?

We need g^2 to be easy to integrate, and $(\sin x)^{1/4} g(x)$ to be easy too. The choice $g(x) = \cos x$ does both: $(\sin x)^{1/4}\cos x$ is a power of $\sin x$ times its derivative. The shape of the answer is a clue as well, since a π in the denominator suggests $\int g^2 = \int_0^{\frac{1}{2}\pi} \cos^2 x\,\mathrm{d}x = \frac{\pi}{4}$. Indeed

$$\int_0^{\frac{1}{2}\pi} (\sin x)^{1/4}\cos x\,\mathrm{d}x = \left[\tfrac{4}{5}(\sin x)^{5/4}\right]_0^{\frac{1}{2}\pi} = \frac{4}{5}$$

and

$$\int_0^{\frac{1}{2}\pi} \cos^2 x\,\mathrm{d}x = \int_0^{\frac{1}{2}\pi} \tfrac{1}{2}(1+\cos 2x)\,\mathrm{d}x = \frac{\pi}{4}.$$

So $(*)$ gives

$$\frac{16}{25} \le J\cdot\frac{\pi}{4}, \qquad \text{that is,} \qquad J \ge \frac{64}{25\pi}.$$

Putting the two halves together,

$$\frac{64}{25\pi} \le \int_0^{\frac{1}{2}\pi} \sqrt{\sin x}\,\mathrm{d}x \le \sqrt{\frac{\pi}{2}}.$$

Aside. Other choices work too, and some do better. Taking $f = (\sin x)^{1/4}$, $g = (\sin x)^{3/4}$ gives $1 \le J \int_0^{\frac{1}{2}\pi} (\sin x)^{3/2}\,\mathrm{d}x$, and a second use of $(*)$ with $f = \sin x$, $g = \sqrt{\sin x}$ shows that $\int_0^{\frac{1}{2}\pi} (\sin x)^{3/2}\,\mathrm{d}x \le \sqrt{\pi/4}$. Together these give the sharper bound $J \ge 2/\sqrt{\pi} \approx 1.128$.

Checks

- (i): as $t \to 0$, the left-hand side is $\tanh(t/2) \approx t/2$, so the bound is tight for small t. For large t the left side tends to 1, while $t/2$ grows without limit.
- (ii): $\int_0^1 \mathrm{e}^{-\frac{1}{2}x^2}\,\mathrm{d}x \approx 0.8556$, while $12(1 - \mathrm{e}^{-1/4})^2 \approx 0.5871$.
- (iii): numerically $J \approx 1.1981$, which sits between $64/(25\pi) \approx 0.8149$ and $\sqrt{\pi/2} \approx 1.2533$. The upper bound is quite close because $\sqrt{\sin x}$ is fairly flat (near 1) over most of the interval, and Schwarz with $f = 1$ is an equality exactly when g is constant.

Common slips

- Getting the direction of the bound wrong. With the unknown integral as $\int fg$, Schwarz can only bound it from above, so a lower bound needs the unknown as $\int f^2$ or $\int g^2$.
- In (i), dividing by $\mathrm{e}^t - 1$ without saying that it is positive.
- Forgetting to take the positive square root, using $J > 0$, in the upper bound of (iii).

Beyond STEP

The Schwarz (Cauchy–Schwarz) inequality says that the inner product $\langle f, g\rangle = \int fg$ satisfies $|\langle f, g\rangle| \le \|f\|\,\|g\|$, with equality exactly when f and g are proportional. It holds in every inner product space, from vectors in $\mathbb{R}^n$ to spaces of functions, and it is what makes "the angle between two functions" meaningful. That idea underlies Fourier series and least-squares approximation. Strang's *Introduction to Linear Algebra* develops inner products and orthogonality, and Abbott's *Understanding Analysis* shows how such inequalities are used to control integrals.

WORKED SOLUTION

STEP II 2017, Question 5

Pure • Parabola Normals and Circles

STEP II 2017, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s2-q5

The key idea

Everything rests on one fact about the parabola $x = at^2$, $y = 2at$: the chord joining the points with parameters t_1 and t_2 has gradient $\dfrac{2}{t_1 + t_2}$. With that, "the normal meets the curve again" and "the angle in a semicircle is a right angle" both become simple equations between parameters, and the geometry turns into short algebra.

Solution

A useful fact: chord gradients

For distinct parameters $t_1 \neq t_2$, the chord joining $(at_1^2, 2at_1)$ and $(at_2^2, 2at_2)$ has gradient

$$\frac{2a(t_1 - t_2)}{a(t_1^2 - t_2^2)} = \frac{2}{t_1 + t_2},$$

cancelling $t_1 - t_2 \neq 0$. (If $t_1 + t_2 = 0$ the chord is vertical.) Letting $t_2 \to t_1$, the tangent at t has gradient $\frac{1}{t}$; directly, $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{2a}{2at} = \dfrac{1}{t}$.

Part (i)

At P the tangent has gradient $\frac{1}{p}$, so the normal has gradient $-p$ (here $p \neq 0$ is used). The point N, with $n \neq p$, lies on this normal exactly when the chord PN has gradient $-p$:

$$\frac{2}{p+n} = -p \iff p + n = -\frac{2}{p} \iff n = -\left(p + \frac{2}{p}\right).$$

(The chord PN cannot be vertical, as the normal at P has finite gradient $-p$.) Note $n \neq p$, since $n - p = -2\left(p + \frac{1}{p}\right)$ and $p + \frac{1}{p} \neq 0$ for real p, so this really is a second point. This is the required result.

Part (ii)

Using $p+n=-\frac{2}{p}$ and $p-n=2p+\frac{2}{p}=\dfrac{2(p^2+1)}{p}$,

$$\begin{aligned}|PN|^2 &= a^2(p^2-n^2)^2+4a^2(p-n)^2\\ &= a^2(p-n)^2\left[(p+n)^2+4\right]\\ &= a^2\cdot\frac{4(p^2+1)^2}{p^2}\cdot\left(\frac{4}{p^2}+4\right)=a^2\cdot\frac{4(p^2+1)^2}{p^2}\cdot\frac{4(p^2+1)}{p^2}\\ &= 16a^2\,\frac{(p^2+1)^3}{p^4},\end{aligned}$$

as required.

To minimise, write $u=p^2>0$ and $f(u)=\dfrac{(u+1)^3}{u^2}$. By the quotient rule,

$$f'(u)=\frac{3(u+1)^2u^2-2u(u+1)^3}{u^4}=\frac{(u+1)^2(3u-2u-2)}{u^3}=\frac{(u+1)^2(u-2)}{u^3}.$$

For $u>0$ the factor $\frac{(u+1)^2}{u^3}$ is positive, so $f'(u)<0$ for $0<u<2$ and $f'(u)>0$ for $u>2$. Hence f decreases then increases, and its minimum on $u>0$ is at $u=2$, i.e. $p^2=2$. Since $|PN|^2=16a^2f(p^2)$, this minimises $|PN|$. (The minimum value is $|PN|^2=16a^2\cdot\frac{27}{4}=108a^2$, so $|PN|=6\sqrt{3}\,a$.)

Part (iii)

Since PN is a diameter and Q is a point of the circle other than P and N, the angle PQN is a right angle (angle in a semicircle). So $QP\perp QN$.

Ask yourself Could QP or QN be vertical, so that we can't multiply gradients?

If QP were vertical then QN would be horizontal, so Q and N would have the same y-coordinate, giving $q=n$, which is impossible as $Q\neq N$. Similarly, QN vertical would force QP horizontal and $q=p$. So both gradients are finite and non-zero, and their product is -1:

$$\frac{2}{q+p}\cdot\frac{2}{q+n}=-1\iff(q+p)(q+n)=-4.$$

Expanding, and using $p+n=-\frac{2}{p}$ and $pn=-p^2-2$ from part (i),

$$q^2+(p+n)q+pn=q^2-\frac{2q}{p}-p^2-2=-4,$$

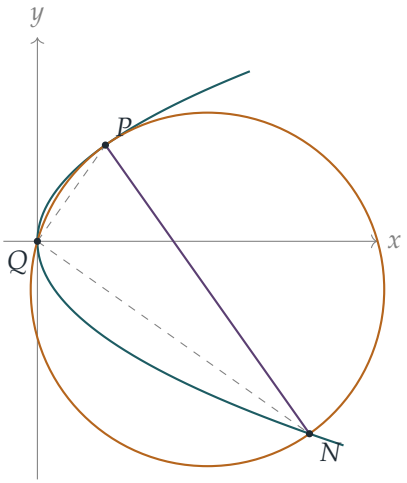
which rearranges to

$$2=p^2-q^2+\frac{2q}{p},$$

as required.

Deduction. Put $q = 0$: the relation becomes $2 = p^2$. Conversely, if $p^2 = 2$ the relation gives $q^2 - \frac{2q}{p} = 0$, i.e. $q = 0$ or $q = \frac{2}{p}$; but $\frac{2}{p} = \frac{p^2}{p} = p$, which is the point P itself, so Q must have $q = 0$. Hence Q is at the origin exactly when $p^2 = 2$, which by part (ii) is exactly when $|PN|$ is at its minimum.

In general, the relation is a quadratic $q^2 - \frac{2}{p}q + (2 - p^2) = 0$ whose roots sum to $\frac{2}{p}$; one root is always $q = p$, so the other is $q = \dfrac{2 - p^2}{p}$, which is zero precisely when $p^2 = 2$.



The figure shows the minimising case $a = 1$, $p = \sqrt{2}$: the normal PN (plum), the circle on PN as diameter (amber), and the right angle at $Q = O$.

Checks

- With $a = 1$, $p = 1$: $n = -3$, so $P = (1, 2)$ and $N = (9, -6)$. Then $|PN|^2 = 64 + 64 = 128$, and the formula gives $16 \cdot 8/1 = 128$.
- With $p = \sqrt{2}$, $a = 1$: $P = (2, 2\sqrt{2})$, $N = (8, -4\sqrt{2})$, and $\overrightarrow{OP} \cdot \overrightarrow{ON} = 16 - 16 = 0$, so the origin does lie on the circle with diameter PN. Here $|PN| = 6\sqrt{3} \approx 10.392$.
- Locating where the circle meets the parabola numerically, for several values of p, gives exactly the parameters n and $\frac{2-p^2}{p}$ (besides P).

Common slips

- Forgetting that $q = p$ also satisfies the final relation, and so wrongly concluding there are two possible positions of Q when $p^2 = 2$.
- Writing the normal gradient as $-\frac{1}{p}$: the tangent gradient is $\frac{1}{p}$, so the normal's is $-p$.

Beyond STEP

A circle and a parabola usually meet in four points, and substituting $x = at^2$, $y = 2at$ into a circle's equation gives a quartic in t with no t^3 term, so the four parameters sum to zero. Here $n + q = -\frac{2}{p} - p + \frac{2}{p} = -p$, so the fourth parameter is p again: the circle touches the parabola at P, as it must, since its radius there lies along the normal. Counting intersections with multiplicity like this is the starting point of algebraic geometry (Bézout's theorem). For more problems in this style, see Siklos, *Advanced Problems in Mathematics*.

WORKED SOLUTION

STEP II 2017, Question 6

Pure • Inequalities and Induction

STEP II 2017, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s2-q6

The key idea

Both parts compare the sum S_n with $2\sqrt{n}$, which is roughly $\int_0^n x^{-1/2}\,\mathrm{d}x$. For the induction in (i), the step from n to $n+1$ comes down to comparing $1/\sqrt{n+1}$ with $2(\sqrt{n+1}-\sqrt{n})$, and rationalising that difference settles it. In (ii), the smallest C is the largest value of the gap between the right-hand side (without C) and S_n. So we study how that gap changes from n to $n+1$, and the given inequality is exactly what tells us the gap always shrinks.

Solution

Part (i)

Let $P(n)$ be the statement $S_n \le 2\sqrt{n}-1$.

Base case. $S_1 = 1$ and $2\sqrt{1}-1 = 1$, so $P(1)$ holds (with equality).

Inductive step. Suppose $P(k)$ holds for some $k \ge 1$. Then

$$S_{k+1} = S_k + \frac{1}{\sqrt{k+1}} \le 2\sqrt{k} - 1 + \frac{1}{\sqrt{k+1}},$$

so it is enough to show that

$$2\sqrt{k} + \frac{1}{\sqrt{k+1}} \le 2\sqrt{k+1}, \qquad \text{that is,} \qquad \frac{1}{\sqrt{k+1}} \le 2\left(\sqrt{k+1}-\sqrt{k}\right).$$

Rationalising,

$$2\left(\sqrt{k+1}-\sqrt{k}\right) = \frac{2}{\sqrt{k+1}+\sqrt{k}} \ge \frac{2}{\sqrt{k+1}+\sqrt{k+1}} = \frac{1}{\sqrt{k+1}},$$

because $\sqrt{k} \le \sqrt{k+1}$ makes the denominator no bigger than $2\sqrt{k+1}$. Hence $S_{k+1} \le 2\sqrt{k+1}-1$, which is $P(k+1)$.

Since $P(1)$ is true and $P(k) \Rightarrow P(k+1)$ for every $k \ge 1$, $S_n \le 2\sqrt{n}-1$ for all positive integers n.

Part (ii): the inequality

For $k \geq 0$ both sides of $(4k+1)\sqrt{k+1} > (4k+3)\sqrt{k}$ are non-negative, and the left-hand side is positive, so the inequality is equivalent to the one between their squares. Now

$$\begin{aligned}(4k+1)^2(k+1) - (4k+3)^2 k &= (16k^2+8k+1)(k+1) - (16k^2+24k+9)k \\ &= (16k^3+24k^2+9k+1) - (16k^3+24k^2+9k) \\ &= 1 > 0.\end{aligned}$$

So $(4k+1)^2(k+1) > (4k+3)^2 k$, and taking positive square roots gives $(4k+1)\sqrt{k+1} > (4k+3)\sqrt{k}$ for all $k \geq 0$, as required.

Part (ii): the smallest C

Define

$$T_n = 2\sqrt{n} + \frac{1}{2\sqrt{n}} - S_n.$$

The inequality $S_n \geq 2\sqrt{n} + \frac{1}{2\sqrt{n}} - C$ is the same as $C \geq T_n$. So it holds for every n exactly when C is at least every T_n, and the smallest such C is the largest value of T_n (if it has one).

Ask yourself How do we find the largest T_n without knowing a formula for S_n?

We look at the change $T_{n+1} - T_n$, in which S_n disappears. Since $S_{n+1} - S_n = 1/\sqrt{n+1}$,

$$\begin{aligned}T_{n+1} - T_n &= 2\sqrt{n+1} - 2\sqrt{n} + \frac{1}{2\sqrt{n+1}} - \frac{1}{2\sqrt{n}} - \frac{1}{\sqrt{n+1}} \\ &= 2\sqrt{n+1} - 2\sqrt{n} - \frac{1}{2\sqrt{n+1}} - \frac{1}{2\sqrt{n}}.\end{aligned}$$

Multiply through by the positive quantity $2\sqrt{n}\sqrt{n+1}$:

$$\begin{aligned}2\sqrt{n}\sqrt{n+1}\left(T_{n+1} - T_n\right) &= 4(n+1)\sqrt{n} - 4n\sqrt{n+1} - \sqrt{n} - \sqrt{n+1} \\ &= (4n+3)\sqrt{n} - (4n+1)\sqrt{n+1}.\end{aligned}$$

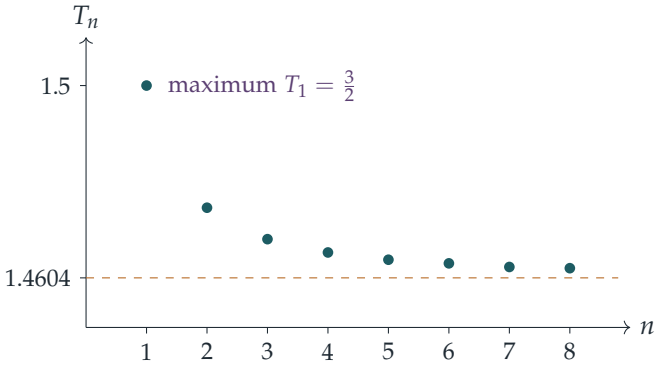
By the inequality we have just proved (with $k = n$), this is negative. Hence $T_{n+1} < T_n$ for every $n \geq 1$: the sequence T_n is strictly decreasing, and its largest value is the first one,

$$T_1 = 2 + \tfrac{1}{2} - 1 = \tfrac{3}{2}.$$

So $S_n \geq 2\sqrt{n} + \frac{1}{2\sqrt{n}} - \frac{3}{2}$ for all $n \geq 1$. No smaller C will do, because at $n = 1$ the inequality reads $1 \geq \frac{5}{2} - C$, which needs $C \geq \frac{3}{2}$. The smallest number is

$$C = \tfrac{3}{2}.$$

The first few values of T_n show how fast it settles down:



(The vertical axis is stretched: it runs from about 1.45 to 1.51.)

Checks

- For $n = 4$: $S_4 = 1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \frac{1}{2} \approx 2.7845$, which lies between $2\sqrt{4} + \frac{1}{4} - \frac{3}{2} = 2.75$ and $2\sqrt{4} - 1 = 3$, as both results require.
- A computer check of the first 200 000 values confirmed both bounds and that T_n decreases; T_n falls from 1.5 towards about 1.46035 and never returns to 1.5.
- The identity $(4k+1)^2(k+1) - (4k+3)^2 k = 1$ is easy to test at $k = 1$: $25 \cdot 2 - 49 = 1$.

Common slips

- Answering with the limit of T_n (about 1.46) instead of its maximum. A C below $\frac{3}{2}$ fails at $n = 1$.
- Squaring an inequality without first noting that both sides are non-negative.
- Forgetting to show that $C = \frac{3}{2}$ is actually the smallest, by exhibiting the case of equality.

Beyond STEP

The sequence T_n decreases to a limit, and that limit is $-\zeta(\frac{1}{2}) \approx 1.46035$, where ζ is the Riemann zeta function, extended beyond the region where its defining series converges. Comparing a sum with an integral and then correcting by terms like $\frac{1}{2\sqrt{n}}$ is the start of the Euler–Maclaurin formula. The underlying fact that a bounded monotonic sequence converges is the monotone convergence theorem of first-year analysis; Abbott's *Understanding Analysis* treats it carefully.

WORKED SOLUTION

STEP II 2017, Question 7

Pure • Logarithmic Differentiation

STEP II 2017, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s2-q7

The key idea

Every power here is defined through logarithms, $a^b = \mathrm{e}^{b \ln a}$, so comparing two powers of x means comparing their logarithms, and since exp is increasing that comparison is equivalent. The one thing to watch is the sign of $\ln x$: it is negative for $0 < x < 1$, and dividing by it reverses inequalities. Taking logarithms also makes the derivatives routine.

Solution

Throughout, $\ln \mathrm{f}(x) = x \ln x$ and $\ln \mathrm{g}(x) = \mathrm{f}(x) \ln x$.

Part (i)

Take $0 < x < 1$, so $\ln x < 0$ and $x - 1 < 0$. Then

$$\ln \mathrm{f}(x) - \ln x = x \ln x - \ln x = (x-1) \ln x > 0,$$

a product of two negative numbers. Since exp is increasing, $\mathrm{f}(x) > x$.

Also $x \ln x < 0$, so $\mathrm{f}(x) = \mathrm{e}^{x \ln x} < 1$. Now compare logarithms again:

$$\begin{aligned} \ln \mathrm{g}(x) - \ln x &= \big(\mathrm{f}(x) - 1\big) \ln x > 0, && \text{since } \mathrm{f}(x) < 1 \text{ and } \ln x < 0, \\ \ln \mathrm{f}(x) - \ln \mathrm{g}(x) &= \big(x - \mathrm{f}(x)\big) \ln x > 0, && \text{since } \mathrm{f}(x) > x \text{ and } \ln x < 0. \end{aligned}$$

So $\ln x < \ln \mathrm{g}(x) < \ln \mathrm{f}(x)$, which gives $x < \mathrm{g}(x) < \mathrm{f}(x)$ for $0 < x < 1$, as required.

For $x > 1$ the same differences apply, but now $\ln x > 0$, $x - 1 > 0$ and $x \ln x > 0$, so $\mathrm{f}(x) > 1$. Then $(x-1) \ln x > 0$ gives $\mathrm{f}(x) > x$, and $(x - \mathrm{f}(x)) \ln x < 0$ gives $\mathrm{g}(x) > \mathrm{f}(x)$. So

$$x < \mathrm{f}(x) < \mathrm{g}(x) \qquad \text{for } x > 1,$$

and at $x = 1$ all three equal 1.

Part (ii)

Differentiating $\ln f(x) = x \ln x$ gives $f'(x)/f(x) = \ln x + 1$, so

$$f'(x) = x^x(\ln x + 1).$$

Since $x^x > 0$, $f'(x) = 0$ exactly when $\ln x = -1$, that is, $x = e^{-1}$. (f' changes from negative to positive there, so this is a minimum, with value $e^{-1/e} \approx 0.692$.)

Part (iii)

As $x \to 0$, $x \ln x \to 0$, and exp is continuous, so

$$f(x) = e^{x \ln x} \to e^0 = 1.$$

Then $\ln g(x) = f(x) \ln x$, where $f(x) \to 1$ and $\ln x \to -\infty$, so $\ln g(x) \to -\infty$ and

$$\lim_{x \to 0} g(x) = 0.$$

Part (iv)

Let $h(x) = x^{-1} + \ln x$ for $x > 0$. Then

$$h'(x) = -\frac{1}{x^2} + \frac{1}{x} = \frac{x-1}{x^2},$$

which is negative for $0 < x < 1$ and positive for $x > 1$. So h decreases then increases, and its least value is $h(1) = 1$. Hence $x^{-1} + \ln x \geq 1$ for all $x > 0$.

Now differentiate $\ln g(x) = f(x) \ln x$ with the product rule, using part (ii):

$$\frac{g'(x)}{g(x)} = f'(x) \ln x + \frac{f(x)}{x} = x^x \left[(\ln x + 1) \ln x + \frac{1}{x} \right] = x^x \left[(\ln x)^2 + \ln x + \frac{1}{x} \right].$$

? **Ask yourself** How does the inequality just proved help with this bracket?

The bracket contains exactly $\frac{1}{x} + \ln x$, which is at least 1. So

$$(\ln x)^2 + \ln x + \frac{1}{x} \;\geq\; (\ln x)^2 + 1 \;>\; 0.$$

Since $g(x) > 0$ and $x^x > 0$, we get

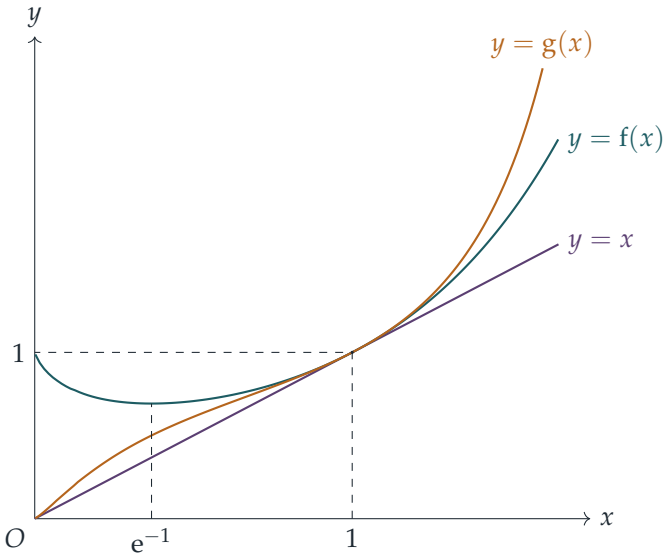
$$g'(x) = g(x)\, x^x \left[(\ln x)^2 + \ln x + \frac{1}{x} \right] > 0 \qquad \text{for all } x > 0,$$

as required.

The sketch

Collecting the facts:

- $y = \mathrm{f}(x)$ starts from 1 as $x \to 0$, falls to its minimum $\mathrm{e}^{-1/\mathrm{e}}$ at $x = \mathrm{e}^{-1}$, then rises through $(1,1)$ and grows very fast. Near 0 it is very steep, since $\mathrm{f}'(x) = x^x(\ln x + 1) \to -\infty$.
- $y = \mathrm{g}(x)$ starts from the origin and increases throughout (part (iv)), passing through $(1,1)$.
- For $0 < x < 1$ the order is $x < \mathrm{g} < \mathrm{f}$; for $x > 1$ it is $x < \mathrm{f} < \mathrm{g}$. So g and f cross at $(1,1)$, and all three curves meet there.
- At $x = 1$: $\mathrm{f}'(1) = 1$ and $\mathrm{g}'(1) = 1 \cdot 1 \cdot (0 + 0 + 1) = 1$, so both curves touch $y = x$ at $(1,1)$.
- Near the origin $\mathrm{g}(x)/x = \mathrm{e}^{(\mathrm{f}(x)-1)\ln x} \to 1$ (as $\mathrm{f}(x) - 1 \approx x \ln x$, the exponent behaves like $x(\ln x)^2 \to 0$), so $y = \mathrm{g}(x)$ leaves the origin along $y = x$, just above it.



Checks

- At $x = \frac{1}{2}$: $\mathrm{f} = \frac{1}{\sqrt{2}} \approx 0.707$ and $\mathrm{g} = (\frac{1}{2})^{0.707} \approx 0.613$, so $0.5 < 0.613 < 0.707$. At $x = 2$: $\mathrm{f} = 4$ and $\mathrm{g} = 2^4 = 16$, so $2 < 4 < 16$.
- $\mathrm{f}(0.3) \approx 0.697$, $\mathrm{f}(\mathrm{e}^{-1}) \approx 0.6922$ and $\mathrm{f}(0.4) \approx 0.693$, consistent with a minimum at e^{-1}.
- The formula for g' gives $\mathrm{g}'(\frac{1}{2}) \approx 0.774$, which matches a difference quotient of g there; a computer check also confirmed $\mathrm{g}' > 0$ on $(0,3)$.

Common slips

- Forgetting that $\ln x < 0$ on $(0,1)$ and so not reversing the inequality when dividing by it; this is exactly why the orderings swap at $x = 1$.
- Differentiating x^x as $x \cdot x^{x-1}$ (treating the index as constant) or as $x^x \ln x$ (treating the base as constant); both pieces are needed.
- Writing the limit of g as $1^{-\infty}$ and guessing; work with $\ln \mathrm{g} = \mathrm{f} \ln x$ instead.

Beyond STEP

The tower $x^{x^{x^{\cdots}}}$ continues this pattern: the odd-height towers x, g, $\ldots$ and the even-height towers f, $\ldots$ interleave for $0 < x < 1$, and the infinite tower converges exactly when $\mathrm{e}^{-\mathrm{e}} \leq x \leq \mathrm{e}^{1/\mathrm{e}}$, a nice exercise in monotone sequences and fixed points. Limits like $x \ln x \to 0$, and the careful definition of a^b via exp and ln, are treated properly in a first analysis course; see Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP II 2017, Question 8

Pure • Vectors and Orthocentre

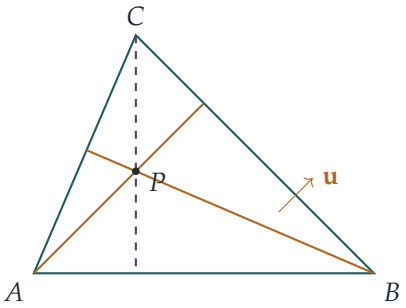
STEP II 2017, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s2-q8

The key idea

This is the vector proof that the three altitudes of a triangle meet at a single point, the *orthocentre*. The trick for finding where two lines $\mathbf{a} + \lambda\mathbf{u}$ and $\mathbf{b} + \mu\mathbf{v}$ meet is to take the scalar product of the equation with a vector perpendicular to $\mathbf{v}$: that kills μ at a stroke. Here such a vector is handed to us, because $\mathbf{v}$ is perpendicular to CA, so $\mathbf{c} - \mathbf{a}$ does the job.



Solution

The first altitude

The line through A perpendicular to BC passes through $\mathbf{a}$ and has direction $\mathbf{u}$ (we are in a plane, so every vector perpendicular to BC is a multiple of $\mathbf{u}$). Its equation is

$$\mathbf{r} = \mathbf{a} + \lambda\mathbf{u}.$$

Similarly the line through B perpendicular to CA is $\mathbf{r} = \mathbf{b} + \mu\mathbf{v}$.

Finding P

These two lines are not parallel: if $\mathbf{u}$ and $\mathbf{v}$ were parallel then BC and CA, being perpendicular to them in the same plane, would be parallel, which is impossible for a triangle. So they meet at exactly one point P, where

$$\mathbf{a}+\lambda\mathbf{u}=\mathbf{b}+\mu\mathbf{v}.$$

Take the scalar product of both sides with $\mathbf{c}-\mathbf{a}$. Since $\mathbf{v}$ is perpendicular to CA, $\mathbf{v}\cdot(\mathbf{c}-\mathbf{a})=0$, and μ disappears:

$$\mathbf{a}\cdot(\mathbf{c}-\mathbf{a})+\lambda\,\mathbf{u}\cdot(\mathbf{c}-\mathbf{a})=\mathbf{b}\cdot(\mathbf{c}-\mathbf{a}),\qquad\text{so}\qquad\lambda\,\mathbf{u}\cdot(\mathbf{c}-\mathbf{a})=(\mathbf{b}-\mathbf{a})\cdot(\mathbf{c}-\mathbf{a}).$$

? **Ask yourself** Are we allowed to divide by $\mathbf{u}\cdot(\mathbf{c}-\mathbf{a})$?

Yes. If $\mathbf{u}\cdot(\mathbf{c}-\mathbf{a})=0$ then $\mathbf{u}$ would be perpendicular to CA as well as to BC; in a plane that forces CA and BC to be parallel, which they are not. Hence

$$\mathbf{p}=\mathbf{a}+\frac{(\mathbf{b}-\mathbf{a})\cdot(\mathbf{c}-\mathbf{a})}{\mathbf{u}\cdot(\mathbf{c}-\mathbf{a})}\,\mathbf{u}.$$

CP is perpendicular to AB

We need $(\mathbf{p}-\mathbf{c})\cdot(\mathbf{b}-\mathbf{a})=0$. Write λ for the coefficient found above. Then

$$(\mathbf{p}-\mathbf{c})\cdot(\mathbf{b}-\mathbf{a})=(\mathbf{a}-\mathbf{c})\cdot(\mathbf{b}-\mathbf{a})+\lambda\,\mathbf{u}\cdot(\mathbf{b}-\mathbf{a}).$$

The useful observation is that, because $\mathbf{u}$ is perpendicular to BC, $\mathbf{u}\cdot(\mathbf{b}-\mathbf{c})=0$, so

$$\mathbf{u}\cdot(\mathbf{b}-\mathbf{a})=\mathbf{u}\cdot(\mathbf{b}-\mathbf{c})+\mathbf{u}\cdot(\mathbf{c}-\mathbf{a})=\mathbf{u}\cdot(\mathbf{c}-\mathbf{a}).$$

Therefore $\lambda\,\mathbf{u}\cdot(\mathbf{b}-\mathbf{a})=\lambda\,\mathbf{u}\cdot(\mathbf{c}-\mathbf{a})=(\mathbf{b}-\mathbf{a})\cdot(\mathbf{c}-\mathbf{a})$, and

$$(\mathbf{p}-\mathbf{c})\cdot(\mathbf{b}-\mathbf{a})=-(\mathbf{c}-\mathbf{a})\cdot(\mathbf{b}-\mathbf{a})+(\mathbf{b}-\mathbf{a})\cdot(\mathbf{c}-\mathbf{a})=0.$$

? **Ask yourself** Does a zero scalar product really give a *line* CP?

Only if $P\neq C$, and this is where "non-right-angled" is used. If $P=C$ then C lies on the line through A perpendicular to BC, so $AC\perp BC$ and the angle at C is a right angle, which is excluded. So $\mathbf{p}-\mathbf{c}\neq\mathbf{0}$, and since $\mathbf{b}-\mathbf{a}\neq\mathbf{0}$, the zero scalar product says exactly that CP is perpendicular to AB. Thus the three altitudes of the triangle all pass through P.

Checks

Take $A=(0,0)$, $B=(5,0)$, $C=(1.5,3.5)$ and $\mathbf{u}=(1,1)$, which is perpendicular to $\mathbf{c}-\mathbf{b}=(-3.5,3.5)$. Then $(\mathbf{b}-\mathbf{a})\cdot(\mathbf{c}-\mathbf{a})=7.5$ and $\mathbf{u}\cdot(\mathbf{c}-\mathbf{a})=5$, so $\lambda=1.5$ and

$P = (1.5, 1.5)$. This lies directly below C, so CP is vertical and perpendicular to AB, and $(\mathbf{p} - \mathbf{b}) \cdot (\mathbf{c} - \mathbf{a}) = (-3.5)(1.5) + (1.5)(3.5) = 0$, so P is on the altitude from B too. The general result was also confirmed by computer on many random triangles. Notice too that the formula for $\mathbf{p}$ is unchanged if $\mathbf{u}$ is replaced by $k\mathbf{u}$, as it must be.

Common slips

- Dotting with $\mathbf{v}$ or $\mathbf{u}$ instead of with $\mathbf{c} - \mathbf{a}$: this leaves both parameters in the equation.
- Dividing by $\mathbf{u} \cdot (\mathbf{c} - \mathbf{a})$ without saying why it is non-zero.
- Missing the step $\mathbf{u} \cdot (\mathbf{b} - \mathbf{a}) = \mathbf{u} \cdot (\mathbf{c} - \mathbf{a})$, which is what makes the final cancellation work.

Beyond STEP

The orthocentre H, centroid G and circumcentre O of any triangle lie on one line, the Euler line, with $\overrightarrow{OH} = 3\overrightarrow{OG}$; with O as origin this is the neat statement $\mathbf{h} = \mathbf{a} + \mathbf{b} + \mathbf{c}$, which you can prove with the same scalar-product ideas. Scalar products of this kind are the starting point of the inner-product spaces studied in a first course on linear algebra; Strang's *Introduction to Linear Algebra* is a good place to see them developed.

WORKED SOLUTION

STEP II 2017, Question 9

Mechanics • Friction and Rigid Bodies

STEP II 2017, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s2-q9

The key idea

Each cylinder is a rigid body acted on by forces at only two places, the plank and the floor, so taking moments about its centre is the natural first move: the two normal reactions pass through the centre, which leaves only the two frictional forces, and they must be equal. After that it is a matter of resolving for the cylinder and for the plank, and then feeding in the friction laws $G \leq \frac{1}{2}R$ and $F \leq \frac{1}{2}N$ at the contacts. The geometry links θ to a and r at the very end.

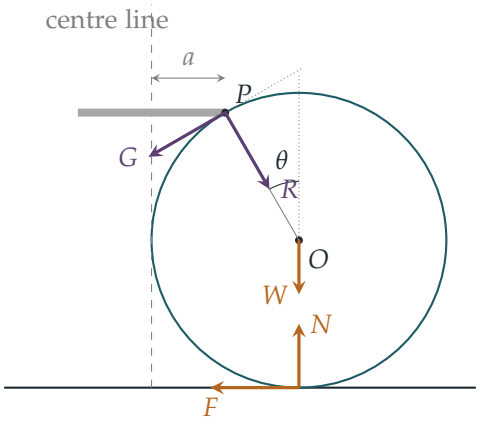
Solution

Model and set-up. Treat the cross-section as two discs of radius r and a horizontal rod, with all contacts rough. The cylinders are a negligible distance apart, so their centres are at horizontal distance r either side of the vertical line of symmetry. The plank has width $2a < 2r$, so it rests with its two long edges on the inner upper parts of the cylinders. By symmetry we need only consider the right-hand cylinder.

Let P be the point where the plank touches it. The tangent at P makes angle θ with the horizontal plank, so the radius to P makes angle θ with the vertical. The forces on the cylinder are:

- its weight W, acting at the centre O;
- at the floor, the normal reaction N upwards and friction F (acting towards the centre line, since the plank pushes the cylinder outwards);
- at P, the normal reaction R along PO, and a frictional force G along the tangent, directed downwards and inwards.

By Newton's third law, the plank feels R along OP (upwards, away from the centre line) and G along the tangent, upwards and outwards.



The dotted line through P is the tangent. Horizontally, P is $r\sin\theta$ inside the centre O, and O is at distance r from the centre line.

Part (i)

Moments about O for the cylinder. W, N and R all pass through O. Both frictional forces are tangential to the circle, so each has moment (force)$\times r$, and they turn in opposite senses: F at the bottom, pointing inwards, and G at P, pointing down and inwards. Hence

$$Fr = Gr \quad \Longrightarrow \quad G = F.$$

Resolving horizontally for the cylinder (outwards positive): R has horizontal component $R\sin\theta$ outwards, G has component $G\cos\theta$ inwards and the floor friction F acts inwards, so

$$R\sin\theta - G\cos\theta - F = 0.$$

With $G = F$ this is

$$R\sin\theta = F(1+\cos\theta),$$

as required.

The deduction. There is no slipping at the plank, so friction there satisfies $G \le \frac{1}{2}R$, that is $F \le \frac{1}{2}R$. Then

$$R\sin\theta = F(1+\cos\theta) \le \tfrac{1}{2}R(1+\cos\theta),$$

and dividing by $\frac{1}{2}R > 0$ (the plank presses on the cylinder) gives $2\sin\theta \le 1+\cos\theta$.

Part (ii)

Resolving vertically for the plank. Each edge receives R, with upward component $R\cos\theta$, and $G = F$ along the tangent upwards, with upward component $F\sin\theta$. So

$$2(R\cos\theta + F\sin\theta) = kW.$$

Using part (i), $R = \dfrac{F(1+\cos\theta)}{\sin\theta}$, so

$$R\cos\theta + F\sin\theta = \frac{F\left[(1+\cos\theta)\cos\theta + \sin^2\theta\right]}{\sin\theta} = \frac{F(1+\cos\theta)}{\sin\theta},$$

since $\cos^2\theta + \sin^2\theta = 1$. Hence

$$\tfrac{1}{2}kW = \frac{(1+\cos\theta)}{\sin\theta}\,F.$$

(The plank's horizontal forces cancel by symmetry, as do its moments.)

Resolving vertically for the cylinder. The reactions from the plank push down by $R\cos\theta + G\sin\theta = \frac{1}{2}kW$, so

$$N = W + \tfrac{1}{2}kW = \left(\frac{2}{k}+1\right)\tfrac{1}{2}kW = \left(1+\frac{2}{k}\right)\left(\frac{1+\cos\theta}{\sin\theta}\right)F.$$

That is the required expression. (This is just the obvious fact that the floor carries half of the total weight $2W + kW$.)

No slipping on the floor. The condition is

$$F \le \tfrac{1}{2}N.$$

From the expression for N,

$$\frac{N}{F} = \left(1+\frac{2}{k}\right)\frac{1+\cos\theta}{\sin\theta}.$$

Part (i) gives $\dfrac{1+\cos\theta}{\sin\theta} \ge 2$, because $\sin\theta > 0$ for acute θ, and $1 + \frac{2}{k} > 1$ because $k > 0$. So $N/F > 2$, i.e. $F < \frac{1}{2}N$. The floor condition therefore holds automatically once the plank does not slip, so it puts no extra restriction on θ.

> **Ask yourself** Why does the floor contact come for free, when both contacts carry the same friction F?

Because the floor presses harder. Comparing the formulas for R and N gives $N = \left(1+\frac{2}{k}\right)R > R$, since the floor also carries the cylinder's own weight. With the same friction at both contacts and a larger normal force at the floor, the plank contact is always the one that slips first.

Part (iii)

Rearrange the inequality from part (i) and square. Both sides are positive, so squaring keeps the direction:

$$4\sin^2\theta \le (1+\cos\theta)^2 \iff 4(1-\cos\theta)(1+\cos\theta) \le (1+\cos\theta)^2.$$

Dividing by $1+\cos\theta > 0$ gives $4 - 4\cos\theta \le 1 + \cos\theta$, i.e. $\cos\theta \ge \frac{3}{5}$. Since θ is acute,

$$\sin\theta = \sqrt{1-\cos^2\theta} \le \sqrt{1-\tfrac{9}{25}} = \tfrac{4}{5}.$$

(Equivalently, with $t = \tan\frac{1}{2}\theta$ the inequality reads $2t \le 1$, and $\sin\theta = \frac{2t}{1+t^2}$ is at most $\frac{4}{5}$ when $t = \frac{1}{2}$.)

The geometry. The edge P of the plank is at horizontal distance a from the centre line. The centre O is at distance r from the centre line, and P lies $r\sin\theta$ closer to it than O does. So

$$a = r - r\sin\theta = r(1-\sin\theta) \ge r\left(1-\tfrac{4}{5}\right) = \tfrac{1}{5}r,$$

which gives $r \le 5a$.

Checks

- At the extreme case $\sin\theta = \frac{4}{5}$, $\cos\theta = \frac{3}{5}$, take $W = 1$ and $k = 1$. The plank equation gives $F = \frac{1}{4}$, then $R = \frac{1}{2}$ and $N = \frac{3}{2}$. So $G/R = \frac{1}{2}$ exactly (the plank is on the point of slipping) while $F/N = \frac{1}{6}$, comfortably below $\frac{1}{2}$, as part (ii) predicts.
- All four equilibrium equations (moments, and resolving for both bodies) were confirmed by computer for 10^5 random choices of W, k and θ. So was the claim that $G \le \frac{1}{2}R$ holds exactly when $\sin\theta \le \frac{4}{5}$, and that $F < \frac{1}{2}N$ always holds then.
- Limiting sense: a wide plank ($a \to r$) sits near the tops, so $\theta \to 0$ and almost no friction is needed. A narrow plank wedges deep between the cylinders, and eventually it slides down between them.

Common slips

- Leaving out the moment equation and so not realising that $G = F$. Without it there are too many unknowns.
- Getting the direction of the plank's friction wrong: the cylinders tend to roll apart, so on the cylinder it acts downwards and inwards.
- Writing $a = r\sin\theta$ rather than $a = r(1-\sin\theta)$: a is measured from the centre line, not from the cylinder's centre.

Beyond STEP

This is a small statically determinate system: three bodies, enough equations to fix every force, and inequalities from friction that mark out the possible configurations. In a first-year course the same ideas reappear as the conditions for rigid-body equilibrium (zero total force and zero total moment), and later in the study of wedging and jamming. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats rigid-body statics and friction carefully.

WORKED SOLUTION

STEP II 2017, Question 10

Mechanics • Work Done Against Resistance

STEP II 2017, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s2-q10

The key idea

Newton's second law gives the driving force at every point, and the work done by the engine is the integral of that force over distance. With constant acceleration, $v\frac{\mathrm{d}v}{\mathrm{d}x} = a$ turns an integral in x into one in v. The only subtlety is that an engine can push but not pull: when the force needed becomes negative, the brakes do that job and the engine contributes nothing, so we integrate only where the driving force is positive.

Solution

Model the car as a particle moving along the x-axis, with driving force F from the engine. While the car is moving forwards the resistance $Av^2 + R$ acts backwards.

First half. Newton's second law gives $F - (Av^2 + R) = ma$, so

$$F = ma + R + Av^2,$$

which is positive throughout. The work done by a force F moving its point of application through $\mathrm{d}x$ is $F\,\mathrm{d}x$, so the work done by the engine over the first half is

$$\int_0^d (ma + R + Av^2)\,\mathrm{d}x,$$

as required. For the change of variable, the acceleration is $v\frac{\mathrm{d}v}{\mathrm{d}x} = a$, so $\mathrm{d}x = \frac{v}{a}\,\mathrm{d}v$. Since $v^2 = 2ax$ (constant acceleration from rest), v runs from 0 at $x = 0$ to $\sqrt{2ad} = w$ at $x = d$. Hence the work is

$$\int_0^w \frac{(ma + R + Av^2)\,v}{a}\,\mathrm{d}v,$$

as required. Evaluating it, and using $w^2 = 2ad$:

$$W_1 = \frac{1}{a}\left[\frac{(ma+R)v^2}{2} + \frac{Av^4}{4}\right]_0^w = \frac{(ma+R)w^2}{2a} + \frac{Aw^4}{4a} = (ma+R)d + Aad^2.$$

Second half. Now the acceleration is $-a$, so $F-(Av^2+R)=-ma$, giving

$$F = Av^2 + R - ma.$$

The car slows from w to 0, with $v\frac{\mathrm{d}v}{\mathrm{d}x} = -a$, so $\mathrm{d}x = -\frac{v}{a}\,\mathrm{d}v$. Over any stretch where $F>0$ and the speed falls from v_1 to v_2 (with $v_1 > v_2$), the engine does work

$$\int_{v_1}^{v_2} F\left(-\frac{v}{a}\right)\mathrm{d}v = \int_{v_2}^{v_1} \frac{(Av^2+R-ma)\,v}{a}\,\mathrm{d}v. \qquad (*)$$

Part (i)

If $R > ma$ then $F = Av^2 + (R-ma) > 0$ for every v, so the engine works throughout the second half. By $(*)$ with $v_1 = w$, $v_2 = 0$:

$$W_2 = \frac{1}{a}\left[\frac{(R-ma)v^2}{2} + \frac{Av^4}{4}\right]_0^w = (R-ma)d + Aad^2.$$

Adding, the $ma\,d$ terms cancel and the total work is

$$W_1 + W_2 = 2Aad^2 + 2Rd,$$

as required.

Part (ii)

Now $R < ma$, so $F = Av^2 - (ma-R)$ is zero exactly when

$$v = u, \qquad \text{where } u^2 = \frac{ma-R}{A} > 0.$$

This speed is actually reached during the second half if $u < w$, that is, if $\frac{ma-R}{A} < 2ad$, which is precisely the condition $R > ma - 2Aad$. So as the car slows from w to 0, F is positive while $v > u$, falls to zero at the speed u, and is negative (braking) for $v < u$. This proves the first claim.

> **Ask yourself** Why can't we just reuse the answer to (i)?

The integral in (i) adds up $F\,\mathrm{d}x$ over the whole second half. Here F is negative for $v < u$, and that stretch would *subtract* from the total. But the engine isn't doing negative work there. It does nothing, and the brakes take over. So we stop the integral at $v = u$. By $(*)$ with $v_1 = w$, $v_2 = u$:

$$\begin{aligned} W_2 &= \frac{1}{a}\left[\frac{(R-ma)v^2}{2} + \frac{Av^4}{4}\right]_u^w \\ &= (R-ma)d + Aad^2 - \frac{1}{a}\left(\frac{(R-ma)u^2}{2} + \frac{Au^4}{4}\right). \end{aligned}$$

Substituting $Au^2 = ma - R$, the bracket is

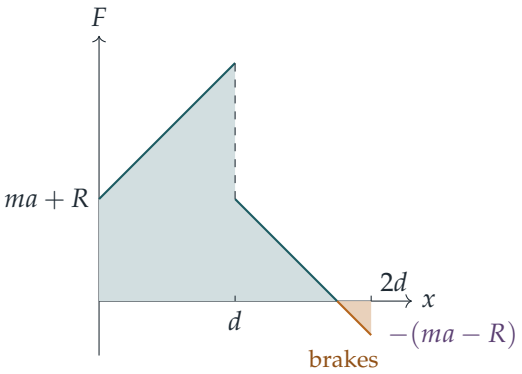
$$-\frac{(ma-R)^2}{2A} + \frac{(ma-R)^2}{4A} = -\frac{(ma-R)^2}{4A},$$

so $W_2 = (R - ma)d + Aad^2 + \frac{(ma-R)^2}{4Aa}$. The first half is unchanged (there $F > 0$ always), so the total is

$$W_1 + W_2 = 2Aad^2 + 2Rd + \frac{(ma - R)^2}{4Aa},$$

as required.

A picture. Since $v^2 = 2ax$ in the first half and $v^2 = 2a(2d - x)$ in the second, F is a *linear* function of x on each half, and the engine's work is the area under the positive part of the graph. In case (ii) the graph dips below the axis near the end. The missing triangle has height $ma - R$ and, since F falls at rate $2Aa$ per unit distance, base $\frac{ma-R}{2Aa}$. Its area $\frac{(ma-R)^2}{4Aa}$ is exactly the extra term.



Checks

- At the boundary $R = ma$ the extra term vanishes, so the answers to (i) and (ii) agree, as they must, since then F only reaches zero at the very end.
- At the other boundary $R = ma - 2Aad$, we have $u = w$, the engine does nothing in the second half, and the total should be $W_1 = (ma + R)d + Aad^2$. Substituting $ma - R = 2Aad$ into the answer to (ii) gives $2Aad^2 + 2Rd + Aad^2 = 2Rd + 3Aad^2$, and $(ma + R)d + Aad^2 = 2Rd + 2Aad^2 + Aad^2$ also, so they agree.
- Numerically, with $m = a = 1$, $A = \frac{1}{2}$, $d = 2$, $R = \frac{1}{2}$ (the case drawn above), the formula in (ii) gives $4 + 2 + \frac{1}{8} = 6.125$, and a direct numerical integration of the positive part of F over the journey gives the same value.
- Units: Aad^2, Rd and $(ma - R)^2/(Aa)$ are all a force times a distance, so they are all energies.

Common slips

- Forgetting that $\mathrm{d}x = -\frac{v}{a}\,\mathrm{d}v$ in the second half, which reverses the limits; a sign error here makes the second-half work negative.

- Integrating the negative part of F in (ii), which just reproduces the answer to (i).
- Not checking that the speed u is actually reached before the car stops: that is what the lower bound on R guarantees.

Beyond STEP

The work done by a force is a line integral, $\int \mathbf{F} \cdot \mathrm{d}\mathbf{r}$, and the change of variable used here is the one-dimensional version of the work–energy theorem. Resistances like Av^2 are not conservative, so the work depends on how the journey is driven, not just where it starts and ends. The engine's "one-sided" behaviour is an early taste of the constraints that appear in optimal control problems, such as finding the most fuel-efficient way to drive a given distance. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats work, energy and velocity-dependent forces carefully.

WORKED SOLUTION

STEP II 2017, Question 11

Mechanics • Projectiles and Energy

STEP II 2017, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

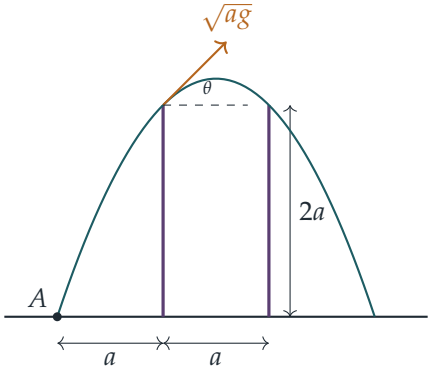
step.maths.org/questions/17-s2-q11

The key idea

Once the particle is level with the top of the first wall, the rest of its flight is a fresh projectile problem: it starts at height $2a$ with a known speed and must still be at least at height $2a$ a horizontal distance a later. Energy gives that speed without knowing the angle of projection. Writing the trajectory in terms of $\tan\theta$ turns "can it clear the second wall?" into "does a quadratic in $\tan\theta$ have a real root?", which handles both parts at once.

Solution

Model each particle as a point mass moving freely under gravity, with no air resistance. Take x horizontal (towards the walls) and y vertically up.



Part (i)

Speed over the first wall. Only gravity does work, so kinetic plus potential energy is constant. Rising from the ground to height $2a$, with speed v at the top of the wall,

$$\tfrac{1}{2}m(5ag) = \tfrac{1}{2}mv^2 + mg(2a) \quad \Longrightarrow \quad v^2 = 5ag - 4ag = ag,$$

so the particle passes over the first wall with speed $\sqrt{ag}$.

The angle. Let the velocity make angle θ above the horizontal as the particle passes over the first wall; $\theta > 0$, because the particle is at height $2a$ again later (over the second wall), so it must still be rising. Measure from the top of the first wall: after horizontal distance X the height above wall-top level is

$$Y = X\tan\theta - \frac{gX^2}{2v^2\cos^2\theta} = X\tan\theta - \frac{X^2}{2a}\,(1+\tan^2\theta),$$

using $v^2 = ag$ and $\sec^2\theta = 1 + \tan^2\theta$. Just clearing the second wall means $Y = 0$ at $X = a$:

$$a\tan\theta - \tfrac{a}{2}(1+\tan^2\theta) = 0 \iff \tan^2\theta - 2\tan\theta + 1 = 0 \iff (\tan\theta - 1)^2 = 0.$$

So $\tan\theta = 1$ and $\theta = 45^\circ$, as required. (Equivalently: the range on level ground from height $2a$ back to $2a$ is $v^2\sin 2\theta/g = a\sin 2\theta$, which equals a only when $\sin 2\theta = 1$.)

Distance of A. The horizontal velocity component never changes, and over the first wall it is $\sqrt{ag}\cos 45^\circ = \sqrt{ag/2}$; the vertical component there is also $\sqrt{ag/2}$, upwards. At A the vertical component u satisfies $u^2 = \frac{1}{2}ag + 2g(2a) = \frac{9}{2}ag$, so $u = 3\sqrt{ag/2}$. (As a check, $\frac{1}{2}ag + \frac{9}{2}ag = 5ag$, matching the launch speed.) The time from A to the first wall is

$$t = \frac{u - \sqrt{ag/2}}{g} = \frac{2\sqrt{ag/2}}{g},$$

and in that time the horizontal distance covered is

$$\sqrt{\tfrac{ag}{2}}\cdot\frac{2\sqrt{ag/2}}{g} = \frac{2\cdot\frac{1}{2}ag}{g} = a.$$

So A is a distance a from the foot of the first wall.

Part (ii)

Now the particle passes over the first wall at height $2a + h$. By energy, its speed w there satisfies

$$w^2 = 5ag - 2g(2a+h) = g(a-2h).$$

The particle moves towards the walls, so its horizontal velocity is non-zero and $w^2 > 0$; hence $h < \frac{1}{2}a$.

Let its direction there make angle ϕ with the horizontal, and write $T = \tan\phi$. Measuring from the particle's position over the first wall, after horizontal distance X its height above that point is

$$Y = XT - \frac{gX^2}{2w^2}(1+T^2) = XT - kX^2(1+T^2)/a^2, \qquad \text{where } k = \frac{ga^2}{2w^2} = \frac{a^2}{2(a-2h)}.$$

At the second wall, $X = a$, so $Y = aT - k(1+T^2)$. To clear that wall the particle must be at height at least $2a$, i.e. $Y \geq -h$:

$$aT - k(1+T^2) + h \geq 0 \iff kT^2 - aT + (k-h) \leq 0.$$

Ask yourself We don't know ϕ. How can we show that *no* direction works?

Since $k > 0$, the quadratic $kT^2 - aT + (k-h)$ opens upwards. It takes a value ≤ 0 for some real T only if its discriminant is non-negative. That discriminant is

$$\Delta = a^2 - 4k(k-h) = a^2 - \frac{a^4}{(a-2h)^2} + \frac{2a^2h}{a-2h}.$$

Multiplying by the positive quantity $(a-2h)^2/a^2$,

$$\frac{(a-2h)^2}{a^2}\Delta = (a-2h)^2 - a^2 + 2h(a-2h) = -2ah.$$

Since $a > 0$ and $h > 0$, this is negative, so $\Delta < 0$. Then $kT^2 - aT + (k-h) > 0$ for every real T, so whatever direction the particle has over the first wall, it is below height $2a$ when it reaches the second wall. It does not clear the second wall.

Note that $h = 0$ gives $\Delta = 0$ and the double root $T = a/(2k) = 1$: exactly the 45° of part (i).

Checks

- In part (i), the components at A are $\sqrt{ag/2}$ and $3\sqrt{ag/2}$, whose squares add to $5ag$. Stepping the motion forward in time from A with $a = 1.7\,\text{m}$ and $g = 9.81\,\text{m}\,\text{s}^{-2}$ gives heights $2.000a$ at both walls.
- The trajectory is symmetric about its highest point, which in part (i) lies halfway between the walls, at horizontal distance $1.5a$ from A. So the particle lands $3a$ from A, a distance a beyond the second wall: the picture is symmetric.
- For part (ii), a computer check over thousands of random launch angles and launch points with speed $\sqrt{5ag}$ found none that passed above the first wall and also cleared the second.

Common slips

- Forgetting that θ in part (i) must be the angle over the first wall, not the angle of projection at A (that one has $\tan\alpha = 3$).
- In part (ii), checking only one particular angle (such as 45°). The claim is about every possible direction, which is why the discriminant is needed.

Beyond STEP

Part (ii) says that, from a point over the first wall, a particle with this energy can only just reach the top of the second wall: that wall-top lies on the boundary of what is reachable. All points reachable from a given point with a given speed lie inside a parabola called the *envelope* or *safety parabola*, and points on it are reached by exactly one trajectory, which is the double root seen here. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats projectiles and energy methods at first-year level.

WORKED SOLUTION

STEP II 2017, Question 12

Probability and Statistics • Poisson Processes

STEP II 2017, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s2-q12

The key idea

Two independent Poisson streams of fish merge into a single Poisson stream whose rate is the sum of the two rates; that is what the convolution in (i) shows. For the timing questions, turn statements about *when* the first fish arrives into statements about *how many* fish have been caught: "Adam's first fish comes after time t" means exactly "Adam catches no fish in $[0,t]$", and the Poisson formula gives that probability straight away.

Solution

Throughout, X and Y are the numbers of fish caught by Adam and Eve in an interval of length T, so $X \sim \text{Po}(\lambda T)$ and $Y \sim \text{Po}(\mu T)$, independently.

Part (i)

For an integer $r \geq 0$, the event $X+Y=r$ splits into the disjoint events $\{X=j,\ Y=r-j\}$ for $j=0,1,\ldots,r$. Using independence,

$$\begin{aligned}\text{P}(X+Y=r) &= \sum_{j=0}^{r} \text{P}(X=j)\,\text{P}(Y=r-j) = \sum_{j=0}^{r} \frac{e^{-\lambda T}(\lambda T)^j}{j!}\cdot\frac{e^{-\mu T}(\mu T)^{r-j}}{(r-j)!} \\ &= \frac{e^{-(\lambda+\mu)T}}{r!}\sum_{j=0}^{r}\binom{r}{j}(\lambda T)^j(\mu T)^{r-j} = \frac{e^{-(\lambda+\mu)T}\big((\lambda+\mu)T\big)^r}{r!},\end{aligned}$$

where we multiplied and divided by $r!$ and then used the binomial theorem. This is the $\text{Po}\big((\lambda+\mu)T\big)$ probability, so the total catch in time T is Poisson with parameter $(\lambda+\mu)T$.

Part (ii)

For $0 \le r \le k$,

$$\mathrm{P}(X = r \mid X+Y = k) = \frac{\mathrm{P}(X=r,\ Y=k-r)}{\mathrm{P}(X+Y=k)} = \frac{\dfrac{e^{-\lambda T}(\lambda T)^r}{r!} \cdot \dfrac{e^{-\mu T}(\mu T)^{k-r}}{(k-r)!}}{\dfrac{e^{-(\lambda+\mu)T}\big((\lambda+\mu)T\big)^k}{k!}},$$

using part (i) for the denominator. The exponentials cancel, and so do the powers of T (there are $r + (k-r) = k$ on top and k underneath), leaving

$$\mathrm{P}(X = r \mid X+Y = k) = \binom{k}{r}\left(\frac{\lambda}{\lambda+\mu}\right)^r\left(\frac{\mu}{\lambda+\mu}\right)^{k-r}.$$

So, given a total of k, Adam's catch is $\mathrm{B}\big(k,\ \frac{\lambda}{\lambda+\mu}\big)$. Each of the k fish is, in effect, independently Adam's with probability $\frac{\lambda}{\lambda+\mu}$. Notice that T has disappeared.

Part (iii)

Let A and E be the times, measured from the start, at which Adam and Eve catch their first fish. Adam's first fish comes after time t exactly when he catches nothing in $[0,t]$, so

$$\mathrm{P}(A > t) = \mathrm{P}\big(\mathrm{Po}(\lambda t) = 0\big) = e^{-\lambda t}, \qquad \text{and similarly} \qquad \mathrm{P}(E > t) = e^{-\mu t}.$$

Hence A has probability density function $\lambda e^{-\lambda t}$ for $t \ge 0$ (differentiate $1 - e^{-\lambda t}$), and A and E are independent because the two catches are.

? **Ask yourself** How do we combine a density for A with a probability for E?

Condition on the value of A: if Adam's first fish comes at time t, he is first precisely when $E > t$, which has probability $e^{-\mu t}$ by independence. Summing over all t,

$$\mathrm{P}(A < E) = \int_0^\infty \lambda e^{-\lambda t}\, e^{-\mu t}\,\mathrm{d}t = \lambda\left[\frac{-e^{-(\lambda+\mu)t}}{\lambda+\mu}\right]_0^\infty = \frac{\lambda}{\lambda+\mu}.$$

(Ties have probability zero, since E has a continuous distribution.) So the first fish is Adam's with probability $\dfrac{\lambda}{\lambda+\mu}$, the same as the binomial parameter in (ii).

Part (iv)

We want $\mathrm{E}(M)$, where $M = \max(A,E)$ is the time by which both have caught a fish. Write $m = \min(A,E)$, the time of the first fish caught by either of them. For any two numbers, max + min is their sum, so

$$M = A + E - m \qquad \Longrightarrow \qquad \mathrm{E}(M) = \mathrm{E}(A) + \mathrm{E}(E) - \mathrm{E}(m).$$

$\mathrm{E}(A)$ *and* $\mathrm{E}(E)$. From the start, Adam's catches form the same kind of process as from the moment of any catch, so the expected wait for his first fish is the same as the given expected gap, $\mathrm{E}(A) = \lambda^{-1}$, and likewise $\mathrm{E}(E) = \mu^{-1}$. (Directly: $\mathrm{E}(A) = \int_0^\infty t\lambda e^{-\lambda t}\,\mathrm{d}t = \lambda^{-1}$, integrating by parts.)

Ask yourself What kind of process is "the first fish caught by either of them"?

By part (i), the combined catch in any interval of length t is $\mathrm{Po}\big((\lambda+\mu)t\big)$. So the combined catch is exactly the same kind of process as Adam's, but with rate $\lambda+\mu$ in place of λ, and its expected waiting time to the first fish is $\mathrm{E}(m) = (\lambda+\mu)^{-1}$. (Directly: $\mathrm{P}(m > t) = \mathrm{P}(A > t)\,\mathrm{P}(E > t) = e^{-(\lambda+\mu)t}$.) Therefore

$$\mathrm{E}(M) = \frac{1}{\lambda} + \frac{1}{\mu} - \frac{1}{\lambda+\mu} = \frac{\lambda^2+\lambda\mu+\mu^2}{\lambda\mu(\lambda+\mu)}.$$

Checks

- If $\lambda = \mu$, part (iii) gives $\frac{1}{2}$, as symmetry demands, and part (iv) gives $\frac{3}{2\lambda}$: wait $\frac{1}{2\lambda}$ for the first fish, then $\frac{1}{\lambda}$ more for the other person, by the same "fresh start" reasoning.
- If $\mu \to \infty$ (Eve catches fish instantly), part (iv) tends to $\frac{1}{\lambda}$, the time Adam needs on his own.
- The answer to (iv) is at least $\max(\frac{1}{\lambda}, \frac{1}{\mu})$, as it must be, since $\frac{1}{\mu} > \frac{1}{\lambda+\mu}$.
- With $\lambda = 1.3$, $\mu = 0.7$, the formulas give 0.65 and 1.6978; a computer simulation of 2×10^5 fishing trips gave 0.651 and 1.700.

Common slips

- In (iv), adding $\frac{1}{\lambda} + \frac{1}{\mu}$, or taking $\max\big(\frac{1}{\lambda}, \frac{1}{\mu}\big)$: the expectation of a maximum is neither of these.
- In (ii), forgetting to divide by $\mathrm{P}(X+Y=k)$, which is what turns the joint probability into a conditional one.

Beyond STEP

What you have met here is the *Poisson process*: waiting times between events are exponential and memoryless, independent Poisson processes superpose into another one, and each event is independently labelled with probability proportional to its rate (the reverse operation, "thinning", also works). These facts are the foundation of queueing theory and continuous-time Markov chains. Grimmett and Welsh, *Probability: An Introduction*, treats the Poisson process and exponential distribution in a way that follows on naturally from this question.

WORKED SOLUTION

STEP II 2017, Question 13

Probability and Statistics • Expectation and Infinite Series

STEP II 2017, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s2-q13

The key idea

In every version the door opens at attempt k exactly when the first $k-1$ attempts fail and the kth succeeds, so the probability is a product of "fail" factors times one "success" factor. What changes between versions is how many keys are in the bag at each draw. Once the probabilities are written down, the expectations are sums: a differentiated geometric series in (i), an arithmetic series in (ii), and in (iii) a telescoping sum that leaves behind a piece of the harmonic series.

Solution

Throughout, let X be the number of the attempt at which the door opens, and let $q = 1 - \frac{1}{n}$.

Part (i)

Each draw is from the full bag of n keys, independently of earlier draws because the key is replaced. So each attempt succeeds with probability $\frac{1}{n}$, and

$$\mathrm{P}(X = k) = q^{k-1} \cdot \frac{1}{n}, \qquad k = 1, 2, 3, \ldots$$

By the binomial expansion, valid since $0 < q < 1$,

$$(1-q)^{-2} = 1 + 2q + 3q^2 + \cdots = \sum_{k=1}^{\infty} k q^{k-1}.$$

Hence

$$\mathrm{E}(X) = \sum_{k=1}^{\infty} k\, q^{k-1} \frac{1}{n} = \frac{1}{n}(1-q)^{-2} = \frac{1}{n} \cdot n^2 = n.$$

The expected number of attempts is n.

Part (ii)

Now the bag loses one wrong key after each failure, so at attempt j there are $n-j+1$ keys. For the door to open at attempt k we need $k-1$ failures and then a success:

$$\mathrm{P}(X=k)=\frac{n-1}{n}\cdot\frac{n-2}{n-1}\cdots\frac{n-k+1}{n-k+2}\cdot\frac{1}{n-k+1}=\frac{1}{n},$$

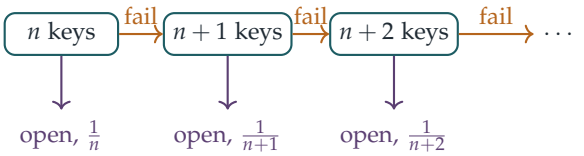
since the product telescopes. (This is also clear directly: drawing keys one by one without replacement puts the correct key in a uniformly random position among n.) So X is uniform on $\{1,2,\ldots,n\}$ and

$$\mathrm{E}(X)=\frac{1}{n}\sum_{k=1}^{n}k=\frac{1}{n}\cdot\frac{n(n+1)}{2}=\frac{n+1}{2}.$$

Part (iii)

The probability. The bag starts with n keys and gains one wrong key after each failure, and always contains exactly one correct key. So at attempt j there are $n+j-1$ keys, and

$$\mathrm{P}(\text{attempt } j \text{ fails} \mid \text{earlier attempts failed})=\frac{n+j-2}{n+j-1}.$$



Multiplying the conditional probabilities for $k-1$ failures and then a success,

$$\begin{aligned}\mathrm{P}(X=k)&=\frac{n-1}{n}\cdot\frac{n}{n+1}\cdots\frac{n+k-3}{n+k-2}\cdot\frac{1}{n+k-1}\\&=\frac{n-1}{n+k-2}\cdot\frac{1}{n+k-1}=\frac{n-1}{(n+k-1)(n+k-2)},\end{aligned}$$

because each numerator cancels the previous denominator, leaving the first numerator $n-1$ and the last fail-denominator $n+k-2$. For $k=1$ there are no fail factors, and the formula gives $\frac{n-1}{n(n-1)}=\frac{1}{n}$, as it should. This is the required result.

The door does open. Before computing the mean it is worth checking that X is finite with probability 1. Since $\dfrac{n-1}{(n+k-1)(n+k-2)}=(n-1)\left(\dfrac{1}{n+k-2}-\dfrac{1}{n+k-1}\right)$, the sum telescopes:

$$\sum_{k=1}^{N}\mathrm{P}(X=k)=(n-1)\left(\frac{1}{n-1}-\frac{1}{n+N-1}\right)\to 1\quad\text{as } N\to\infty.$$

The expectation. We need $\sum k\,\mathrm{P}(X=k)$, so we split $\dfrac{k}{(n+k-1)(n+k-2)}$ into partial fractions. Writing

$$k = A(n+k-2) + B(n+k-1)$$

and putting $k = 1-n$ gives $1-n = -A$, so $A = n-1$; putting $k = 2-n$ gives $B = 2-n$. Hence

$$k\,\mathrm{P}(X=k) = (n-1)\left(\frac{n-1}{n+k-1} - \frac{n-2}{n+k-2}\right).$$

Ask yourself This doesn't telescope neatly, because the coefficients $n-1$ and $n-2$ differ. So what is left over?

Let S_N be the sum of the first N terms. In the first fraction the denominators run over $m = n, \ldots, n+N-1$; in the second over $m = n-1, \ldots, n+N-2$. On the common range $m = n, \ldots, n+N-2$ each $\frac{1}{m}$ appears with coefficient $(n-1)-(n-2) = 1$, and the two ends are left over:

$$S_N = (n-1)\left(\sum_{m=n}^{n+N-2} \frac{1}{m} + \frac{n-1}{n+N-1} - \frac{n-2}{n-1}\right).$$

The last two terms are bounded (the middle one is positive and the last is a fixed number), while

$$\sum_{m=n}^{n+N-2} \frac{1}{m} = \sum_{m=1}^{n+N-2} \frac{1}{m} - \sum_{m=1}^{n-1} \frac{1}{m} \to \infty$$

as $N \to \infty$, since the harmonic series diverges and we subtract only a fixed finite amount. As $n-1 > 0$, $S_N \to \infty$. All the terms $k\,\mathrm{P}(X=k)$ are positive, so the series defining $\mathrm{E}(X)$ diverges to infinity: the expected number of attempts is infinite, even though the door is certain to open eventually.

Checks

- For $n = 2$ the three answers are 2, $\frac{3}{2}$ and ∞. Version 2 is the most efficient, as it must be, since it never retries a key; version 1 is in between.
- In (iii) with $n = 3$ the formula gives $\frac{1}{3}, \frac{1}{6}, \frac{1}{10}$ for $k = 1, 2, 3$; directly, $\frac{1}{3}$, $\frac{2}{3}\cdot\frac{1}{4} = \frac{1}{6}$ and $\frac{2}{3}\cdot\frac{3}{4}\cdot\frac{1}{5} = \frac{1}{10}$. A computer simulation of 2×10^5 games gave frequencies 0.334, 0.168 and 0.099.
- Also for $n = 3$, the partial sum S_N with $N = 10^4$ is about 15.6, and it keeps growing like $2\ln N$: there is no limit.

Common slips

- In (iii), taking the bag size at attempt k to be $n+k$ rather than $n+k-1$: only $k-1$ keys have been added before the kth draw.

- Concluding that the door might never open because the mean is infinite. The probabilities sum to 1; it is only the tail, decaying like $1/k^2$, that is too heavy for $\sum k\,\mathrm{P}(X = k)$ to converge.
- Telescoping the partial fractions as if both coefficients were equal and getting a finite answer.

Beyond STEP

Version 3 gives a random variable that is finite with probability 1 but has infinite mean, because $\mathrm{P}(X = k)$ behaves like $1/k^2$. Heavy-tailed distributions like this, and the famous St Petersburg paradox, show why first-year probability courses insist that $\mathrm{E}(X)$ exists only when $\sum k\,\mathrm{P}(X = k)$ converges. The bag that gains a wrong key after each failure is a relative of Pólya's urn, a standard model of "rich get richer" processes. Grimmett and Welsh, *Probability: An Introduction*, covers expectation of discrete random variables and these subtleties.

WORKED SOLUTION

STEP III 2017, Question 1

Pure • Telescoping Series Bounds

STEP III 2017, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s3-q1

The key idea

The identity in (i) writes each reciprocal binomial coefficient as a constant times a *difference* of consecutive terms, so the sum telescopes. In (ii), $1/n^3$ is awkward to sum but reciprocals of binomial coefficients are not: $\binom{n+1}{3}$ and $\binom{n+2}{5}$ are polynomials in n that behave like multiples of n^3 and n^5, so we trap $1/n^3$ between expressions we can sum exactly using (i).

Solution

Part (i)

Write $\binom{m}{k} = \dfrac{m!}{k!\,(m-k)!}$. Then

$$\begin{aligned}\frac{1}{\binom{n+r-1}{r}} - \frac{1}{\binom{n+r}{r}} &= \frac{r!\,(n-1)!}{(n+r-1)!} - \frac{r!\,n!}{(n+r)!} \\ &= \frac{r!\,(n-1)!}{(n+r)!}\big[(n+r)-n\big] = \frac{r\cdot r!\,(n-1)!}{(n+r)!},\end{aligned}$$

where we took out the common factor $r!\,(n-1)!/(n+r)!$, using $(n+r)! = (n+r)\,(n+r-1)!$ and $n! = n\,(n-1)!$. Multiplying by $\frac{r+1}{r}$ gives

$$\frac{r+1}{r}\left(\frac{1}{\binom{n+r-1}{r}} - \frac{1}{\binom{n+r}{r}}\right) = \frac{(r+1)!\,(n-1)!}{(n+r)!} = \frac{1}{\binom{n+r}{r+1}},$$

since $(n+r)-(r+1) = n-1$. This is the required identity.

The sum. Put $a_n = 1/\binom{n+r-1}{r}$, so that the identity reads $1/\binom{n+r}{r+1} = \frac{r+1}{r}(a_n - a_{n+1})$. Summing from $n=1$ to N, everything cancels except the first and last terms:

$$\sum_{n=1}^{N} \frac{1}{\binom{n+r}{r+1}} = \frac{r+1}{r}\left(a_1 - a_{N+1}\right) = \frac{r+1}{r}\left(1 - \frac{1}{\binom{N+r}{r}}\right),$$

because $a_1 = 1/\binom{r}{r} = 1$. As $N \to \infty$, $\binom{N+r}{r} \geq N + r \to \infty$ (for $r \geq 1$), so

$$\sum_{n=1}^{\infty} \frac{1}{\binom{n+r}{r+1}} = \frac{r+1}{r}.$$

The deduction. With $r = 2$ the sum from $n = 1$ is $\frac{3}{2}$. The $n = 1$ term is $1/\binom{3}{3} = 1$, so

$$\sum_{n=2}^{\infty} \frac{1}{\binom{n+2}{3}} = \frac{3}{2} - 1 = \frac{1}{2}.$$

Part (ii)

First inequality. For $n \geq 3$,

$$\frac{1}{\binom{n+1}{3}} = \frac{3!}{(n+1)n(n-1)} = \frac{3!}{n^3 - n} > \frac{3!}{n^3},$$

since $0 < n^3 - n < n^3$ (the left inequality because $n \geq 2$).

Second inequality. We have $\binom{n+1}{3} = \frac{1}{6}n(n^2-1)$ and $\binom{n+2}{5} = \frac{1}{120}(n+2)(n+1)n(n-1)(n-2) = \frac{1}{120}n(n^2-1)(n^2-4)$. So

$$\frac{20}{\binom{n+1}{3}} - \frac{1}{\binom{n+2}{5}} = \frac{120}{n(n^2-1)} - \frac{120}{n(n^2-1)(n^2-4)} = \frac{120\,(n^2-5)}{n(n^2-1)(n^2-4)}.$$

For $n \geq 3$ all the factors n, n^2-1, n^2-4 are positive, so this is less than $120/n^3 = 5!/n^3$ if and only if

$$n^2(n^2-5) < (n^2-1)(n^2-4), \quad \text{that is,} \quad n^4 - 5n^2 < n^4 - 5n^2 + 4,$$

which is true.

? **Ask yourself** Which sums from part (i) do we need, and where do they start?

We need $\sum_{n\geq 3} 1/\binom{n+1}{3}$ and $\sum_{n\geq 3} 1/\binom{n+2}{5}$. In the first, put $m = n-1$: it becomes $\sum_{m\geq 2} 1/\binom{m+2}{3} = \frac{1}{2}$, which is exactly the deduction in (i). In the second, put $m = n-2$: it becomes $\sum_{m\geq 1} 1/\binom{m+4}{5}$, which is (i) with $r = 4$, giving $\frac{5}{4}$.

All the series involved converge (they are positive and bounded by convergent ones, as the inequalities themselves show). Summing a strict inequality between convergent series term by term gives a strict inequality, because the sum of the positive differences is at least its first term, which is positive. Write $T = \sum_{n\geq 3} 1/n^3$.

Upper bound. Summing the first inequality over $n \geq 3$:

$$6T < \sum_{n=3}^{\infty} \frac{1}{\binom{n+1}{3}} = \frac{1}{2}, \qquad \text{so} \qquad T < \frac{1}{12}.$$

Lower bound. Summing the second inequality over $n \geq 3$:

$$120\,T > 20 \cdot \frac{1}{2} - \frac{5}{4} = \frac{35}{4}, \qquad \text{so} \qquad T > \frac{35}{480} = \frac{7}{96}.$$

Finally, $\sum_{n\geq 1} 1/n^3 = 1 + \frac{1}{8} + T$ and $1 + \frac{1}{8} = \frac{108}{96}$. Hence

$$\frac{108+7}{96} = \frac{115}{96} < \sum_{n=1}^{\infty} \frac{1}{n^3} < \frac{108}{96} + \frac{8}{96} = \frac{116}{96},$$

as required.

Checks

- Test the identity with $n = r = 1$: $1/\binom{2}{2} = 1$ and $2\,(1/\binom{1}{1} - 1/\binom{2}{1}) = 2 \cdot \frac{1}{2} = 1$.
- With $r = 1$ the sum is $\sum 1/\binom{n+1}{2} = \sum \frac{2}{n(n+1)} = 2$, the familiar telescoping sum, matching $\frac{r+1}{r} = 2$.
- Numerically $\sum 1/n^3 \approx 1.20206$, while $\frac{115}{96} \approx 1.19792$ and $\frac{116}{96} \approx 1.20833$, so the bounds hold and pin the value down to within about 0.01.

Common slips

- Getting the starting index wrong after shifting: $\sum_{n\geq 3} 1/\binom{n+1}{3}$ starts at $\binom{4}{3}$, which is the $n = 2$ term of the deduction in (i), not the $n = 1$ term.
- Forgetting the terms $1 + \frac{1}{8}$ for $n = 1, 2$, which the inequalities do not cover.

Beyond STEP

The number $\sum 1/n^3$ is $\zeta(3)$, Apéry's constant; unlike $\sum 1/n^2 = \pi^2/6$ it has no known closed form, and Apéry's 1978 proof that it is irrational used rapidly convergent series not unlike the ones here. Comparing a series with a telescoping one, and summing inequalities term by term, are standard tools of a first analysis course. For a careful treatment of convergence and comparison tests, see Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP III 2017, Question 2

Pure • Rotations in the Complex Plane

STEP III 2017, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s3-q2

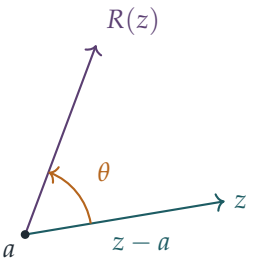
The key idea

A rotation about a is easiest to describe by moving the origin to a: subtract a, multiply by $e^{i\theta}$, then add a back. Every rotation therefore has the form $z \mapsto e^{i\alpha}z + k$, and conversely such a map is a rotation by α about its fixed point (provided $e^{i\alpha} \neq 1$). So to identify SR we compose the two formulae and find the fixed point. The half-angle factorisation $1 - e^{i\alpha} = -2i\, e^{i\alpha/2} \sin \frac{1}{2}\alpha$ turns the answer into the required shape.

Solution

Part (i)

The vector from the centre a to P is $z - a$. Rotating anticlockwise by θ about a rotates this vector by θ, which in complex numbers is multiplication by $e^{i\theta}$; the length is unchanged since $|e^{i\theta}| = 1$.



So the image w of P satisfies $w - a = e^{i\theta}(z - a)$, that is

$$w = e^{i\theta}z + a\left(1 - e^{i\theta}\right),$$

as required. In the same way $S(z) = e^{i\phi}z + b\left(1 - e^{i\phi}\right)$.

Part (ii)

Apply R first and then S:

$$\begin{aligned} SR(z) &= \mathrm{e}^{\mathrm{i}\phi}\left(\mathrm{e}^{\mathrm{i}\theta}z + a\left(1-\mathrm{e}^{\mathrm{i}\theta}\right)\right) + b\left(1-\mathrm{e}^{\mathrm{i}\phi}\right) \\ &= \mathrm{e}^{\mathrm{i}(\theta+\phi)}z + a\,\mathrm{e}^{\mathrm{i}\phi}\left(1-\mathrm{e}^{\mathrm{i}\theta}\right) + b\left(1-\mathrm{e}^{\mathrm{i}\phi}\right). \end{aligned}$$

By part (i), a rotation by $\theta+\phi$ about c is $z \mapsto \mathrm{e}^{\mathrm{i}(\theta+\phi)}z + c\left(1-\mathrm{e}^{\mathrm{i}(\theta+\phi)}\right)$. The coefficients of z already agree, so SR is this rotation exactly when

$$c\left(1-\mathrm{e}^{\mathrm{i}(\theta+\phi)}\right) = a\,\mathrm{e}^{\mathrm{i}\phi}\left(1-\mathrm{e}^{\mathrm{i}\theta}\right) + b\left(1-\mathrm{e}^{\mathrm{i}\phi}\right). \qquad (*)$$

? **Ask yourself** How can we turn the brackets $1-\mathrm{e}^{\mathrm{i}\alpha}$ into sines of half-angles?

Take out a factor $\mathrm{e}^{\mathrm{i}\alpha/2}$ to make the bracket symmetric:

$$1-\mathrm{e}^{\mathrm{i}\alpha} = \mathrm{e}^{\mathrm{i}\alpha/2}\left(\mathrm{e}^{-\mathrm{i}\alpha/2} - \mathrm{e}^{\mathrm{i}\alpha/2}\right) = -2\mathrm{i}\,\mathrm{e}^{\mathrm{i}\alpha/2}\sin\tfrac{1}{2}\alpha.$$

Using this with $\alpha = \theta+\phi$, θ and ϕ, equation $(*)$ becomes

$$-2\mathrm{i}\,c\,\mathrm{e}^{\mathrm{i}(\theta+\phi)/2}\sin\tfrac{1}{2}(\theta+\phi) = -2\mathrm{i}\,a\,\mathrm{e}^{\mathrm{i}\phi}\mathrm{e}^{\mathrm{i}\theta/2}\sin\tfrac{1}{2}\theta - 2\mathrm{i}\,b\,\mathrm{e}^{\mathrm{i}\phi/2}\sin\tfrac{1}{2}\phi.$$

Dividing through by $-2\mathrm{i}\,\mathrm{e}^{\mathrm{i}(\theta+\phi)/2}$ (never zero) gives

$$c\sin\tfrac{1}{2}(\theta+\phi) = a\,\mathrm{e}^{\mathrm{i}\phi/2}\sin\tfrac{1}{2}\theta + b\,\mathrm{e}^{-\mathrm{i}\theta/2}\sin\tfrac{1}{2}\phi.$$

If $\theta+\phi \neq 2n\pi$ then $\sin\frac{1}{2}(\theta+\phi) \neq 0$, so this determines a unique c, and with that c the map SR is the rotation by $\theta+\phi$ about c. This is the required result. (The condition matters: if $\sin\frac{1}{2}(\theta+\phi) = 0$ there is no fixed point to rotate about, in general.)

The case $\theta+\phi = 2\pi$. Now $\mathrm{e}^{\mathrm{i}(\theta+\phi)} = 1$ and $\mathrm{e}^{\mathrm{i}\phi} = \mathrm{e}^{-\mathrm{i}\theta}$, so

$$\begin{aligned} SR(z) &= z + a\,\mathrm{e}^{-\mathrm{i}\theta}\left(1-\mathrm{e}^{\mathrm{i}\theta}\right) + b\left(1-\mathrm{e}^{-\mathrm{i}\theta}\right) \\ &= z + a\left(\mathrm{e}^{-\mathrm{i}\theta} - 1\right) + b\left(1-\mathrm{e}^{-\mathrm{i}\theta}\right) \\ &= z + (b-a)\left(1-\mathrm{e}^{-\mathrm{i}\theta}\right). \end{aligned}$$

So SR is a *translation* by $(b-a)\left(1-\mathrm{e}^{-\mathrm{i}\theta}\right)$, a distance $2|b-a|\,|\sin\frac{1}{2}\theta|$. It is the identity when $a=b$ (rotating by θ and then by $2\pi-\theta$ about the same point) or when θ is a multiple of 2π; otherwise it is a genuine translation with no fixed point.

Part (iii)

Swapping the roles of (a,θ) and (b,ϕ) in the formula for SR,

$$RS(z) = \mathrm{e}^{\mathrm{i}(\theta+\phi)}z + b\,\mathrm{e}^{\mathrm{i}\theta}\left(1-\mathrm{e}^{\mathrm{i}\phi}\right) + a\left(1-\mathrm{e}^{\mathrm{i}\theta}\right).$$

Both maps multiply z by $\mathrm{e}^{\mathrm{i}(\theta+\phi)}$, so $RS = SR$ exactly when the constant terms agree. Their difference is

$$\begin{aligned} SR(z) - RS(z) &= a\big(1-\mathrm{e}^{\mathrm{i}\theta}\big)\big(\mathrm{e}^{\mathrm{i}\phi}-1\big) + b\big(1-\mathrm{e}^{\mathrm{i}\phi}\big)\big(1-\mathrm{e}^{\mathrm{i}\theta}\big) \\ &= (b-a)\big(1-\mathrm{e}^{\mathrm{i}\theta}\big)\big(1-\mathrm{e}^{\mathrm{i}\phi}\big). \end{aligned}$$

A product is zero only if a factor is zero, so $RS = SR$ if and only if

- $a = b$ (the rotations have the same centre), or
- $\mathrm{e}^{\mathrm{i}\theta} = 1$, i.e. θ is a multiple of 2π, so R is the identity, or
- $\mathrm{e}^{\mathrm{i}\phi} = 1$, i.e. ϕ is a multiple of 2π, so S is the identity.

In words: two genuine rotations commute only when they share a centre.

Checks

- With $a = b$ the formula in (ii) gives $c\sin\frac{1}{2}(\theta+\phi) = a\big(\mathrm{e}^{\mathrm{i}\phi/2}\sin\frac{1}{2}\theta + \mathrm{e}^{-\mathrm{i}\theta/2}\sin\frac{1}{2}\phi\big)$, and expanding the bracket gives $\sin\frac{1}{2}\theta\cos\frac{1}{2}\phi + \cos\frac{1}{2}\theta\sin\frac{1}{2}\phi = \sin\frac{1}{2}(\theta+\phi)$ (the imaginary parts cancel), so $c = a$, as it must.
- With $\phi = 0$ (S the identity), $c\sin\frac{1}{2}\theta = a\sin\frac{1}{2}\theta$, so $c = a$.
- Take $a = 0$, $b = 1$, $\theta = \phi = \frac{1}{2}\pi$. Then $R(z) = \mathrm{i}z$, $S(z) = \mathrm{i}z + 1 - \mathrm{i}$, so $SR(z) = -z + 1 - \mathrm{i}$, a half-turn about $\frac{1}{2}(1-\mathrm{i})$. The formula gives $c\cdot 1 = \mathrm{e}^{-\mathrm{i}\pi/4}\cdot\frac{1}{\sqrt{2}} = \frac{1}{2}(1-\mathrm{i})$. Agreed.
- All three results were also confirmed by computer on thousands of random centres, angles and points, rotating geometrically (by modulus and argument) rather than with the formulae; the discrepancies were below 10^{-12}.

Common slips

- Composing in the wrong order: SR means R first, so $R(z)$ is substituted into S.
- Dividing by $1-\mathrm{e}^{\mathrm{i}(\theta+\phi)}$ without noting that it is non-zero precisely because $\theta+\phi \neq 2n\pi$.
- In (iii), forgetting the trivial cases where one rotation is the identity.

Beyond STEP

The maps $z \mapsto \alpha z + \beta$ with $|\alpha| = 1$ are exactly the orientation-preserving isometries of the plane, and they form a group: rotations and translations together, closed under composition. Part (iii) is a small instance of the fact that this group is not commutative, and the commutator $RSR^{-1}S^{-1}$ turns out to be a translation. Needham's *Visual Complex Analysis* develops this geometric view of complex multiplication beautifully, and Allenby's *Rings, Fields and Groups* covers the group theory.

WORKED SOLUTION

STEP III 2017, Question 3

Pure • Roots of Polynomials

STEP III 2017, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s3-q3

This question has a full guided treatment in *Beyond STEP*, Section 6.4, *Solving the Quartic*: https://beyondstep.keithmcnulty.org/ch06/quartic.html

The key idea

The coefficients of any polynomial are symmetric functions of its roots, so the coefficient A of the cubic is minus the sum of its three roots, and that sum is a familiar symmetric function of α, β, γ, δ. For the specific quartic, the cubic has an integer root, so it factorises completely. Knowing $\alpha\beta + \gamma\delta$ then splits the quartic into two quadratic factors, one with roots α, β and one with roots γ, δ, and the remaining coefficients pin down those factors.

Solution

Finding A

Throughout we use Vieta's relations for the quartic:

$$\alpha+\beta+\gamma+\delta = -p, \qquad \alpha\beta+\alpha\gamma+\alpha\delta+\beta\gamma+\beta\delta+\gamma\delta = q,$$
$$\alpha\beta\gamma+\alpha\beta\delta+\alpha\gamma\delta+\beta\gamma\delta = -r, \qquad \alpha\beta\gamma\delta = s.$$

The cubic is monic with roots $y_1 = \alpha\beta + \gamma\delta$, $y_2 = \alpha\gamma + \beta\delta$ and $y_3 = \alpha\delta + \beta\gamma$, so

$$y^3 + Ay^2 + \cdots = (y - y_1)(y - y_2)(y - y_3),$$

and comparing coefficients of y^2 gives $A = -(y_1 + y_2 + y_3)$. But $y_1 + y_2 + y_3$ contains each of the six products of two different roots exactly once, so it equals q. Hence

$$A = -q.$$

Part (i)

With $p = 0$, $q = 3$, $r = -6$, $s = 10$ the cubic has coefficients

$$A = -3, \qquad pr - 4s = -40, \qquad 4qs - p^2 s - r^2 = 120 - 0 - 36 = 84,$$

so it is

$$y^3 - 3y^2 - 40y + 84 = 0.$$

Any integer root divides 84. Trying small divisors, $y = 2$ gives $8 - 12 - 80 + 84 = 0$, so $y - 2$ is a factor:

$$y^3 - 3y^2 - 40y + 84 = (y - 2)(y^2 - y - 42) = (y - 2)(y - 7)(y + 6).$$

The roots are 2, 7 and -6; the largest is 7, so

$$\alpha\beta + \gamma\delta = 7.$$

Part (ii)

Expanding,

$$(\alpha + \beta)(\gamma + \delta) = \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta,$$

which is the sum of all six pair products with $\alpha\beta$ and $\gamma\delta$ removed. So

$$(\alpha + \beta)(\gamma + \delta) = q - (\alpha\beta + \gamma\delta) = 3 - 7 = -4.$$

Next, $\alpha\beta$ and $\gamma\delta$ have sum 7 and product $\alpha\beta\gamma\delta = s = 10$. They are therefore the roots of

$$t^2 - 7t + 10 = (t - 5)(t - 2) = 0,$$

and since $\alpha\beta > \gamma\delta$ we get $\alpha\beta = 5$ and $\gamma\delta = 2$.

Part (iii)

Let $u = \alpha + \beta$. Since $p = 0$, the sum of all four roots is zero, so $\gamma + \delta = -u$, and part (ii) gives $-u^2 = -4$, so $u = \pm 2$.

? **Ask yourself** Which sign of u is correct, and how can we tell?

We haven't yet used r. Grouping the four triple products in pairs,

$$\alpha\beta\gamma + \alpha\beta\delta + \alpha\gamma\delta + \beta\gamma\delta = \alpha\beta(\gamma + \delta) + \gamma\delta(\alpha + \beta) = 5(-u) + 2u = -3u.$$

This equals $-r = 6$, so $u = -2$. (The other sign would give $r = +6$, a different quartic.) Thus

$$\alpha + \beta = -2, \quad \alpha\beta = 5, \qquad \gamma + \delta = 2, \quad \gamma\delta = 2.$$

So α, β are the roots of $x^2 + 2x + 5 = 0$, giving $x = -1 \pm 2\mathrm{i}$, and γ, δ are the roots of $x^2 - 2x + 2 = 0$, giving $x = 1 \pm \mathrm{i}$. The quartic

$$x^4 + 3x^2 - 6x + 10 = (x^2 + 2x + 5)(x^2 - 2x + 2) = 0$$

therefore has roots

$$x = -1 + 2\mathrm{i}, \quad -1 - 2\mathrm{i}, \quad 1 + \mathrm{i}, \quad 1 - \mathrm{i}.$$

Checks

- Multiply out $(x^2+2x+5)(x^2-2x+2)$: the x^3 terms cancel and we get $x^4+3x^2-6x+10$, exactly the quartic.
- With these roots, $\alpha\gamma+\beta\delta$ and $\alpha\delta+\beta\gamma$ (in some order) should be the other two roots of the cubic, 2 and -6. Their sum is $q-7=-4=2+(-6)$, as it should be.
- The formula $A=-q$ was also confirmed by computer on a thousand quartics with randomly chosen complex roots.

Common slips

- Taking $A=q$: the y^2 coefficient is *minus* the sum of the roots.
- Stopping at $u=\pm 2$ and giving both possibilities: only one sign is consistent with $r=-6$.
- Expecting real roots: this quartic is positive for all real x, so all four roots are complex.

Beyond STEP

The cubic here is the *resolvent cubic* of the quartic, and it is the heart of Ferrari's method: solving any quartic reduces to solving a cubic, then quadratics. Why this works is explained by Galois theory, which studies how permutations of the roots act on expressions like $\alpha\beta+\gamma\delta$ (each of the three is fixed by a subgroup of eight of the 24 permutations). Allenby's *Rings, Fields and Groups* is a good place to start on these ideas.

WORKED SOLUTION

STEP III 2017, Question 4

Pure • Geometric Mean of Functions

STEP III 2017, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s3-q4

The key idea

The geometric mean is just an exponential of an ordinary average: $\ln F(y)$ is the mean value of $\ln f$ over $[0, y]$. So every property of F comes from a property of logarithms (change of base, $\ln(fg) = \ln f + \ln g$) combined with linearity of the integral. For the converse in (iv), the condition becomes an equation involving an integral, and differentiating it turns it into a simple differential equation.

Solution

Throughout, write

$$\ln F(y) = \frac{1}{y}\int_0^y \ln f(x)\,\mathrm{d}x \qquad (y > 0),$$

which is the definition with logarithms taken of both sides.

Part (i)

Since $a > 0$ and $a \neq 1$, $\log_a$ is defined and $\ln a \neq 0$, and the change-of-base rule gives $\log_a f(x) = \dfrac{\ln f(x)}{\ln a}$. Also $a^t = \mathrm{e}^{t\ln a}$ for any real t. Hence

$$\begin{aligned} a^{\frac{1}{y}\int_0^y \log_a f(x)\,\mathrm{d}x} &= \exp\left(\ln a \cdot \frac{1}{y}\int_0^y \frac{\ln f(x)}{\ln a}\,\mathrm{d}x\right) \\ &= \exp\left(\frac{1}{y}\int_0^y \ln f(x)\,\mathrm{d}x\right) = F(y), \end{aligned}$$

where we took the constant $\frac{1}{\ln a}$ outside the integral and cancelled it against $\ln a$. This is the required result. (The condition $a \neq 1$ is exactly what lets us divide by $\ln a$.)

Part (ii)

Since $f(x) > 0$ and $g(x) > 0$, we have $h(x) > 0$, so H is defined, and $\ln h(x) = \ln f(x) + \ln g(x)$. By linearity of the integral,

$$\ln H(y) = \frac{1}{y}\int_0^y \big(\ln f(x) + \ln g(x)\big)\,\mathrm{d}x = \ln F(y) + \ln G(y) = \ln\big(F(y)G(y)\big).$$

As ln is one-to-one, $H(y) = F(y)G(y)$.

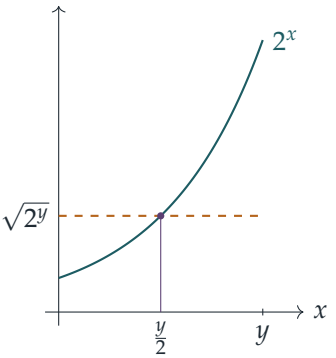
Part (iii)

Here $f(x) = b^x > 0$ and $\ln f(x) = x \ln b$, so

$$\ln F(y) = \frac{1}{y}\int_0^y x \ln b\,\mathrm{d}x = \frac{1}{y}\cdot\frac{y^2}{2}\ln b = \frac{y}{2}\ln b.$$

Therefore $F(y) = \mathrm{e}^{\frac{y}{2}\ln b} = b^{y/2} = \sqrt{b^y}$, as required.

The picture below shows what this says for $b = 2$: the geometric mean of 2^x over $[0, y]$ is the value of 2^x at the midpoint $\frac{y}{2}$, because $\ln 2^x$ is linear and the average of a linear function is its midpoint value.



Part (iv)

We assume, as the question implicitly does, that f is continuous, so that the integrals exist and we may use the fundamental theorem of calculus. Put $\phi(x) = \ln f(x)$, which is defined because $f(x) > 0$, and is continuous. The hypothesis $F(y) = \sqrt{f(y)}$ says, after taking logarithms, that

$$\frac{1}{y}\int_0^y \phi(x)\,\mathrm{d}x = \tfrac{1}{2}\phi(y) \quad \text{that is,} \quad \int_0^y \phi(x)\,\mathrm{d}x = \frac{y}{2}\,\phi(y) \qquad \text{for all } y > 0. \qquad (*)$$

Ask yourself How can we get rid of the integral, and are we allowed to?

Differentiate with respect to y. The left-hand side of $(*)$ is differentiable, with derivative $\phi(y)$, by the fundamental theorem of calculus. So the right-hand side is differentiable

too, and therefore so is $\phi(y) = \frac{2}{y} \times$ (left-hand side) for $y > 0$. Differentiating $(*)$ using the product rule on the right:

$$\phi(y) = \tfrac{1}{2}\phi(y) + \tfrac{y}{2}\phi'(y) \quad \Longrightarrow \quad y\,\phi'(y) - \phi(y) = 0.$$

Since $y > 0$ we may divide by y^2 and recognise a quotient-rule derivative:

$$\frac{\mathrm{d}}{\mathrm{d}y}\left(\frac{\phi(y)}{y}\right) = \frac{y\phi'(y) - \phi(y)}{y^2} = 0.$$

So $\phi(y)/y$ is a constant, k say, on the interval $y > 0$ (a function with zero derivative on an interval is constant, by the mean value theorem). Hence $\ln f(x) = kx$ and

$$f(x) = \mathrm{e}^{kx} = b^x, \qquad \text{where } b = \mathrm{e}^k > 0.$$

This holds for every $x > 0$; the value at the single point $x = 0$ doesn't affect any of the integrals, and by continuity $f(0) = 1 = b^0$ anyway. Together with (iii), this shows that the functions whose geometric mean is $\sqrt{f(y)}$ are exactly the functions b^x.

Checks

- In (iii) with $b = 4$, $y = 2$: $\ln F(2) = \frac{1}{2}\int_0^2 x \ln 4 \,\mathrm{d}x = \ln 4$, so $F(2) = 4 = \sqrt{4^2}$; a numerical integration agrees.
- In (iv), try a function that is not of the form b^x, such as $f(x) = 3 \cdot 2^x$. Then $F(y) = 3 \cdot 2^{y/2}$ by (ii) and (iii), while $\sqrt{f(y)} = \sqrt{3} \cdot 2^{y/2}$, so the condition fails, as it must. Our argument would give $\phi(y)/y = \ln 2 + \frac{\ln 3}{y}$, which is not constant.
- Parts (i) and (ii) were confirmed by computer for random positive functions, values of y and bases a on both sides of 1.

Common slips

- In (iv), differentiating $(*)$ but forgetting the product rule on $\frac{y}{2}\phi(y)$, which loses the ϕ' term entirely.
- Concluding $f(x) = b^x$ from (iii) alone: (iii) shows that b^x has the property, not that nothing else does. Part (iv) is the converse and needs its own argument.
- In (i), not noting where $a \neq 1$ is needed: it is what makes $\log_a$ meaningful and $\ln a \neq 0$.

Beyond STEP

The continuous geometric mean is the integral version of $(x_1 x_2 \cdots x_n)^{1/n}$, and Jensen's inequality for the concave function ln shows that it never exceeds the ordinary mean $\frac{1}{y}\int_0^y f(x)\,\mathrm{d}x$, the continuous AM–GM inequality. Part (iv) is a small example of a

functional equation solved by turning it into a differential equation, which requires care about differentiability; in analysis courses you learn exactly when the fundamental theorem of calculus allows this. For the rigorous background, see Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP III 2017, Question 5

Pure • Polar Curves

STEP III 2017, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s3-q5

The key idea

Write the curve parametrically, with θ as the parameter: $x = f(\theta)\cos\theta$, $y = f(\theta)\sin\theta$. The gradient then comes from the chain rule, and "meeting at right angles" becomes "the product of the gradients is -1". In the last part the family of curves $r = a(1+\sin\theta)$ passes through every point of C, so the right-angle condition holds all along C. That turns it into a first-order differential equation for f.

Solution

The gradient

On the curve, $x = f(\theta)\cos\theta$ and $y = f(\theta)\sin\theta$. Differentiating with respect to θ by the product rule,

$$\frac{\mathrm{d}x}{\mathrm{d}\theta} = f'(\theta)\cos\theta - f(\theta)\sin\theta, \qquad \frac{\mathrm{d}y}{\mathrm{d}\theta} = f'(\theta)\sin\theta + f(\theta)\cos\theta.$$

By the chain rule, and then dividing top and bottom by $\cos\theta$ (assuming $\cos\theta \neq 0$),

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{f'(\theta)\sin\theta + f(\theta)\cos\theta}{f'(\theta)\cos\theta - f(\theta)\sin\theta} = \frac{f'(\theta)\tan\theta + f(\theta)}{f'(\theta) - f(\theta)\tan\theta}.$$

Curves meeting at right angles

At a meeting point both curves are at the same θ. Write $t = \tan\theta$, and write f, f', g, g' for the values there. If neither tangent is vertical, the curves are perpendicular exactly when the product of the gradients is -1:

$$(f't + f)(g't + g) = -(f' - ft)(g' - gt).$$

Expanding both sides, the terms in $f'gt$ and $fg't$ cancel, which leaves

$$f'g't^2 + fg + f'g' + fgt^2 = 0 \iff (1+t^2)(f'g' + fg) = 0.$$

Since $1 + t^2 = \sec^2\theta > 0$, this gives $f'(\theta)g'(\theta) + f(\theta)g(\theta) = 0$, as required.

> **Ask yourself** What if one tangent is vertical, or $\cos\theta = 0$, so that a gradient or $\tan\theta$ doesn't exist?

Use direction vectors instead of gradients. The tangent to $r = f(\theta)$ has direction $\big(f'\cos\theta - f\sin\theta,\ f'\sin\theta + f\cos\theta\big)$, and similarly for g. Their scalar product is

$$(f'\cos\theta - f\sin\theta)(g'\cos\theta - g\sin\theta) + (f'\sin\theta + f\cos\theta)(g'\sin\theta + g\cos\theta) = f'g' + fg,$$

because the cross terms cancel and $\cos^2\theta + \sin^2\theta = 1$. So perpendicularity is the same condition in every case.

Finding $f(\theta)$

For $r = a(1 + \sin\theta)$ we have $g'(\theta) = a\cos\theta$.

> **Ask yourself** The condition only holds where the curves meet. Why can we use it at every point of C?

Take any point of C, at angle θ with $f(\theta) > 0$ and $\sin\theta \neq -1$. The choice $a = f(\theta)/(1 + \sin\theta) > 0$ gives a member of the family that passes through that point at the same θ. By hypothesis this curve meets C at right angles, so

$$f'(\theta)\, a\cos\theta + f(\theta)\, a(1 + \sin\theta) = 0.$$

Dividing by $a > 0$, every such θ satisfies

$$f'(\theta)\cos\theta + f(\theta)(1 + \sin\theta) = 0.$$

Multiply by $1 - \sin\theta$ and use $1 - \sin^2\theta = \cos^2\theta$:

$$\cos\theta\left[f'(\theta)(1 - \sin\theta) + f(\theta)\cos\theta\right] = 0.$$

Where $\cos\theta \neq 0$, the bracket vanishes. By the quotient rule the bracket is $(1 - \sin\theta)^2$ times the derivative of $f(\theta)/(1 - \sin\theta)$:

$$\frac{\mathrm{d}}{\mathrm{d}\theta}\left(\frac{f(\theta)}{1 - \sin\theta}\right) = \frac{f'(\theta)(1 - \sin\theta) + f(\theta)\cos\theta}{(1 - \sin\theta)^2} = 0.$$

So $f(\theta) = k(1 - \sin\theta)$ for a constant k, on the interval $-\frac{1}{2}\pi < \theta < \frac{1}{2}\pi$. (Equivalently, we could separate variables in $f'/f = -(1 + \sin\theta)/\cos\theta = -\cos\theta/(1 - \sin\theta)$.) The curve passes through $r = 4$, $\theta = -\frac{1}{2}\pi$, so by continuity $k(1 + 1) = 4$, giving $k = 2$. On $\frac{1}{2}\pi < \theta < \frac{3}{2}\pi$ the same argument applies, and $\theta = \frac{3}{2}\pi$ is the same direction as $-\frac{1}{2}\pi$, so again $k = 2$. Hence

$$\boxed{f(\theta) = 2(1 - \sin\theta)}$$

This is another cardioid, the reflection in the x-axis of $r = 2(1 + \sin\theta)$.

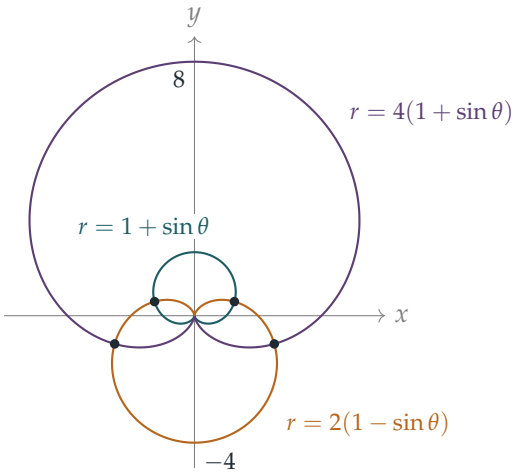
The sketch

The curve $r = a(1+\sin\theta)$ is a cardioid with its cusp at the origin, symmetric about the y-axis. It reaches up to $(0, 2a)$ and crosses the x-axis at $(\pm a, 0)$. The curve $r = 2(1-\sin\theta)$ is a cardioid of the same shape opening downwards: its cusp is at the origin, it crosses the x-axis at $(\pm 2, 0)$ and it reaches down to $(0, -4)$.

The meeting points come from solving $a(1+\sin\theta) = 2(1-\sin\theta)$, which gives $\sin\theta = \frac{2-a}{2+a}$:

- $a = 1$: $\sin\theta = \frac{1}{3}$, $r = \frac{4}{3}$, at the points $\left(\pm\frac{8\sqrt{2}}{9}, \frac{4}{9}\right)$;
- $a = 4$: $\sin\theta = -\frac{1}{3}$, $r = \frac{8}{3}$, at the points $\left(\pm\frac{16\sqrt{2}}{9}, -\frac{8}{9}\right)$.

All three curves also pass through the origin.



Checks

- At $\theta = 0$ the curve $r = 2(1-\sin\theta)$ is at $(2, 0)$, and $r = 2(1+\sin\theta)$ passes through the same point. There $f = 2$, $f' = -2$, $g = 2$, $g' = 2$, so $f'g' + fg = -4 + 4 = 0$, as it should be.
- At $\theta = -\frac{1}{2}\pi$, $f = 2(1+1) = 4$, as required.
- The tangents at $\left(\frac{8\sqrt{2}}{9}, \frac{4}{9}\right) \approx (1.257, 0.444)$ were confirmed perpendicular by computer, as were the tangents for many random values of a.

Common slips

- Using the right-angle condition only at one meeting point. The point of "for each positive a" is that the condition then holds at every point of C.

- Forgetting that $f(\theta)/(1-\sin\theta)$ is undefined at $\theta = \frac{1}{2}\pi$, and so being careless about where the constant is fixed.
- Drawing $r = 2(1-\sin\theta)$ opening upwards: since $r = 4$ at $\theta = -\frac{1}{2}\pi$, the curve lies mainly below the x-axis.

Beyond STEP

A family of curves that meets another family at right angles everywhere is called a pair of *orthogonal trajectories*. Here the cardioids $r = a(1+\sin\theta)$ and $r = b(1-\sin\theta)$ form such a pair. The idea appears in first-year differential equations and again in physics, where field lines cross equipotentials at right angles. Boyce and DiPrima, *Elementary Differential Equations*, covers how to find orthogonal trajectories. Needham, *Visual Complex Analysis*, shows how conformal maps produce such families.

WORKED SOLUTION

STEP III 2017, Question 6

Pure • Integration by Substitution

STEP III 2017, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

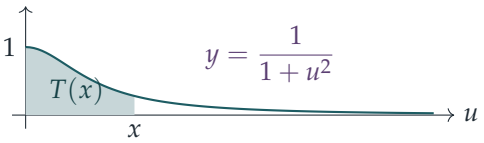
step.maths.org/questions/17-s3-q6

This question has a full guided treatment in *Beyond STEP*, Section 4.2, *Building Trigonometry from an Integral*: https://beyondstep.keithmcnulty.org/ch04/trig-from-integral.html

The key idea

We are building the inverse tangent from nothing but its integral, so every fact must come from a substitution. The substitution $u \mapsto 1/u$ swaps the ranges $(0, x)$ and $(1/x, \infty)$, and the Möbius substitution $v = \frac{u+a}{1-au}$ leaves the integrand-with-differential $\frac{\mathrm{d}u}{1+u^2}$ unchanged, so it simply shifts the limits of integration. Choosing the numbers in the final part so that the shifted limit lands back on a value we already know produces an equation we can solve.

Solution



$T(x)$ is the shaded area; T_∞ is the whole area under the curve for $u \geq 0$.

Part (i)

Substitute $u = 1/t$ in the integral for $T(x)$, so $\mathrm{d}u = -t^{-2}\,\mathrm{d}t$; as u runs from 0 to x, t runs from ∞ down to $1/x$. Then

$$\begin{aligned} T(x) &= \int_{\infty}^{1/x} \frac{1}{1+t^{-2}}\left(-\frac{1}{t^2}\right)\mathrm{d}t = \int_{1/x}^{\infty} \frac{1}{t^2+1}\,\mathrm{d}t \\ &= \int_0^{\infty} \frac{\mathrm{d}t}{1+t^2} - \int_0^{1/x} \frac{\mathrm{d}t}{1+t^2} = T_\infty - T(x^{-1}). \end{aligned}$$

(The integral starting at 0 is a proper integral, because T_∞ is finite, so splitting the range is legitimate.) This is the required result.

Part (ii)

The derivative. By the quotient rule, for $u \neq a^{-1}$,

$$\frac{\mathrm{d}v}{\mathrm{d}u} = \frac{(1-au)+a(u+a)}{(1-au)^2} = \frac{1+a^2}{(1-au)^2}.$$

On the other hand,

$$1+v^2 = \frac{(1-au)^2+(u+a)^2}{(1-au)^2} = \frac{1+a^2u^2+u^2+a^2}{(1-au)^2} = \frac{(1+a^2)(1+u^2)}{(1-au)^2}.$$

Dividing by $1+u^2$ gives $\dfrac{1+v^2}{1+u^2} = \dfrac{1+a^2}{(1-au)^2} = \dfrac{\mathrm{d}v}{\mathrm{d}u}$, as required.

The addition formula. Take $a > 0$ and $0 < x < \frac{1}{a}$. For $0 \le u \le x$ we have $1-au \ge 1-ax > 0$, so v is defined and continuous on the whole range, and $\frac{\mathrm{d}v}{\mathrm{d}u} > 0$, so v increases from $v(0) = a$ to $v(x) = \frac{x+a}{1-ax}$.

? **Ask yourself** Why does it matter that $x < \frac{1}{a}$?

If x reached $\frac{1}{a}$, then v would blow up inside the range of integration and the substitution would be meaningless. With $x < \frac{1}{a}$ it is a smooth, increasing change of variable on $[0, x]$.

The derivative result says exactly that $\dfrac{1}{1+u^2} = \dfrac{1}{1+v^2}\dfrac{\mathrm{d}v}{\mathrm{d}u}$, so

$$\begin{aligned} T(x) = \int_0^x \frac{\mathrm{d}u}{1+u^2} &= \int_a^{\frac{x+a}{1-ax}} \frac{\mathrm{d}v}{1+v^2} \\ &= \int_0^{\frac{x+a}{1-ax}} \frac{\mathrm{d}v}{1+v^2} - \int_0^a \frac{\mathrm{d}v}{1+v^2} = T\left(\frac{x+a}{1-ax}\right) - T(a). \end{aligned}$$

Both arguments of T here are positive, so everything is defined.

The first deduction. Write $X = \frac{x+a}{1-ax}$. By part (i) applied to x and then to a,

$$T(x^{-1}) = T_\infty - T(x) = T_\infty - T(X) + T(a) = T_\infty - T(X) + T_\infty - T(a^{-1}),$$

which is $T(x^{-1}) = 2T_\infty - T\left(\frac{x+a}{1-ax}\right) - T(a^{-1})$.

The second deduction. Given $b > 0$ and $y > \frac{1}{b}$, put $x = \frac{1}{y}$ and $a = \frac{1}{b}$. Then $x > 0$, $a > 0$, and $y > \frac{1}{b}$ is the same as $x < b = \frac{1}{a}$, so the previous result applies. Multiplying top and bottom by by,

$$\frac{x+a}{1-ax} = \frac{\frac{1}{y} + \frac{1}{b}}{1 - \frac{1}{by}} = \frac{b+y}{by-1},$$

and $x^{-1} = y$, $a^{-1} = b$, so

$$T(y) = 2T_\infty - T\left(\frac{y+b}{by-1}\right) - T(b).$$

Part (iii)

$T(\sqrt{3})$**.** Take $y = b = \sqrt{3}$ in the last result; the condition $y > \frac{1}{b}$ holds since $\sqrt{3} > \frac{1}{\sqrt{3}}$. Here

$$\frac{y+b}{by-1} = \frac{2\sqrt{3}}{3-1} = \sqrt{3},$$

so $T(\sqrt{3}) = 2T_\infty - T(\sqrt{3}) - T(\sqrt{3})$, that is $3T(\sqrt{3}) = 2T_\infty$ and

$$T(\sqrt{3}) = \tfrac{2}{3}T_\infty.$$

$T(1)$ **first.** Part (i) with $x = 1$ gives $T(1) = T_\infty - T(1)$, so $T(1) = \frac{1}{2}T_\infty$.

Ask yourself Which value of a makes $\frac{x+a}{1-ax}$ equal to 1 when $x = a$?

We need $\frac{2a}{1-a^2} = 1$, i.e. $a^2 + 2a - 1 = 0$, whose positive root is $a = \sqrt{2} - 1$.

$T(\sqrt{2}-1)$**.** So take $x = a = \sqrt{2} - 1 > 0$ in the addition formula. The condition $x < \frac{1}{a}$ is $\sqrt{2} - 1 < \sqrt{2} + 1$, which holds. Since $a^2 = 3 - 2\sqrt{2}$,

$$\frac{x+a}{1-ax} = \frac{2(\sqrt{2}-1)}{1-(3-2\sqrt{2})} = \frac{2(\sqrt{2}-1)}{2(\sqrt{2}-1)} = 1,$$

so $T(\sqrt{2}-1) = T(1) - T(\sqrt{2}-1)$, giving

$$T(\sqrt{2}-1) = \tfrac{1}{2}T(1) = \tfrac{1}{4}T_\infty.$$

Checks

Of course T is really arctan and $T_\infty = \frac{\pi}{2}$, which we weren't allowed to use but can certainly check against: $\frac{\pi}{3} = \frac{2}{3} \cdot \frac{\pi}{2}$ and $\frac{\pi}{8} = \frac{1}{4} \cdot \frac{\pi}{2}$. Evaluating the integrals numerically (Simpson's rule) gives $T_\infty \approx 1.570796$, $T(\sqrt{3}) \approx 1.047198$ and $T(\sqrt{2}-1) \approx 0.392699$, matching $\frac{2}{3}T_\infty$ and $\frac{1}{4}T_\infty$. The addition formula of part (ii) is the familiar $\tan(A+B)$ formula in disguise, with $x = \tan A$ and $a = \tan B$.

Common slips

- Forgetting to reverse the limits after $u = 1/t$: the minus sign from $\mathrm{d}u = -t^{-2}\,\mathrm{d}t$ is exactly what flips $\int_{\infty}^{1/x}$ to $\int_{1/x}^{\infty}$.
- Quietly using $\tan\frac{\pi}{3} = \sqrt{3}$ or arctan facts in part (iii): the question forbids it, and the point is to get everything from parts (i) and (ii).
- Not checking the conditions $x < \frac{1}{a}$ and $y > \frac{1}{b}$ for the particular numbers used.

Beyond STEP

Defining a function by an integral and deriving all its properties from that definition is how analysis courses build ln, exp, arctan and even π rigorously: here π could simply be defined as $2T_{\infty}$. The substitution $v = \frac{u+a}{1-au}$ is a Möbius transformation, and the fact that it preserves $\frac{\mathrm{d}u}{1+u^2}$ reflects the rotations of a circle. For the analysis side, see Abbott, *Understanding Analysis*; for the geometry of Möbius maps, Needham, *Visual Complex Analysis*.

WORKED SOLUTION

STEP III 2017, Question 7

Pure • Tangents to an Ellipse

STEP III 2017, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s3-q7

The key idea

The parameter t turns the condition "the tangent at T passes through (X, Y)" into a quadratic equation in t. Each root is a point of contact, so counting tangents from a point is the same as counting real roots, which is a discriminant question. In part (ii) the two contact parameters p and q are exactly the two roots of that quadratic, so their sum and product come straight from the coefficients.

Solution

The point lies on the ellipse

Substituting the coordinates of T,

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{(1-t^2)^2 + 4t^2}{(1+t^2)^2} = \frac{1 + 2t^2 + t^4}{(1+t^2)^2} = 1,$$

so T lies on the ellipse for every real t.

Part (i)

Differentiating with respect to t,

$$\frac{\mathrm{d}x}{\mathrm{d}t} = \frac{-4at}{(1+t^2)^2}, \qquad \frac{\mathrm{d}y}{\mathrm{d}t} = \frac{2b(1-t^2)}{(1+t^2)^2}.$$

Rather than divide these (awkward when $t = 0$), use the standard tangent to the ellipse at (x_0, y_0), namely $\frac{xx_0}{a^2} + \frac{yy_0}{b^2} = 1$; its normal direction $\left(\frac{x_0}{a^2}, \frac{y_0}{b^2}\right)$ is perpendicular to the velocity above, since $\frac{1-t^2}{a} \cdot (-4at) + \frac{2t}{b} \cdot 2b(1-t^2) = 0$. At T the tangent L is therefore

$$\frac{x(1-t^2)}{a(1+t^2)} + \frac{2ty}{b(1+t^2)} = 1 \iff bx(1-t^2) + 2aty = ab(1+t^2).$$

Putting $(x, y) = (X, Y)$ and collecting powers of t:

$$bX - bXt^2 + 2aYt - ab - abt^2 = 0 \iff (a + X)bt^2 - 2aYt + b(a - X) = 0,$$

as required.

Two tangents. Read the other way round: the tangent at the point with parameter t passes through (X, Y) exactly when t satisfies this equation. As $X \neq -a$ and $b \neq 0$, the coefficient of t^2 is non-zero, so it is a genuine quadratic, with discriminant

$$4a^2Y^2 - 4b^2(a + X)(a - X) = 4\big(a^2Y^2 - (a^2 - X^2)b^2\big).$$

If $a^2Y^2 > (a^2 - X^2)b^2$ this is positive, so there are two distinct real roots $t_1 \neq t_2$. Different values of t give different points of the ellipse (the map $t \mapsto T$ is one-to-one; with $t = \tan\frac{\theta}{2}$ it is the familiar $(a\cos\theta, b\sin\theta)$ for $-\pi < \theta < \pi$). Different points of contact give different tangent lines, because a tangent to an ellipse meets it only at its point of contact. So there are two distinct tangents through (X, Y).

Ask yourself Could we have missed a tangent that the parameter t does not reach?

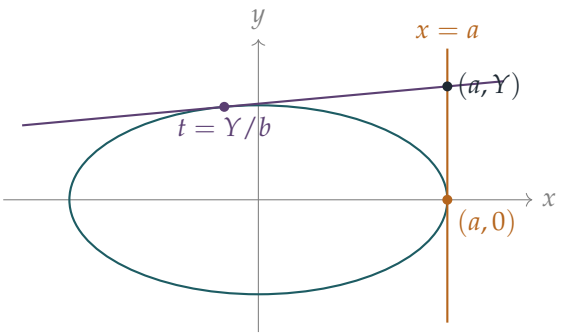
The only point of the ellipse not of the form $(*)$ is $(-a, 0)$ (the limit $t \to \infty$), whose tangent is $x = -a$. That line passes through (X, Y) only if $X = -a$, which is excluded. So the quadratic accounts for every tangent through (X, Y).

Interpretation. Dividing the condition by $a^2b^2 > 0$ gives

$$\frac{Y^2}{b^2} > 1 - \frac{X^2}{a^2} \iff \frac{X^2}{a^2} + \frac{Y^2}{b^2} > 1,$$

which says (X, Y) lies *outside* the ellipse. So from any point outside the ellipse we can draw two tangents to it, just as for a circle.

The case $X^2 = a^2$. Now the condition reads $a^2Y^2 > 0$, i.e. $Y \neq 0$: the point lies on one of the vertical lines $x = \pm a$, but not at the vertex $(\pm a, 0)$. One tangent is simply the vertical line $x = \pm a$ itself, which touches the ellipse at $(\pm a, 0)$. The other touches the ellipse somewhere else: for $X = a$ the quadratic becomes $2abt^2 - 2aYt = 0$, with roots $t = 0$ (the vertical tangent) and $t = Y/b \neq 0$; for $X = -a$ it becomes $-2aYt + 2ab = 0$, giving $t = b/Y$, and the second tangent is $x = -a$, the one at the missing point $(-a, 0)$. The sketch shows the case $X = a$, $Y > 0$.



So the result still holds when $X^2 = a^2$: there are two distinct tangents through (X, Y) whenever $Y \neq 0$, i.e. whenever the point is outside the ellipse.

Part (ii)

The tangent at P passes through (X, Y), so by part (i) p is a root of

$$(a + X)bt^2 - 2aYt + b(a - X) = 0,$$

and likewise q. Since $X \neq -a$ this is a quadratic, and $p \neq q$ (as $P \neq Q$), so p and q are its two roots. Hence

$$pq = \frac{b(a - X)}{b(a + X)} = \frac{a - X}{a + X} \quad \Longrightarrow \quad (a + X)pq = a - X,$$

as required, and

$$p + q = \frac{2aY}{b(a + X)}.$$

The final part

Setting $x = 0$ in the tangent $bx(1 - t^2) + 2aty = ab(1 + t^2)$ gives $2aty = ab(1 + t^2)$. The tangents do meet the y-axis, so $p, q \neq 0$ (at $t = 0$ the tangent is $x = a$), and

$$y_1 = \frac{b(1 + p^2)}{2p}, \qquad y_2 = \frac{b(1 + q^2)}{2q}.$$

Writing $\frac{1+p^2}{p} = p + \frac{1}{p}$,

$$y_1 + y_2 = \frac{b}{2}\left(p + q + \frac{1}{p} + \frac{1}{q}\right) = \frac{b}{2}(p + q)\left(1 + \frac{1}{pq}\right) = \frac{b(p + q)(1 + pq)}{2pq}.$$

From the results above, $1 + pq = \dfrac{2a}{a + X}$, so, using $pq = \dfrac{a - X}{a + X} \neq 0$,

$$y_1 + y_2 = \frac{b}{2} \cdot \frac{2aY}{b(a + X)} \cdot \frac{2a}{a + X} \cdot \frac{a + X}{a - X} = \frac{2a^2Y}{(a + X)(a - X)} = \frac{2a^2Y}{a^2 - X^2}.$$

Setting this equal to $2b$ and multiplying by $(a^2 - X^2)/(2a^2b)$ (non-zero, as $X^2 \neq a^2$):

$$\frac{Y}{b} = \frac{a^2 - X^2}{a^2} = 1 - \frac{X^2}{a^2} \iff \frac{X^2}{a^2} + \frac{Y}{b} = 1,$$

as required. So the intersection point lies on a parabola through $(\pm a, 0)$ and $(0, b)$.

Checks

- With $a = 2$, $b = 1$ and $(X, Y) = (4, 1)$ (outside, since $4 + 1 > 1$), the quadratic is $6t^2 - 4t - 2 = 0$, with roots $t = 1$ and $t = -\frac{1}{3}$: the tangents at $(0, 1)$ and $(1.6, -0.6)$. Both, $y = 1$ and $2x - 3y = 5$, do pass through $(4, 1)$. Here $pq = -\frac{1}{3} = \frac{2-4}{2+4}$ and $p + q = \frac{2}{3} = \frac{4}{6}$, as the formulas predict.

- At the centre $(0,0)$ the discriminant is $-4a^2b^2 < 0$: no tangents, as expected from inside the ellipse.
- Many random ellipses and pairs of points P, Q were checked by computer, intersecting the tangents directly; the formulas for pq, $p+q$ and y_1+y_2 all agreed, and when $y_1+y_2=2b$ the intersection satisfied $X^2/a^2+Y/b=1$.

Common slips

- Finding the gradient as $\frac{\mathrm{d}y/\mathrm{d}t}{\mathrm{d}x/\mathrm{d}t}$ and forgetting that it fails at $t=0$, where the tangent is vertical.
- Treating $(a+X)bt^2-2aYt+b(a-X)=0$ as a quadratic when $X=-a$; this is exactly why the case $X^2=a^2$ needs separate treatment.
- Stopping at "the discriminant is positive" without saying why two roots give two *different* lines.

Beyond STEP

The rational parametrisation $\left(\frac{1-t^2}{1+t^2}, \frac{2t}{1+t^2}\right)$ of a conic is the start of algebraic geometry: it is why every conic with one rational point has infinitely many, and it underlies the classification of Pythagorean triples. The pair of tangents from an external point leads to the idea of the *polar* line and to projective geometry, where the awkward missing point at $t=\infty$ becomes an ordinary point. Siklos, *Advanced Problems in Mathematics*, has more questions of this type.

WORKED SOLUTION

STEP III 2017, Question 8

Pure • Summation by Parts

STEP III 2017, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s3-q8

The key idea

The opening identity is the discrete version of integration by parts ("summation by parts"): it trades a sum involving differences of b for one involving differences of a. To use it on a trigonometric sum, we pick b_m so that $b_{m+1} - b_m$ produces the terms we want; the sum-to-product formula turns a difference of sines or cosines into a single product. Choosing a_m to be 1 or m then makes the other side easy.

Solution

The identity

The quantity $a_{n+1}b_{n+1} - a_1b_1$ telescopes:

$$a_{n+1}b_{n+1} - a_1b_1 = \sum_{m=1}^{n} \left(a_{m+1}b_{m+1} - a_m b_m\right).$$

Adding and subtracting $a_m b_{m+1}$ inside each bracket,

$$a_{m+1}b_{m+1} - a_m b_m = a_m(b_{m+1} - b_m) + b_{m+1}(a_{m+1} - a_m).$$

Summing over $m = 1, \ldots, n$ and rearranging gives exactly the required identity.

Throughout the trigonometric parts we assume $\sin \frac{1}{2}x \neq 0$ (that is, x is not a multiple of 2π), so that $\operatorname{cosec} \frac{1}{2}x$ is defined.

Part (i)

With $b_m = \sin mx$, the sum-to-product formula with $A = (m+1)x$, $B = mx$ gives

$$b_{m+1} - b_m = \sin(m+1)x - \sin mx = 2\cos\left(m + \tfrac{1}{2}\right)x \, \sin \tfrac{1}{2}x.$$

Now take $a_m = 1$ for every m, so that $a_{m+1} - a_m = 0$ and the final sum in the identity vanishes. The identity becomes

$$2\sin\tfrac{1}{2}x \sum_{m=1}^{n} \cos\bigl(m+\tfrac{1}{2}\bigr)x = b_{n+1} - b_1 = \sin(n+1)x - \sin x.$$

Dividing by $2\sin\frac{1}{2}x \neq 0$,

$$\sum_{m=1}^{n} \cos\bigl(m+\tfrac{1}{2}\bigr)x = \tfrac{1}{2}\bigl(\sin(n+1)x - \sin x\bigr)\operatorname{cosec}\tfrac{1}{2}x,$$

as required.

Part (ii)

The factor m suggests $a_m = m$, so that $a_{m+1} - a_m = 1$. We then need $b_{m+1} - b_m$ to be a multiple of $\sin mx$.

Ask yourself Which choice of b_m makes $b_{m+1} - b_m$ a multiple of $\sin mx$?

In part (i), the difference of two sines at $(m+1)x$ and mx gave a cosine at the *midpoint* $(m+\frac{1}{2})x$. To get $\sin mx$ we want a difference of cosines whose arguments have midpoint mx, so we take $b_m = \cos\bigl(m-\frac{1}{2}\bigr)x$. Using $\cos A - \cos B = -2\sin\frac{A+B}{2}\sin\frac{A-B}{2}$ (which also follows from the first note in the question),

$$b_{m+1} - b_m = \cos\bigl(m+\tfrac{1}{2}\bigr)x - \cos\bigl(m-\tfrac{1}{2}\bigr)x = -2\sin mx\,\sin\tfrac{1}{2}x.$$

Write $S = \sum_{m=1}^{n} m\sin mx$. With $a_m = m$ the identity gives

$$-2\sin\tfrac{1}{2}x\,S = (n+1)\cos\bigl(n+\tfrac{1}{2}\bigr)x - \cos\tfrac{1}{2}x - \sum_{m=1}^{n}\cos\bigl(m+\tfrac{1}{2}\bigr)x.$$

The last sum is exactly part (i). Substituting it and multiplying through by $-2\sin\frac{1}{2}x$:

$$4\sin^2\tfrac{1}{2}x\,S = 2\sin\tfrac{1}{2}x\cos\tfrac{1}{2}x - 2(n+1)\cos\bigl(n+\tfrac{1}{2}\bigr)x\,\sin\tfrac{1}{2}x + \sin(n+1)x - \sin x.$$

Now simplify the two products:

- $2\sin\frac{1}{2}x\cos\frac{1}{2}x = \sin x$, which cancels the $-\sin x$;
- by $2\cos A\sin B = \sin(A+B) - \sin(A-B)$ with $A = (n+\frac{1}{2})x$, $B = \frac{1}{2}x$,

$$2\cos\bigl(n+\tfrac{1}{2}\bigr)x\,\sin\tfrac{1}{2}x = \sin(n+1)x - \sin nx.$$

Hence

$$\begin{aligned} 4\sin^2\tfrac{1}{2}x\,S &= -(n+1)\bigl(\sin(n+1)x - \sin nx\bigr) + \sin(n+1)x \\ &= -n\sin(n+1)x + (n+1)\sin nx. \end{aligned}$$

Dividing by $4\sin^2\frac{1}{2}x$,

$$\sum_{m=1}^{n} m\sin mx = \left(-\tfrac{n}{4}\sin(n+1)x + \tfrac{n+1}{4}\sin nx\right)\operatorname{cosec}^2\tfrac{1}{2}x,$$

so

$$p = -\frac{n}{4}, \qquad q = \frac{n+1}{4}.$$

Checks

- For $n = 1$ the formula gives $\frac{1}{4}(2\sin x - \sin 2x)\operatorname{cosec}^2\frac{1}{2}x = \frac{1}{2}\sin x\,(1-\cos x)\operatorname{cosec}^2\frac{1}{2}x = \sin x$, using $1-\cos x = 2\sin^2\frac{1}{2}x$. That is the single term $1\cdot\sin x$.
- Numerically, with $n = 3$ and $x = 1$: $\sin 1 + 2\sin 2 + 3\sin 3 \approx 3.08343$, and $\left(-\frac{3}{4}\sin 4 + \sin 3\right)\operatorname{cosec}^2\frac{1}{2} \approx 3.08343$ too.
- Part (ii) can also be found by differentiating $\sum_{m=1}^{n}\cos mx$ with respect to x, which gives $-S$; this is a useful independent check.

Common slips

- Choosing $b_m = \cos mx$ in part (ii): the difference is then a multiple of $\sin(m+\frac{1}{2})x$, not $\sin mx$, and the method stalls.
- Sign errors in $\cos A - \cos B$: it is $-2\sin\frac{A+B}{2}\sin\frac{A-B}{2}$, with a minus sign.
- Forgetting to state that $\sin\frac{1}{2}x \neq 0$ before dividing by it.

Beyond STEP

Summation by parts (Abel summation) is the discrete twin of integration by parts, and it is a workhorse of first-year analysis. Combined with bounds like part (i), which shows the partial sums of $\cos(m+\frac{1}{2})x$ stay bounded, it proves Dirichlet's test: for example, $\sum\frac{\sin mx}{m}$ converges for every x. Sums like part (ii) also appear in Fourier series, where such closed forms are the starting point for studying convergence. Abbott's *Understanding Analysis* treats Abel summation and Dirichlet's test clearly.

WORKED SOLUTION

STEP III 2017, Question 9

Mechanics • Connected Particles and Energy

STEP III 2017, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

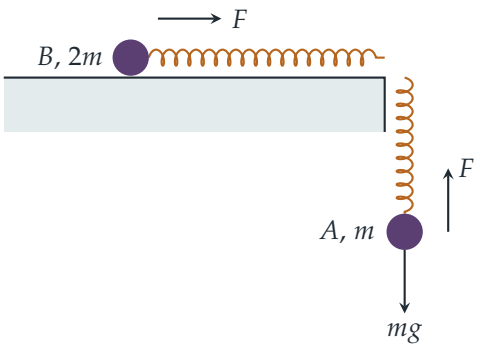
step.maths.org/questions/17-s3-q9

The key idea

The spring tension is unknown and changes with time, but it acts with equal size on both particles. So if we add the two equations of motion in the right proportions, the tension cancels and leaves something we can integrate straight away. In the second part, that integrated relation tells us where both particles are at time T and how their speeds are linked. Conservation of energy then gives a second equation, and the two together fix the speed of B.

Solution

We model A and B as particles and the spring as light. The table is smooth, and the spring runs smoothly over the edge of the table. Once released, A falls vertically and B slides along the table towards the edge. Let the tension in the spring be F at time t. While $x < a$, B is still on the table.



The relation $y + 2x = \frac{1}{2}gt^2$

Take y positive downwards for A and x positive towards the edge for B. By Newton's second law:

$$\begin{aligned} \text{for } A \text{ (vertically):} \quad & m\ddot{y} = mg - F, \\ \text{for } B \text{ (horizontally):} \quad & 2m\ddot{x} = F. \end{aligned}$$

(For B, the weight and the normal reaction cancel, and there's no friction.) Adding the two equations gets rid of F:

$$m\ddot{y} + 2m\ddot{x} = mg \quad \Longrightarrow \quad \ddot{y} + 2\ddot{x} = g.$$

Both particles start from rest with $x = y = 0$. Integrating once gives $\dot{y} + 2\dot{x} = gt$, with no constant because both velocities are zero at $t = 0$. Integrating again gives

$$y + 2x = \tfrac{1}{2}gt^2,$$

again with no constant, which is the required result.

Ask yourself We don't know F, since it depends on the extension $y - x$, which keeps changing. How can we still say anything?

We never need F. It appears with opposite signs in the two equations once we multiply them by the right factors (here, just add them). The combination $y + 2x$ is $1/m$ times the momentum-like quantity $my + 2mx$, and the only external force driving it is A's weight.

The speed of B at time T

At time $T = \sqrt{6a/g}$, particle B reaches the edge, so $x = a$. Since $\frac{1}{2}gT^2 = \frac{1}{2}g \cdot \frac{6a}{g} = 3a$, the first part gives

$$y + 2a = 3a \quad \Longrightarrow \quad y = a.$$

So the extension of the spring at time T is $y - x = a - a = 0$. The spring is at its natural length again, and it stores no elastic energy.

Write u and v for the speeds of B and A at time T. The once-integrated relation $\dot{y} + 2\dot{x} = gt$ gives, at $t = T$,

$$v + 2u = gT = \sqrt{6ag}. \tag{1}$$

Energy. The table is smooth and the spring is light, so mechanical energy is conserved. At the start, both particles are at rest and the spring is unstretched. At time T, A has dropped a distance a and the spring is unstretched again, so the loss of potential energy mga has all become kinetic energy:

$$\tfrac{1}{2}mv^2 + \tfrac{1}{2}(2m)u^2 = mga \quad \Longrightarrow \quad v^2 + 2u^2 = 2ga. \tag{2}$$

Ask yourself Two equations, one of them quadratic: shouldn't we expect two possible speeds?

Substituting $v = \sqrt{6ag} - 2u$ from (1) into (2):

$$\begin{aligned} 6ag - 4\sqrt{6ag}\,u + 4u^2 + 2u^2 &= 2ga \\ 6u^2 - 4\sqrt{6ag}\,u + 4ag &= 0 \\ 3u^2 - 2\sqrt{6ag}\,u + 2ag &= 0. \end{aligned}$$

The discriminant is $4 \cdot 6ag - 4 \cdot 3 \cdot 2ag = 0$, so the quadratic is a perfect square and there's only one root:

$$\left(\sqrt{3}\,u - \sqrt{2ag}\right)^2 = 0 \quad \Longrightarrow \quad u = \sqrt{\frac{2ag}{3}}.$$

So the speed of B when it reaches the edge is $\sqrt{2ag/3}$, as required. From (1), $v = \sqrt{6ag} - 2\sqrt{2ag/3} = \sqrt{2ag/3}$ as well. At that moment the two particles move with the same speed, which fits with the spring being at its natural length.

Checks

- The repeated root has a meaning. For fixed total 'momentum' $v + 2u$, the kinetic energy $\frac{1}{2}mv^2 + mu^2$ is smallest when $v = u$, and that smallest value is exactly mga here. So energy and (1) can only both hold if $u = v$, and then (1) gives $3u = \sqrt{6ag}$ directly.
- Solving the motion in full (which the question tells us not to do) gives $\omega T = 2\pi$, where $\omega^2 = 3\lambda/(2am)$: the extension goes through exactly one full oscillation and comes back to zero. A step-by-step numerical integration with $a = 1.3\,\mathrm{m}$ and $g = 9.8\,\mathrm{m\,s^{-2}}$ gives $x(T) = y(T) = a$ and both speeds equal to $2.9143\,\mathrm{m\,s^{-1}} = \sqrt{2ag/3}$.
- Dimensions: $\sqrt{ag}$ is a speed, and $\sqrt{a/g}$ is a time.

Common slips

- Forgetting that the spring has elastic energy in general. The argument works only because $x = y = a$ at time T, which makes the extension zero. Show this explicitly.
- Writing the energy equation alone and trying to solve it for u. It has two unknowns, so we need relation (1) as well.

Beyond STEP

The quantity $my + 2mx$ behaves like a momentum: internal forces (here the spring tension) cancel in it, and only external forces change it. That's the start of the mechanics

of systems of particles. In general you split the motion into a centre-of-mass part and a relative part, and here the relative coordinate $y - x$ performs simple harmonic motion. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops these ideas carefully.

WORKED SOLUTION

STEP III 2017, Question 10

Mechanics • Rotating Rigid Body

STEP III 2017, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s3-q10

The key idea

While the particle does not slip, rod and particle turn together as one rigid body about the hinge, so conservation of energy gives $\dot\theta$ straight away, and differentiating gives $\ddot\theta$. The particle then moves on a circle about P, so its acceleration has the familiar radial and transverse components. Newton's second law for the particle alone, resolved along and perpendicular to the rod, gives the friction and normal reaction, and slipping starts when friction reaches μ times the reaction.

Solution

The energy equation

Model the particle as a point mass and the hinge as smooth. The moment of inertia of the uniform rod about its end P is $\frac{1}{3}m(3a)^2 = 3ma^2$, and the particle, a distance ℓ from P, contributes $m\ell^2$. While there is no slipping, the system is a rigid body turning about P with angular speed $\dot\theta$, and its kinetic energy is

$$\tfrac{1}{2}\left(3ma^2 + m\ell^2\right)\dot\theta^2.$$

When the rod has turned through θ, its centre (at distance $\frac{3}{2}a$ from P) has fallen $\frac{3}{2}a\sin\theta$ and the particle has fallen $\ell\sin\theta$. The hinge reaction does no work (its point of application is fixed), and friction and the normal reaction are internal to the system and do no net work when there is no slipping. Starting from rest, conservation of energy gives

$$\tfrac{1}{2}m\left(3a^2 + \ell^2\right)\dot\theta^2 = mg\left(\tfrac{3}{2}a + \ell\right)\sin\theta,$$

which, multiplied by $2/m$, is

$$(3a^2 + \ell^2)\dot\theta^2 = g(3a + 2\ell)\sin\theta,$$

as required.

The angular acceleration

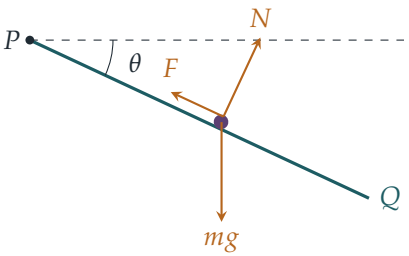
Differentiating with respect to t,

$$2(3a^2+\ell^2)\dot\theta\,\ddot\theta = g(3a+2\ell)\cos\theta\,\dot\theta.$$

For $t>0$ we have $\dot\theta\neq 0$, so we may divide by $2\dot\theta$ (the result at $t=0$ follows by continuity, or directly by taking moments about P):

$$\ddot\theta = \frac{g(3a+2\ell)\cos\theta}{2(3a^2+\ell^2)}.$$

Forces on the particle



The rod lies at angle θ below the horizontal. The rod pushes on the particle with a normal reaction N perpendicular to the rod (away from it, since the particle sits on top) and a friction force F along the rod, which we take to act towards P. The particle moves on a circle of radius ℓ about P, so its acceleration has a component $\ell\ddot\theta$ perpendicular to the rod in the direction of increasing θ (down and away from the rod's upper face), and a component $\ell\dot\theta^2$ along the rod towards P.

Perpendicular to the rod, in the direction of increasing θ, the weight has component $mg\cos\theta$ and N acts the other way:

$$mg\cos\theta - N = m\ell\ddot\theta.$$

Along the rod, towards P, the weight has component $-mg\sin\theta$ (it pulls the particle away from P):

$$F - mg\sin\theta = m\ell\dot\theta^2.$$

The normal reaction

Using $\ddot\theta$ from above,

$$\begin{aligned} N &= mg\cos\theta - m\ell\ddot\theta = mg\cos\theta\left(1-\frac{\ell(3a+2\ell)}{2(3a^2+\ell^2)}\right)\\ &= mg\cos\theta\,\frac{6a^2+2\ell^2-3a\ell-2\ell^2}{2(3a^2+\ell^2)} = \frac{3mga(2a-\ell)\cos\theta}{2(3a^2+\ell^2)}. \end{aligned}$$

Since $\ell<2a$, the factor $2a-\ell$ is positive, and $\cos\theta>0$ for acute θ. So $N>0$; in particular the normal reaction is non-zero, and the particle stays in contact with the rod, while θ is acute.

The point of slipping

From the second equation and the energy equation,

$$F = mg\sin\theta + m\ell\dot{\theta}^2 = mg\sin\theta\left(1 + \frac{\ell(3a+2\ell)}{3a^2+\ell^2}\right)$$
$$= mg\sin\theta\,\frac{3a^2+3a\ell+3\ell^2}{3a^2+\ell^2} = \frac{3mg(a^2+a\ell+\ell^2)\sin\theta}{3a^2+\ell^2}.$$

This is positive for $0 < \theta < \frac{1}{2}\pi$, so friction does act towards P, as assumed: without it the particle would slide outwards along the rod.

Ask yourself Why is the condition for slipping $F = \mu N$, rather than an inequality?

While the particle grips the rod, friction can supply any force up to μN, so we need $F \le \mu N$. At $\theta = 0$ we have $F = 0 < \mu N$. As θ increases, F grows (like $\sin\theta$) and N shrinks (like $\cos\theta$), so F/N, which is a positive multiple of $\tan\theta$, increases steadily. The particle is on the point of slipping at the first angle where $F = \mu N$:

$$\frac{3mg(a^2+a\ell+\ell^2)\sin\theta}{3a^2+\ell^2} = \mu\,\frac{3mga(2a-\ell)\cos\theta}{2(3a^2+\ell^2)}.$$

Cancelling $3mg/(3a^2+\ell^2)$ and dividing by $\cos\theta \neq 0$,

$$\tan\theta = \frac{\mu a(2a-\ell)}{2(\ell^2+a\ell+a^2)},$$

as required.

If $\ell > 2a$

The formula for N was derived assuming the particle moves with the rod. If $\ell > 2a$, it gives $N < 0$ at $\theta = 0$, which is impossible: the rod can only push on the particle, not pull it. So the particle cannot move with the rod, even at the instant of release.

To see what does happen, consider the rod on its own. Taking moments about P at the moment of release,

$$3ma^2\,\ddot{\theta} = mg\cdot\tfrac{3}{2}a \quad\Longrightarrow\quad \ddot{\theta} = \frac{g}{2a},$$

so the point of the rod beneath the particle starts with downward acceleration $\ell\ddot{\theta} = \frac{\ell}{2a}g$, which is greater than g when $\ell > 2a$. The rod falls away faster than the particle can fall freely. So at the moment of release the particle leaves the rod (the reaction between them is zero) and begins to fall freely under gravity, while the rod swings down on its own.

Checks

- Taking moments about P for the whole system gives $(3ma^2+m\ell^2)\ddot{\theta} = mg\cos\theta\,(\frac{3}{2}a+\ell)$, the same $\ddot{\theta}$ as differentiating the energy equation.

- The boundary case $\ell = 2a$ fits both pictures: $N = 0$ at release, and the rod's point at distance $2a$ starts with acceleration exactly g.
- With $a = \ell = 1$ and $\mu = 1$, the slipping angle has $\tan\theta = \frac{1}{6}$, and at that angle the formulas above give $F/N = 6\tan\theta = 1 = \mu$. The formulas for N and F were also confirmed by computer, by stepping the motion numerically and computing the particle's acceleration directly from its position.

Common slips

- Using the moment of inertia about the centre, $\frac{1}{12}m(3a)^2$, instead of about the hinge; the moment of inertia of a rod of length L about an end is $\frac{1}{3}mL^2$.
- Forgetting the centripetal term $m\ell\dot{\theta}^2$ in the friction equation; it is what makes the numerator $a^2 + a\ell + \ell^2$ rather than a^2.
- Getting the sign of $\ell\ddot{\theta}$ wrong in the normal equation: the rod is accelerating away from the particle, so N is less than $mg\cos\theta$, not more.

Beyond STEP

The last part is a classic surprise: a point on a falling hinged rod can accelerate faster than g, which is why a coin placed near the end of a falling ruler (or a chimney toppling over) gets left behind. The systematic way to find constraint forces like N and F is to write the equations of motion in polar coordinates, and Lagrangian mechanics later shows how to get the motion without them at all. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats rigid-body rotation and these polar-coordinate accelerations carefully.

WORKED SOLUTION

STEP III 2017, Question 11

Mechanics • Conservation of Momentum

STEP III 2017, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s3-q11

This question has a full guided treatment in *Beyond STEP*, Section 10.3, *Rockets and Guns*: https://beyondstep.keithmcnulty.org/ch10/rockets.html

The key idea

No external horizontal force acts on the system, so horizontal momentum is conserved at every firing. The odd-looking ground speed $v - V$ is really saying something simple: the projectile always leaves at speed v *relative to the truck*. For the kinetic energy, don't recompute everything from scratch each time; use $(*)$ to simplify the change in energy at one firing, and then add up.

Solution

Model the truck, guns and projectiles as particles moving along one line, and take the forward direction (the way the truck moves) as positive. A projectile fired when the truck has speed V has velocity $-(v - V) = V - v$, that is, velocity $-v$ relative to the truck. The firings are internal to the system and the track is smooth, so the total horizontal momentum is conserved at each firing.

Part (i)

Before firing everything is at rest, so the momentum is zero. With $V = 0$, each projectile leaves with velocity $-v$, and the truck (mass M) moves forwards with speed u:

$$Mu - nmv = 0 \quad \Longrightarrow \quad u = \frac{nmv}{M}.$$

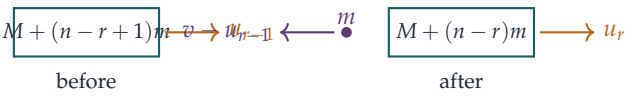
The system started with no kinetic energy, so the energy gained is

$$K = \tfrac{1}{2}Mu^2 + \tfrac{1}{2}nmv^2 = \tfrac{1}{2}\,\frac{n^2m^2v^2}{M} + \tfrac{1}{2}nmv^2 = \tfrac{1}{2}nmv^2\left(1 + \frac{nm}{M}\right),$$

as required.

Part (ii)

Consider the rth firing. Just before it, the truck carries $n-r+1$ unfired projectiles, so the moving mass is $M+(n-r+1)m$ with speed u_{r-1}. Just afterwards, mass $M+(n-r)m$ moves with speed u_r and one projectile of mass m has velocity $u_{r-1}-v$. The projectiles fired earlier are unaffected and can be left out.



Conservation of momentum gives

$$\big(M+(n-r+1)m\big)u_{r-1} = \big(M+(n-r)m\big)u_r + m(u_{r-1}-v).$$

Write the left-hand side as $\big(M+(n-r)m\big)u_{r-1} + mu_{r-1}$. The mu_{r-1} terms then cancel, leaving

$$\big(M+(n-r)m\big)(u_r-u_{r-1}) = mv, \qquad \text{so} \qquad u_r-u_{r-1} = \frac{mv}{M+(n-r)m},$$

which is $(*)$.

Adding $(*)$ for $r=1,\dots,n$, the left-hand side telescopes and $u_0=0$, so

$$u_n = \sum_{r=1}^{n} \frac{mv}{M+(n-r)m}.$$

Each denominator satisfies $M+(n-r)m \geq M$, so each term is at most $\dfrac{mv}{M}$. The inequality is strict for $r=1$, because $n>1$ makes $(n-1)m>0$. Hence

$$u_n < n\cdot\frac{mv}{M} = u.$$

Ask yourself Why should firing one at a time be worse?

In the early firings the gun has to push not only the truck but also all the projectiles still on board, so each shot gives a smaller increase in speed. The term $r=n$ (the last shot, with nothing left on board) is the only one that matches a share of the simultaneous firing.

Part (iii)

At the rth firing, only the truck (with its remaining load) and the projectile just fired change their velocities. So

$$\begin{aligned} K_r - K_{r-1} &= \tfrac{1}{2}\big(M+(n-r)m\big)u_r^2 + \tfrac{1}{2}m(u_{r-1}-v)^2 - \tfrac{1}{2}\big(M+(n-r+1)m\big)u_{r-1}^2 \\ &= \tfrac{1}{2}\big(M+(n-r)m\big)(u_r^2-u_{r-1}^2) + \tfrac{1}{2}m\big[(v-u_{r-1})^2 - u_{r-1}^2\big] \\ &= \tfrac{1}{2}\big(M+(n-r)m\big)(u_r-u_{r-1})(u_r+u_{r-1}) + \tfrac{1}{2}m(v^2-2vu_{r-1}). \end{aligned}$$

By $(*)$, $\big(M+(n-r)m\big)(u_r-u_{r-1})=mv$, so the first term is $\frac{1}{2}mv(u_r+u_{r-1})$. Therefore

$$\begin{aligned}K_r-K_{r-1}&=\tfrac{1}{2}mvu_r+\tfrac{1}{2}mvu_{r-1}+\tfrac{1}{2}mv^2-mvu_{r-1}\\&=\tfrac{1}{2}mv^2+\tfrac{1}{2}mv(u_r-u_{r-1}),\end{aligned}$$

as required.

Adding this for $r=1,\ldots,n$, and using $K_0=0$ and $u_0=0$, both sides telescope:

$$K_n=\tfrac{1}{2}nmv^2+\tfrac{1}{2}mv(u_n-u_0)=\tfrac{1}{2}nmv^2+\tfrac{1}{2}mvu_n.$$

Finally, compare with part (i). Since $u=nmv/M$,

$$K=\tfrac{1}{2}nmv^2+\tfrac{1}{2}\,\frac{n^2m^2v^2}{M}=\tfrac{1}{2}nmv^2+\tfrac{1}{2}nmv\,u.$$

From part (ii), $u_n<u$, and $u<nu$ because $u>0$ and $n>1$. As $mv>0$,

$$K_n=\tfrac{1}{2}nmv^2+\tfrac{1}{2}mvu_n<\tfrac{1}{2}nmv^2+\tfrac{1}{2}mvu<\tfrac{1}{2}nmv^2+\tfrac{1}{2}nmvu=K.$$

So $K_n<K$: firing one at a time releases less kinetic energy.

Checks

- Take $n=3$, $M=10$, $m=1$, $v=1$. Then $u=0.3$, while $(*)$ gives $u_3=\frac{1}{12}+\frac{1}{11}+\frac{1}{10}=\frac{181}{660}\approx 0.274$, which is smaller, as it should be. The energies are $K=1.95$ and $K_3=\frac{2161}{1320}\approx 1.637$.
- Working out the momentum and the kinetic energy of every body directly after each shot, for thousands of random choices of n, M, m and v, agrees exactly with $(*)$ and with the formula for K_n; this was confirmed by computer.
- Limiting case: if $m\ll M$, every term of $(*)$ is close to mv/M, so $u_n\approx u$, and both energies are close to $\frac{1}{2}nmv^2$. When the truck is very heavy, the order of firing hardly matters.

Common slips

- Forgetting that the unfired projectiles ride with the truck, and using mass M instead of $M+(n-r+1)m$ before the rth shot.
- Giving the projectile velocity $-v$ instead of $u_{r-1}-v$: the question fixes its speed relative to the ground as $v-V$, not v.
- Including the kinetic energy of earlier projectiles in K_r-K_{r-1}. They are unaffected and cancel.

Beyond STEP

This is a discrete version of rocket motion. If the n shots are replaced by a continuous stream of exhaust, the sum for u_n becomes an integral and gives the Tsiolkovsky rocket equation, $\Delta u = v \ln(m_0/m_1)$. A rocket is in the position of the truck firing one shot at a time: its fuel can't all be burnt in a single instant, so it pays the price that part (ii) uncovers. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats variable-mass systems and rockets carefully.

WORKED SOLUTION

STEP III 2017, Question 12

Probability and Statistics • Joint Discrete Distributions

STEP III 2017, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s3-q12

The key idea

Everything follows from two standard sums, $\sum_{r=1}^{n} r = \frac{1}{2}n(n+1)$ and $\sum_{r=1}^{n} r^2 = \frac{1}{6}n(n+1)(2n+1)$. First fix k by making the probabilities add up to 1. Then find the marginal distribution by summing over y. For the covariance we need $\mathrm{E}(X)$ and $\mathrm{E}(XY)$. The symmetry of $x+y$ in x and y halves the work.

Solution

Part (i)

The probabilities over all n^2 pairs (x,y) must add up to 1. By symmetry, $\sum_{x,y} x = \sum_{x,y} y$. Each of these equals $n\sum_{x=1}^{n} x = \frac{1}{2}n^2(n+1)$, since every x appears once for each of the n values of y. So

$$1 = k\sum_{x=1}^{n}\sum_{y=1}^{n}(x+y) = k\cdot 2\cdot \tfrac{1}{2}n^2(n+1) = k\,n^2(n+1), \qquad \text{so} \qquad k = \frac{1}{n^2(n+1)}.$$

For the marginal distribution of X we sum over y, with x fixed:

$$\begin{aligned}\mathrm{P}(X=x) &= k\sum_{y=1}^{n}(x+y) = k\Big(nx + \tfrac{1}{2}n(n+1)\Big) = \frac{kn\,(2x+n+1)}{2}\\ &= \frac{n+1+2x}{2n(n+1)},\end{aligned}$$

which is the required result. Since $x+y$ is symmetric in x and y, Y has the same distribution: $\mathrm{P}(Y=y) = \dfrac{n+1+2y}{2n(n+1)}$.

Ask yourself To show that X and Y are *not* independent, do we have to check every pair (x,y)?

No. Independence needs $\mathrm{P}(X=x, Y=y) = \mathrm{P}(X=x)\,\mathrm{P}(Y=y)$ for *every* pair, so one pair where it fails is enough. Take $x=y=1$:

$$\mathrm{P}(X=1,Y=1) = 2k = \frac{2}{n^2(n+1)}, \qquad \mathrm{P}(X=1)\,\mathrm{P}(Y=1) = \frac{(n+3)^2}{4n^2(n+1)^2}.$$

These are equal exactly when $8(n+1) = (n+3)^2$, that is $n^2 - 2n + 1 = 0$, or $(n-1)^2 = 0$. So they are equal only for $n=1$. Since $n \geq 2$, the two probabilities differ, and X and Y are **not independent**.

Part (ii)

We use $\mathrm{Cov}(X,Y) = \mathrm{E}(XY) - \mathrm{E}(X)\,\mathrm{E}(Y)$, and note that $\mathrm{E}(Y) = \mathrm{E}(X)$ because the two variables have the same distribution.

The mean. Using the marginal from part (i),

$$\begin{aligned}\mathrm{E}(X) &= \frac{1}{2n(n+1)} \sum_{x=1}^{n} \big((n+1)x + 2x^2\big) = \frac{1}{2n(n+1)} \Big(\tfrac{1}{2}n(n+1)^2 + \tfrac{1}{3}n(n+1)(2n+1)\Big) \\ &= \frac{1}{2}\Big(\frac{n+1}{2} + \frac{2n+1}{3}\Big) = \frac{7n+5}{12}.\end{aligned}$$

The mean of the product. Here $xy(x+y) = x^2y + xy^2$, and the two parts give equal double sums by symmetry. Also, the double sum of x^2y factorises as a product of single sums:

$$\begin{aligned}\mathrm{E}(XY) &= k \sum_{x=1}^{n}\sum_{y=1}^{n} xy(x+y) = 2k\Big(\sum_{x=1}^{n} x^2\Big)\Big(\sum_{y=1}^{n} y\Big) = 2k \cdot \frac{n(n+1)(2n+1)}{6} \cdot \frac{n(n+1)}{2} \\ &= \frac{n^2(n+1)^2(2n+1)}{6\,n^2(n+1)} = \frac{(n+1)(2n+1)}{6}.\end{aligned}$$

The covariance. Putting these together over the common denominator 144:

$$\begin{aligned}\mathrm{Cov}(X,Y) &= \frac{(n+1)(2n+1)}{6} - \frac{(7n+5)^2}{144} = \frac{24(2n^2+3n+1) - (49n^2+70n+25)}{144} \\ &= \frac{-n^2+2n-1}{144} = -\frac{(n-1)^2}{144}.\end{aligned}$$

Since $n \geq 2$, $(n-1)^2 > 0$, so $\mathrm{Cov}(X,Y) = -\dfrac{(n-1)^2}{144} < 0$. The covariance is negative, as required.

It makes sense that the covariance is negative. The joint probability depends only on $x+y$, which encourages a large total. If X is already large, Y does not need to be large as well to make the total large, so Y has slightly less reason to be big.

Checks

- Take $n=2$. Then $k = \frac{1}{12}$, and the four probabilities are $\frac{2}{12}, \frac{3}{12}, \frac{3}{12}, \frac{4}{12}$, which add up to 1. The marginals are $\mathrm{P}(X=1) = \frac{5}{12}$ and $\mathrm{P}(X=2) = \frac{7}{12}$, matching the formula.

So $\mathrm{E}(X) = \frac{19}{12}$ and $\mathrm{E}(XY) = \frac{2+6+6+16}{12} = \frac{5}{2}$. This gives $\mathrm{Cov}(X,Y) = \frac{5}{2} - \frac{361}{144} = -\frac{1}{144}$, which agrees with $-(n-1)^2/144$.

- The marginal probabilities add up to 1: $\sum_x (n+1+2x) = n(n+1) + n(n+1) = 2n(n+1)$.
- The value of $(n-1)^2$ that makes the independence test fail is the same one that makes the covariance zero. In both, $n = 1$ is the degenerate case where $X = Y = 1$ with certainty.
- A computer check of the exact distributions for $n = 2, \ldots, 29$ confirmed k, the marginals, $\mathrm{E}(X)$, $\mathrm{E}(XY)$ and the covariance formula.

Common slips

- Forgetting that $\sum_{x,y} x$ has a factor n from the sum over y. This gives $k = 1/(n(n+1))$, which is wrong.
- Trying to show non-independence "in general" rather than with one explicit pair. A single counterexample is both enough and the clearest argument.
- Thinking that a nonzero covariance is needed to show dependence. Here it does show dependence, but zero covariance would *not* prove independence.

Beyond STEP

A joint distribution of the form $f(x,y) = g(x+y)$ is a simple model of variables that are linked through their total. The sign of the covariance then depends on the shape of g. Here g is linear, and the covariance is a small negative quantity. At university, joint distributions, marginals, conditional distributions and correlation are studied together, with continuous versions using double integrals in place of double sums. Grimmett and Welsh, *Probability: An Introduction*, covers this material in a style close to STEP.

WORKED SOLUTION

STEP III 2017, Question 13

Probability and Statistics • Expectation and Variance

STEP III 2017, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s3-q13

The key idea

Write $X - x$ as $(X - \mu) + (\mu - x)$. The first piece has mean zero, so when we square and take expectations the cross term vanishes. That leaves the variance plus a constant, which is the same identity as "$\mathrm{E}(X^2) = \mathrm{Var}(X) + \mu^2$". For the uniform case, Y is a function of X that is not one-to-one, so the safest way to its density is through its cumulative distribution function.

Solution

The function V

For any fixed number x,

$$\begin{aligned} \mathrm{V}(x) = \mathrm{E}\big((X-x)^2\big) &= \mathrm{E}\big(((X-\mu)+(\mu-x))^2\big) \\ &= \mathrm{E}\big((X-\mu)^2\big) + 2(\mu-x)\,\mathrm{E}(X-\mu) + (\mu-x)^2 \\ &= \sigma^2 + (x-\mu)^2, \end{aligned}$$

using linearity of expectation (here $\mu - x$ is a constant) and $\mathrm{E}(X - \mu) = 0$.

The mean of Y

Since $\mathrm{V}(x) = \sigma^2 + (x - \mu)^2$ holds for every real x, we may put $x = X$:

$$Y = \mathrm{V}(X) = \sigma^2 + (X-\mu)^2.$$

? **Ask yourself** Isn't $\mathrm{V}(X)$ just a number, since V was defined by an expectation?

No: V is an ordinary function of a real variable (a quadratic in x). The expectation was taken with x held fixed. Feeding the random variable X into that quadratic gives a new random variable. So

$$\mathrm{E}(Y) = \sigma^2 + \mathrm{E}\big((X-\mu)^2\big) = \sigma^2 + \sigma^2 = 2\sigma^2,$$

which is $(*)$.

The uniform case

For X uniform on $[0,1]$, $\mu = \frac{1}{2}$ and $\sigma^2 = \int_0^1 x^2\,\mathrm{d}x - \frac{1}{4} = \frac{1}{3} - \frac{1}{4} = \frac{1}{12}$. Hence

$$\mathrm{V}(x) = \tfrac{1}{12} + \left(x - \tfrac{1}{2}\right)^2 = x^2 - x + \tfrac{1}{3}.$$

Distribution of Y. Now $Y = \frac{1}{12} + (X - \frac{1}{2})^2$. As X runs over $[0,1]$, $|X - \frac{1}{2}|$ runs over $[0, \frac{1}{2}]$, so Y takes values in $[\frac{1}{12}, \frac{1}{12} + \frac{1}{4}] = [\frac{1}{12}, \frac{1}{3}]$.

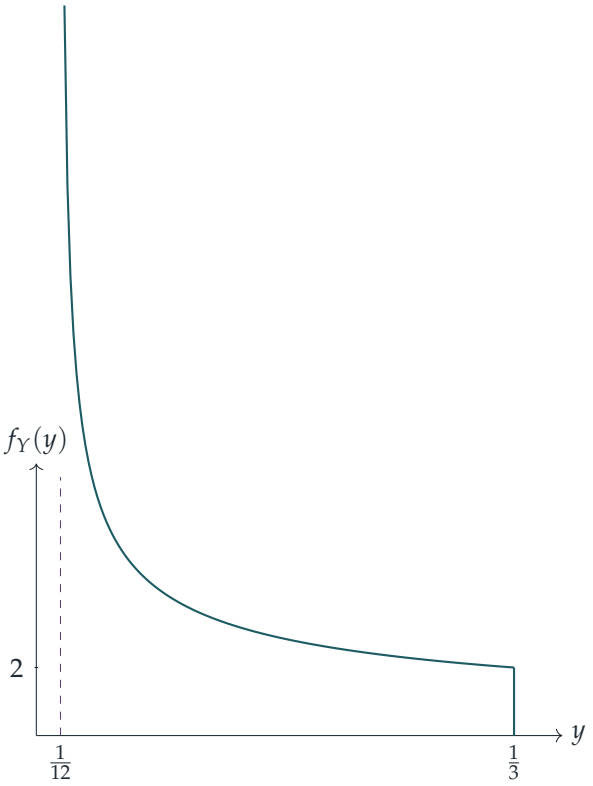
The map $X \mapsto Y$ is two-to-one (the points $\frac{1}{2} \pm h$ give the same Y), so we work with the distribution function. For $\frac{1}{12} \le y \le \frac{1}{3}$,

$$\begin{aligned} \mathrm{P}(Y \le y) &= \mathrm{P}\big((X - \tfrac{1}{2})^2 \le y - \tfrac{1}{12}\big) = \mathrm{P}\big(\tfrac{1}{2} - s \le X \le \tfrac{1}{2} + s\big), \quad \text{where } s = \sqrt{y - \tfrac{1}{12}}, \\ &= 2s = 2\sqrt{y - \tfrac{1}{12}}, \end{aligned}$$

because $0 \le s \le \frac{1}{2}$, so the interval lies inside $[0,1]$ and has length $2s$. (At $y = \frac{1}{3}$ this gives $2 \cdot \frac{1}{2} = 1$, as it should.) Differentiating,

$$f_Y(y) = \frac{1}{\sqrt{y - \frac{1}{12}}} \quad \text{for } \tfrac{1}{12} < y \le \tfrac{1}{3}, \qquad f_Y(y) = 0 \text{ otherwise.}$$

The density is unbounded near $y = \frac{1}{12}$, but it is still integrable: Y is most likely to be close to its minimum, because $(X - \frac{1}{2})^2$ is flat near $X = \frac{1}{2}$.



Checking $(*)$. Using the density, and substituting $t = y - \frac{1}{12}$ (so t runs from 0 to $\frac{1}{4}$),

$$\mathrm{E}(Y) = \int_{1/12}^{1/3} \frac{y}{\sqrt{y - \frac{1}{12}}}\,\mathrm{d}y = \int_0^{1/4} \left(t^{1/2} + \tfrac{1}{12}t^{-1/2}\right)\mathrm{d}t = \left[\tfrac{2}{3}t^{3/2} + \tfrac{1}{6}t^{1/2}\right]_0^{1/4}$$
$$= \tfrac{2}{3}\cdot\tfrac{1}{8} + \tfrac{1}{6}\cdot\tfrac{1}{2} = \tfrac{1}{12} + \tfrac{1}{12} = \tfrac{1}{6}.$$

Since $2\sigma^2 = 2\cdot\frac{1}{12} = \frac{1}{6}$, this agrees with $(*)$.

Checks

- The density integrates to 1: $\int_0^{1/4} t^{-1/2}\,\mathrm{d}t = 2\cdot\frac{1}{2} = 1$.
- $\mathrm{V}(x)$ is smallest at $x = \mu$, with minimum value σ^2: the mean is the point that minimises expected squared distance. In the uniform case $\mathrm{V}(\frac{1}{2}) = \frac{1}{12}$, the bottom of the range of Y.
- A computer simulation with 400 000 uniform values gave a sample mean of Y of about 0.1667, matching $\frac{1}{6}$.

Common slips

- Treating $\mathrm{V}(X)$ as the constant σ^2, which gives $\mathrm{E}(Y) = \sigma^2$.
- Using the one-to-one change-of-variable formula $f_Y = f_X \left|\mathrm{d}x/\mathrm{d}y\right|$ with only one branch, which gives half the correct density. Both $X = \frac{1}{2} + s$ and $X = \frac{1}{2} - s$ contribute.

Beyond STEP

$\mathrm{V}(x) = \sigma^2 + (x - \mu)^2$ is the simplest bias–variance decomposition. If you estimate a quantity by x, the mean squared error splits into the spread of the data plus the squared bias. The result $\mathrm{E}(Y) = 2\sigma^2$ says that the expected squared distance between two independent copies of X is $2\sigma^2$. Statisticians use this fact again and again, for example when explaining why the sample variance divides by $n - 1$. Blitzstein and Hwang, *Introduction to Probability*, covers both ideas, and also finding the density of a non-monotonic function of a random variable.

Beyond STEP

STEP is the hardest mathematics examination most school students ever sit. It is also one of the best: its questions are short, unfamiliar and deep, and the best of them are small pieces of university mathematics in disguise.

This book takes the most interesting STEP questions, from the specimen papers that preceded the first examination in 1987 to the papers of 2025, and uses each one as a doorway. Every question is solved carefully, with the questions a good student asks along the way, and then the book steps through the door: where the idea comes from, what it becomes at university, and what to read next.

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